

DEMONSTRATION OF BIORETENTION FOR TREATMENT OF AGRICULTURAL RUNOFF

Final Report

for

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by

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LIST OF ACRONYMS

Acronym	Definition
ADPs	Antecedent Dry Periods
BMPs	Best Management Practices
CBA	Completely Biochar Amended
CEC	Cation Exchange Capacity
COD	Chemical Oxygen Demand
CRF	Controlled Release Fertilizer
DI	Deionized Water
DO	Dissolved Oxygen
DON	Dissolved Organic Nitrogen
DWT	Drinking Water Treatment
EBCT	Empty Bed Contact Time
EPA	Environmental Protection Agency
FDACS	Florida Department of Agriculture and Consumer Services
FIB	Fecal Indicator Bacteria
HLR	Hydraulic Loading Rate
HRT	Hydraulic Residence Time
IWSZ	Internal Water Storage Zone
MRT	Mean Residence Time
NLR	Nutrient Loading Rate
NO _x	The sum of NO ₃ ⁻ -N plus NO ₂ ⁻ -N
NRR	Nutrient Removal Rate
NS	Not (Statistically) Significant
P	Phosphorus
PBA	Partially Biochar Amended
PSD	Pore Size Distribution
SA	Specific Surface Area
TAN	Total Ammonia Nitrogen (sum of NH ₄ ⁺ -N + NH ₃ -N)
TIN	Total Inorganic Nitrogen (sum of NH ₄ ⁺ -N + NO ₂ ⁻ -N + NO ₃ ⁻ -N)
TP	Total Phosphorus
UIUC	University of Illinois Urbana-Champaign

INTRODUCTION

Background

Agriculture plays a vital role in Florida's economy, generating \$7.35 billion worth of products in 2017 (USDA-NASS 2019), including \$5.7 billion in nursery and greenhouse crops. Florida agriculture is also a major non-point source of pollutants to ground and surface waters, including solids, organic matter, nutrients (e.g., N and P) and fecal indicator bacteria (FIB) (FDEP 2017). Extensive irrigation and fertilization are required for production of marketable plants in Florida (Sharma et al., 2008). Nursery irrigation runoff contains high concentrations of nitrate (NO_3^-) and phosphate (PO_4^{3-}), which vary depending on the season and irrigation and fertilizer management practices (Wilson et al., 2006). These nutrients can migrate to groundwater or are carried with runoff to rivers, springs and marine ecosystems.

In 2022, FDACS Office of Agricultural Water Policy (OAWP) contracted with University of South Florida (USF) investigators, Ergas and Nachabe, to develop and test biochar enhanced bioretention systems as potential best management practices (BMPs) for removal of nutrients from nursery runoff. Bioretention systems are passive bioreactors that remove suspended solids, organic carbon, nutrients and FIB in stormwater through filtration, adsorption and biodegradation (Lopez-Ponnada et al., 2020). These systems can be modified to incorporate an upturned outlet (Figure 1), resulting in the formation of a saturated internal water storage zone (IWSZ). During infiltration, runoff percolates through an unsaturated layer where nitrification ($\text{NH}_4^+ \Rightarrow \text{NO}_3^-$) is promoted by aerobic conditions. Nitrate generated in the unsaturated layer migrates to the IWSZ, which can be amended with a solid, slow-release electron donor such as wood chips to promote the activity of denitrifying bacteria ($\text{NO}_3^- \Rightarrow \text{N}_2$).

The innovation of our project was to incorporate biochar into the bioretention media to enhance nutrient removal from nursery runoff. Biochar is a low-cost by-product of pyrolysis of organic waste material, such as wood waste, livestock manure or crop residues, which is produced under oxygen limited conditions. Biochar has a high cation exchange capacity (CEC) and specific surface area (SA) and has been widely used as an amendment for agricultural soils. Although prior studies have investigated biochar amended bioretention for treatment of agricultural and urban runoff (Rahman et al., 2020; 2021; 2024), there is a lack of information on the use of this technology for treatment of nursery runoff.

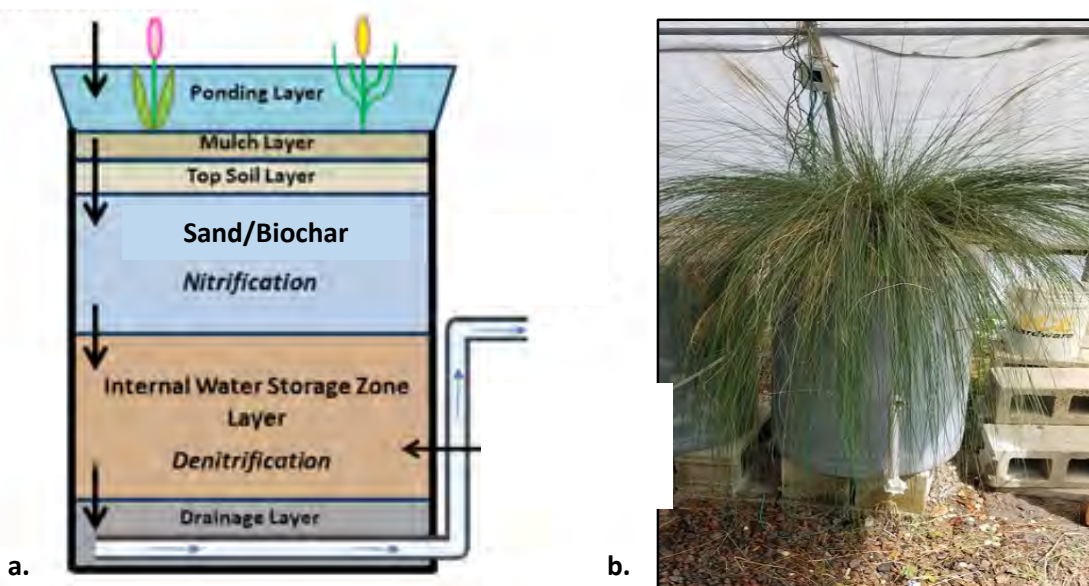


Figure 1: Modified bioretention systems: a) schematic, b) photo of Phase I pilot system.

Unlike N, the flux of P to the atmosphere is negligible and most commercially available biochars have limited adsorption capacity for PO_4^{3-} due to their negative surface charge (Yao et al., 2012). Therefore, biochar amended bioretention systems are expected to ultimately leach phosphate from the medium. A number of different low-cost adsorbent materials have been used to remove phosphate from agricultural runoff (Altamira-Algarra et al., 2022), including modified biochars, industrial and agricultural wastes and synthetic materials. However, research was needed on the best approach for bioretention media amendment to improve P removal from nursery runoff.

Phase I Summary

Phase I of this research (FDACS 025851) investigated a novel engineered bioretention medium - sand amended with biochar to improve the performance of bioretention systems treating runoff from dairy farms (Rahman et al., 2020; 2021; 2024). In bench-scale column studies, we showed that biochar amended bioretention systems achieved significantly higher total nitrogen (TN) and *E. coli* removals and better hydraulic performance than conventional sand media bioretention systems (Rahman et al., 2020; 2021). Bench-scale studies were also used to determine the optimal biochar amendment rate for subsequent pilot studies. In pilot-scale studies at the USF botanical gardens (Figure 1b), side-by-side bioretention experiments were conducted with semi-synthetic dairy runoff at varying hydraulic loading rates (HLRs) to compare the performance of modified bioretention systems with four different configurations: 1) sand media (S), 2) sand media planted with Muhly Grass (*Muhlenbergia*) (SP), 3) sand/biochar media (B), 4) sand/biochar media planted with Muhly Grass (BP) (Rahman et al., 2024). The biochar amended bioretention system with plants achieved > 90% total ammonia nitrogen (TAN) removal, >95% TN removal, > 89% TP removal and > 6 log *E. coli* removal. The results showed that the biochar helps to

retain NH_4^+ , dissolved organic nitrogen (DON) and dissolved organic carbon (DOC), and enhanced both nitrification in the aerobic zone and denitrification in the IWSZ. Biochar also improved infiltration capacity, increased plant growth in BP compared with SP (Figure 1), which increases plant nutrient uptake and rhizosphere microbial activity. Note that the Phase I study did not investigate P removal or look at removal of other types of agricultural runoff, other than dairy runoff, which is high in ammonium, organic carbon and FIB.

Phase II Research Objectives

Our Phase II project built on our Phase I project to contribute to OAWP's mission to develop, adopt and assist with the implementation of agricultural BMPs to protect and conserve water resources. Phase II addressed the need to demonstrate bioretention systems to stakeholders at an actual field site for treatment of nursery runoff. There is tremendous variability in the characteristics of agricultural runoff including hydraulic and pollutant loading rates, temperature and antecedent dry periods (ADP). These variables impact performance and design of a bioretention system, such as the hydraulic residence time (HRT) and the biodegradation kinetics of the key microbial communities such as nitrifiers and denitrifiers. Additional research was also needed to understand the potential for P adsorption and recovery from nursery runoff. Therefore, the objectives of the Phase II project were divided into the following tasks:

- 1) Design, construct and evaluate biochar enhanced bioretention systems for removal of nutrients from plant nursery runoff;
- 2) Conduct on-farm demonstration of the technology, particularly for producers that participate in FDACS BMP program;
- 3) Investigate the use of modified biochar for recovery of P from plant nursery runoff.
- 4) Evaluate the potential to use spent modified biochar as a slow-release fertilizer and soil amendment.
- 5) Develop design guidelines to contribute to the catalogue of BMPS provided by FDACS to facilitate technology transfer and implementation.

Note that after preliminary research was carried out on P adsorption with modified biochar for Task 3, the scopes for Tasks 3 and 4 were broadened to evaluate additional readily available adsorbent materials for P adsorption and recovery, not just modified biochar adsorbents.

MATERIALS AND METHODS

The Phase II project was carried out from January 2022 through June 2024 as shown in Table 1.

Table 1: Project overview showing deliverables and tasks.

Deliverable #	Purpose	Tasks/Deliverables
1	Site Selection and Runoff Characteristics	• Select field site for study • Characterize runoff flow and chemical composition • Design two pilot-scale bioretention units
2	Construct Field-Scale Bioretention Systems	• Submit photographs of the completed construction
3	Bioretention Monitoring and Data Analysis	• Develop sampling and analytical protocols and quality assurance/quality control (QA/QC) plans • Monitor bioretention units under varying conditions: – Effect of biochar amendment in the IWSZ – Effect of HLR – Effect of outlet elevation – Effect of plants
4	P Removal and Recovery Using Adsorbents	• Determine P capacity of commercial biochar, modified biochar, and readily available phosphorus adsorbents • Determine the potential of P recovery from saturated adsorbents
5	Create a Bioretention Design Spreadsheet Model	• Recommend media materials, particle size composition, permeability and porosity • Provide guidelines on bioretention design • Provide information on cost estimation
6	Research Dissemination	• Presentations at conferences • MS thesis • Peer reviewed publication • Webinar and workshop at field site • Undergraduate and graduate student training

Deliverable #1: Site Selection and Runoff Characteristics:

- FDACS staff identified two nurseries within 25 miles of the USF campus that were participating in FDACS BMP program. The USF team visited Harrell's Nursery in Plant City

and Holmberg Nursery in Lithia in January 2022 and met with the nursery managers to discuss the project. Holmberg Farm was selected as the study site.

- An initial runoff characterization was performed at Holmberg Farm. Samples were collected every 10 to 15 minutes during irrigation events and transported to the USF laboratories to test the chemical composition (TIN, TP, $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, $\text{PO}_4^{3-}\text{-P}$, COD). Irrigation flow rates were also measured during the characterization studies.

Site Selection

FDACS staff identified two nurseries within 25 miles of the USF campus who were participating in FDACS BMP program. The USF team visited Harrell's Nursery in Plant City and Holmberg Nursery in Lithia in January 2022 and met with the nursery managers to discuss the project. Holmberg Nursery was selected as the study site due to the strong support and willingness to participate from the nursery manager. Holmberg Nursery is a large commercial nursery that grows and distributes citrus trees, flowering plants, and fruit plants. It is located at 13430 Hobson Simmons Rd, Lithia, FL 33547 (Figure A1). Runoff from greenhouses used for citrus production could be captured from a culvert and then treated in a bioretention basin, a favorable feature for this study (Figure A2).

Runoff Characterization

The initial runoff characterization for the Holmberg farm was performed in 2022 before the bioretention systems were built, using open channel hydraulic modeling procedures. The diameter, water depth and velocity of the fluid in the drainage culvert were measured at three locations at 30-minute intervals throughout the irrigation periods to understand how flow rate changes with time (Figure A3). Fluid velocity was measured using a Hach FH950 (Loveland, CO) portable velocity meter. In addition, runoff samples were collected during four irrigation events and transported to the USF laboratories to test the chemical composition (TP, $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, $\text{PO}_4^{3-}\text{-P}$, COD). Runoff flow rate and chemical composition results shown in Table A1 were used to develop the preliminary design for the bioretention basin.

In 2023, after the construction of two bioretention systems, additional runoff characterization was carried out during three irrigation events. For each event, sampling started when runoff started flowing in the culvert, approximately one hour after irrigation commenced in the greenhouse. For each event, three influent and effluent samples (6 total) were collected, every 10 to 15 minutes, stored on ice, and brought to the USF laboratory for water quality analysis (TP, $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, $\text{PO}_4^{3-}\text{-P}$, pH, COD, Alkalinity). Results of runoff water quality are shown in Table A2. The velocity was measured using an OTT MF pro (Kempton, Germany) water flow meter. The area of runoff was calculated by measuring the height of the runoff in the culvert, assuming a circular channel with a 61 cm diameter. The flow rate was calculated by multiplying the measured runoff velocity by the channel cross-sectional area. A steady state (75 minutes – 220 minutes) average runoff flow rate of 1.57 cfs (44 L/s) was calculated based on these results (Figure A14).

Preliminary characterization showed that the runoff was high in NO_3^- -N and low in NH_4^+ -N (Table A1, Table A2), indicating nutrient leaching from the slow release fertilizer being used (Florikan). The manufacturer's information on the fertilizer mentions that it provides slow release of 16% nitrogen and 5% available phosphate. The chemical oxygen demand (COD) of the runoff was relatively low (~55 mg/L). Therefore, the bioretention design was modified from the Phase I system to include an IWSZ containing wood chips to promote denitrification.

Deliverable #2: Construct Field-Scale Bioretention Systems

- Two 90-gallon (341 L) molded polyethylene tanks (0.91m x 0.61m x 0.61m) were used to construct the field-scale bioretention systems at Holmberg Farm. The first tank was completely biochar amended (CBA) in both the IWSZ and unsaturated layer, and the second tank was partially biochar amended (PBA), with biochar included only in the unsaturated layer. The bioretention systems were fed influent runoff for 1 month to acclimate microbial communities prior to testing.

Materials

Proprietary high permeability sand media (HPS) was generously donated by a stormwater management company. The sand was washed with tap water to remove dust and soil. Biochar was generously donated by Biochar Now (Berthoud, CO). Biochar was sieved with a #8 sieve using a RO-TAP™ Sieve Shaker to achieve particles > 0.09 in. Oak wood chips were obtained from a local tree service. Oak woodchips were hand sorted to achieve particle size between 1-4 in. Gravel was obtained from Sefner Rock & Gravel (Tampa, FL). Porosity tests were performed on biochar + HPS (1:1 by vol), and biochar + HPS + wood chip (2:2:1 by vol) following ASTM protocols.

Two Muhly grass (*Muhlenbergia capillaris*) plants were transplanted into the bioretention systems on 12/12/2023 (Figure A23 and Figure A24). The two muhly grass plants were purchased from Holmberg Farms (Lithia, FL), where the project is located. Muhly Grass is a native Florida perennial, that attracts wildlife and has favorable light and moisture requirements, growth rate and mature plant height and spread.

Pilot Scale Bioretention Systems

Two 90-gallon (341 L) molded polyethylene tanks (0.91m x 0.61m x 0.61m) were purchased from United States Plastics Corp. (Lima, OH). Each tank was supported by an external frame constructed from 2x4 lumber. The outlet was configured with an underdrain and “upturned elbow” using 2.5 cm internal diameter PVC pipes (Figure A5). The upturned elbow was configured to allow testing of varying outlet elevation; the lower outlet was 44 cm high, and the upper outlet was 69 cm high. The heights of the IWSZ were designed to match the top of the 46 cm layers that contained woodchips (54 cm high) (Figure A7), or the top of the 25 cm layer of HPS + biochar (79 cm high) (Figure A8). A PVC ball valve on the lower outlet could be closed to allow for adjustments to the upper outlet height as needed (Figure A10).

One bioretention unit had biochar amendments in both the IWSZ and unsaturated layer and is referred to as the completely biochar amended unit, or CBA unit. One bioretention unit had biochar amendment only in the unsaturated layer and is referred to as the partially biochar amended unit, or PBA unit. The CBA unit consisted of the following layers (from bottom to top): 5.1 cm gravel drainage layer, 2.5 cm HPS drainage layer, 46 cm HPS + biochar + Woodchip layer (2:2:1 by volume), 25 cm HPS + biochar layer, 2.5 cm HPS layer, 10 cm free board (Figure 2a). The PBA unit consisted of the following layers (from bottom to top): 5 cm gravel drainage layer, 2.5 cm HPS drainage layer, 46 cm HPS + Woodchip layer (4:1 by volume), 25 cm HPS + biochar layer, 2.5 cm HPS layer, 10 cm free board (Figure 2b). In both units, gravel and HPS layers at the bottom of the system were designed to prevent clogging in the PVC pipes (Figure A6) and a HPS layer was placed at the top to prevent biochar from floating to the surface (Figure A9).

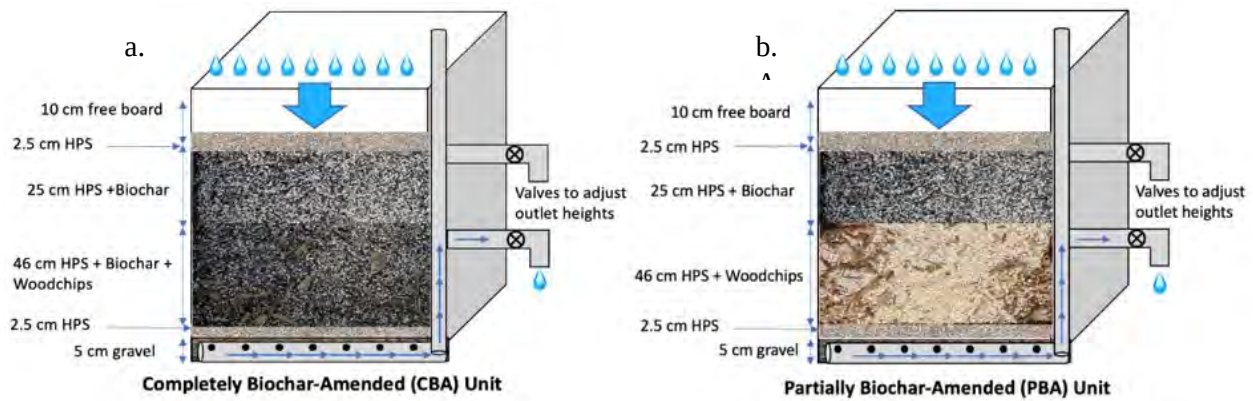


Figure 2: Design schematics for the a) completely biochar amended (CBA) unit and b) partially biochar amended (PBA) unit.

Auxiliary equipment included a screen, pump and float switch (Danner Manufacturing, Inc., Islandia, NY). The system was powered by a Jackery (Freemont, CA) power supply and solar panels (Figure 3). The float switch, power supply and solar panels allowed for automatic operation of the bioretention systems whenever irrigation was initiated. The feed pipe was also equipped with a float valve that allowed irrigation runoff to bypass the bioretention systems once the tanks were full. Portable power stations were enclosed in a Husky Five gallon professional-duty storage container (Pacific, MO) to prevent weather or insects from destroying the electrical wiring.

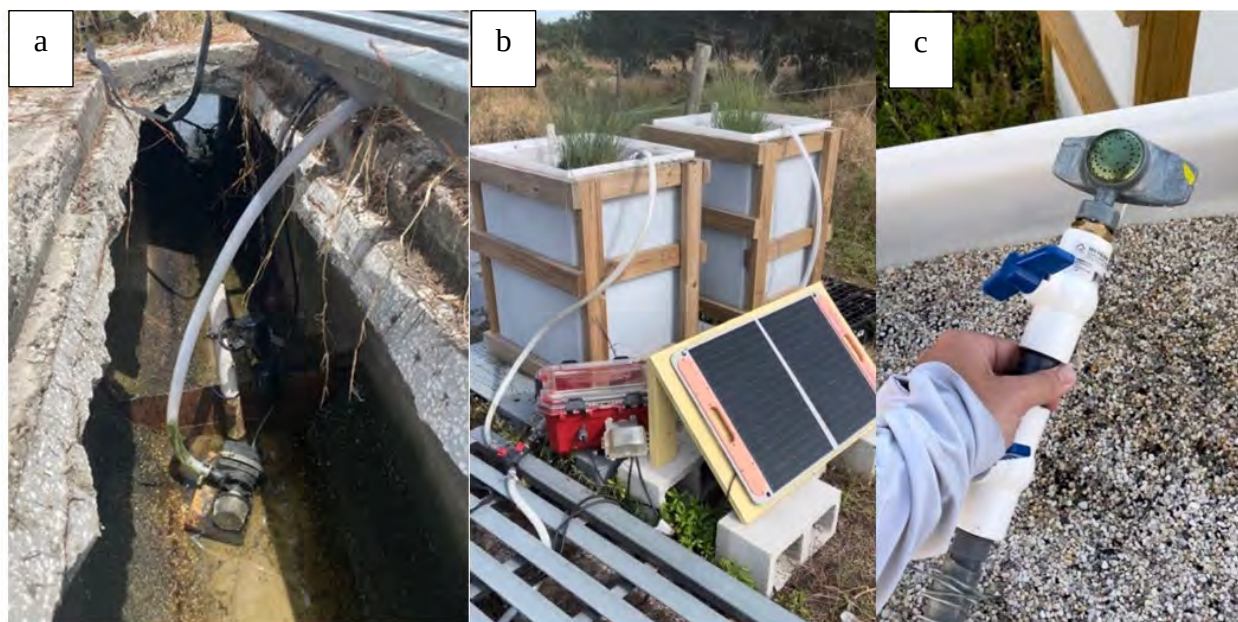


Figure 3: Field bioretention setup: a) pump, weir, and float switch in collection slough, b) pilot-scale bioretention systems, power supply, and solar panel, c) valves and sprinkler for flow control and dispersion.

Deliverable #3: Bioretention Monitoring and Data Analysis

- Tracer studies were conducted on both bioretention systems (CBA and PBA) with varying outlet heights (54 cm vs. 79 cm) using potassium chloride (KCl) as a conservative tracer. This allowed us to investigate the influence of biochar amendment and outlet elevation on HRT.
- The two bioretention systems were initially monitored during 15 runoff events with varying configurations to investigate the effect of HLR (low, medium, and high), outlet elevation (lower and upper), and plant presence (with and without Muhly grass) on nutrient removal. Each study was carried out in triplicate (3 events with same configuration). During each event, influent runoff and effluent samples from both systems were collected every 30 minutes for 3 hours, resulting in 21 samples.
- A One-Way analysis of variance (ANOVA) test with a significant threshold set at 95% confidence interval (α value of 0.05) was performed to determine the significance of HLR, outlet height, and plant presence on nutrient removal between the data sets. Three additional storm events (16–18) were studied using the optimal bioretention design configuration for nutrient removal, which was determined from the analysis of the data collected during the first 15 storm events (1–15).

Tracer Studies

Tracer studies were performed on both bioretention systems at both outlet heights with a constant HLR, and without Muhly grass. To start the tracer study, a 1 L slug of 20 g/L potassium chloride (KCl) solution was sprayed onto the bioretention surface. The systems were then fed runoff until

the effluent conductivity was equal to the influent conductivity. Effluent conductivity was measured every 5 minutes, and influent conductivity was measured every 30 minutes. Figure A26 to Figure A31 show the tracer concentration over time for each test.

Bioretention Monitoring

The target HLR was calculated using Equation 1, assuming that the bioretention surface area would be designed as a percentage of the greenhouse area. The runoff flow rate was based on the results of the influent runoff characterization study. The irrigated greenhouse area was 49,000 m² (12 acres). This resulted in target HLRs of 0.11 cm/min, 0.22 cm/min, and 0.55 cm/min for bioretention areas of 5%, 2.5% and 1% of the greenhouse area, respectively.

$$HLR = \frac{44 \frac{L}{s}}{(49,000 \text{ m}^2) * \left(\frac{\% \text{ of area}}{100}\right)} * \frac{\text{m}^3}{1000 \text{ L}} * \frac{100 \text{ cm}}{\text{m}} * \frac{1 \text{ min}}{60 \text{ s}} \quad (1)$$

The bioretention systems were monitored over the course of 18 runoff events (Table 2) to evaluate the effect of media materials, outlet elevation, HLR, and plant presence. During each event, influent and effluent samples were collected every 30 minutes for 3 hours (21 samples total). Sampling would start when runoff was observed in the culvert. Influent flow rates were measured every 30 minutes to ensure that systems were operating at the target HLR.

Table 2: Summary of bioretention monitoring runoff events.

Event #	Date	HLR (cm/min)	Estimated % area covered*	Outlet Height	Muhly grass?
1	6-Sep-23	0.11	5.0	Lower	No
2	12-Sep-23	0.11	5.0	Lower	No
3	26-Sep-23	0.11	5.0	Lower	No
4	18-Oct-23	0.22	2.5	Lower	No
5	25-Oct-23	0.22	2.5	Lower	No
6	31-Oct-23	0.22	2.5	Lower	No
7	3-Nov-23	0.55	1.0	Lower	No
8	8-Nov-23	0.55	1.0	Lower	No
9	16-Nov-23	0.55	1.0	Lower	No
10	20-Nov-23	0.55	1.0	Upper	No
11	29-Nov-23	0.55	1.0	Upper	No
12	6-Dec-23	0.55	1.0	Upper	No
13	22-Jan-24	0.55	1.0	Lower	Yes
14	31-Jan-24	0.55	1.0	Lower	Yes
15	8-Feb-24	0.55	1.0	Lower	Yes
16	20-Mar-24	0.22	2.5	Upper	Yes
17	10-Apr-24	0.22	2.5	Upper	Yes
18	18-Apr-24	0.22	2.5	Upper	Yes

*based on Equation 1. HLR: Hydraulic Loading Rate

Analytical Methods

A summary of the analytical methods used for water quality analysis is provided in Table 3. Concentrations of NO_2^- , NO_3^- , NH_4^+ , SO_4^{2-} , and PO_4^{3-} were measured after samples were filtered through a 0.45 μm membrane filter (Fisher Scientific, Waltham, MA) using a Metrohm 850 Ion Chromatography System (Herisau, Switzerland). TIN was calculated as the sum of NO_2^- -N, NO_3^- -N, and NH_4^+ -N. TP was measured using Hach (Loveland CO) TNT test kits and a Lovibond spectrophotometer (Dortmund, Germany). COD concentrations were measured using Lovibond test kits and a Lovibond spectrophotometer. Standard curves for the COD and TP measurements are shown in Figure A35 and Figure A36, respectively. pH and Alkalinity were measured using calibrated Orion 5-Star meter and probe (Thermo Scientific, Beverly, MA). All samples were refrigerated when stored and brought to room temperature before any measurements. Dissolved oxygen (DO) concentrations were measured in the field during events 8-12 using a YSI pPoquatro multiparameter meter (Yellow Springs, OH). Conductivity was measured using a YSI Proquatro multiparameter meter (Yellow Springs, OH) during the tracer study. Conductivity was converted to KCl concentration through a standard curve for the tracer, shown in Figure A25. Method Detection Limits (MDLs) are determined using Standard Method protocols (APHA et al., 2018).

Table 3: Water quality parameters, measurement methods and quality assurance procedures.

Test	Instrument or Kit used	Standard method*	MDL or range	Calibration frequency	Max sample storage time
TP	Hach TNT 844	4500-P J 4500-P E	0.05 mg/L	Annually	3 days
PO_4^{3-}	-	4500-P E	0.05 mg/L	Annually	48 hrs
COD	Lovibond™ MD600 COD	5220 D	6.77 mg/L	Annually	3 days
NO_3^- -N NO_2^- -N NH_4^+ -N PO_4^{3-} SO_4^{2-}	Metrohm 850 IC system	4110 B	0.01mg/L 0.04mg/L 0.2mg/L 0.02mg/L Not determined	Every eluent change	48 hr
pH	Orion meter	4500-H ⁺ B	0-14 ($\pm$ 0.2 pH units)	Once per day with pH 4, 7, & 10 buffers	24 hr
Alkalinity	Dosimeter Plus Titrator	2320 B	20mg/L as CaCO_3	Once per day	24 hr

*APHA (2018) *Standard methods for examination water and wastewater*. American Public Health Association. 23rd Edn. APHA, AWWA, WPCF, Washington D.C, USA. 2018.

Deliverable #4: P Removal and Recovery Using Adsorbents

- Initial screening studies were conducted with eight (8) adsorbent materials, including two (2) untreated biochars, one (1) proprietary modified biochar, two (2) recycled industrial materials, and three (3) commercially available phosphorus adsorbents. Materials were tested in small batch experiments to assess P adsorption capacity and kinetics.
- Based on the small batch results, the P removal kinetics of four adsorbent materials were tested further in batch and column tests.
- Materials from the four adsorbent materials recovered from the column experiments were oven-dried and tested for potential P recovery by carrying out desorption experiments.

Screening Studies

Eight (8) adsorbents were initially screened for P adsorption capacity and uptake kinetics: Sunshine Organics and Compost (Jacksonville FL) biochar, Biochar Now (Berthoud, CO) biochar, Crushed Mixed Seashells, PO₄Sponge manufactured by MetaMateria (Columbus, OH), Phosflow distributed by PET Water Solutions (Milwaukee, WI), proprietary HPS described above, modified ‘designer’ biochar produced at the University of Illinois Urbana-Champaign (UIUC), and iron-based drinking water treatment (DWT) residuals obtained from the David Tippin Water Treatment Facility in Tampa, FL. The proprietary HPS was the same material used in the bioretention design, which was provided by a stormwater management company. PO₄Sponge and Phosflow are commercially available proprietary phosphate adsorbents. Sunshine Organics Biochar, Crushed Mixed Seashells, and DWT residuals are recycled industrial materials. UIUC biochar is also a proprietary phosphate adsorbent but is not commercially available and has only been used for research (Yang et. al, 2021). Since Sunshine Organics Biochar is produced in Jacksonville, Florida, from waste wood materials, it was selected as the biochar for testing in removal and recovery. Biochar Now, the type of biochar used in the bioretention design, was also tested in a small-batch kinetic experiment to determine if there was a difference between the two untreated commercially available biochar products.

Small-batch P adsorption kinetic and capacity experiments were carried out using phosphate solutions with variable concentrations created with a phosphate monobasic potassium phosphate (KH₂PO₄) and potassium phosphate dibasic (K₂HPO₄), resulting in a neutral pH. Small-batch kinetic experiments were performed by placing 0.25 g of the adsorbent in 25 mL of 5 mg PO₄-P/L. Centrifuge tubes were placed horizontally on a shaker at variable times of 1 minute, 30 minutes, and 24 hours. After shaking, aliquots of supernatant were filtered through a 0.45 µm filters and tested for dissolved phosphate using the Metrohm 850 Ion Chromatography system. Small-batch adsorption capacity experiments were conducted by placing media in 40 mL of solution with variable P concentrations (0, 2.5, 5, 10, 20, and 40 mg PO₄-P/L) for 24 hours. After shaking, aliquots of supernatant were filtered through a 0.45 µm filters and tested for inorganic dissolved phosphate (PO₄-P) using Standard Method 4500-P E.

Large Batch and Column Studies

Based on the small-batch experiments, the kinetics of four selected adsorbents were tested in additional batch and column experiments to compare the difference in kinetics in a batch and flow-through system. The materials tested were: 1) Phosflow, 2) HPS, 3) Drinking Water Treatment Residuals, 4) UIUC modified biochar. For the ‘large-batch’ kinetic experiment, 50 g of each adsorbent was continuously stirred in 2-L synthetic runoff. Samples of the adsorbate were collected over 48 hours. For the column studies, selected adsorbents were packed into a 500 mL Ko-flow acrylic column with a length of 40 cm and a diameter of 5.1 cm. Synthetic runoff was pumped in an up-flow direction through the adsorbent materials using a peristaltic pump. Flow rate was measured gravimetrically. Samples in the ‘large-batch’ kinetic and column studies were tested for inorganic phosphate using Standard Method 4500-P E.

Data Analysis

Phosphorus adsorbed (q , mg of adsorbate /g of adsorbent) was used in all adsorption experiments and was calculated using Equation 2 for small batch experiments. Where V is the volume of adsorbate solution (L), m is the dry mass of the adsorbent (g), C_0 is the initial adsorbate concentration (mg/L), and C is the final adsorbate concentration (mg/L). For large batch and column experiments q is calculated over time (q_t) using adsorbate concentration measurements over time (C_t). For flow through experiments, V from Equation 2 is substituted with flow rate (Q) multiplied by the time step between sampling C_t (Δt) resulting in Equation 3.

$$q = V * \frac{(C_0 - C)}{m} \quad (2)$$

$$q_t = (Q \Delta t) \frac{(C_{0,avg} - C_t)}{m} \quad (3)$$

The capacity of adsorbents was determined by plotting q as a function of C after 24 hours. Data points were visually inspected over the range of C , if a linear trend was observed, a linear relationship (Equation 4) was fit with K_L , as the isotherm parameter. Although most adsorbents exhibit non-linear behavior over large fluctuations in concentrations, a linear assumption is suitable for low-concentration ranges of environmental pollutants (Chu, 2023). If a non-linear trend was observed, a Freundlich isotherm (Equation 5) was fit with K and n as isotherm parameters (Freundlich, 1906). Isotherm parameters were fit by minimizing the residual sum of squares of q between the data and isotherm equations.

$$q = K_L C \quad (4)$$

$$q = K C^n \quad (5)$$

Adsorption kinetics were determined using the large batch study data by plotting q as a function of time and fitting pseudo-first-order (PFO, Equation 6) and pseudo-second-order (PSO, Equation 7) rate kinetic equations to the data. Where K_1 and K_2 are rate constants, and q_{eq} is the maximum phosphorus adsorbed at equilibrium. Kinetic parameters were fitted by minimizing the residual sum of squares of q between the data and rate equations

$$\frac{dq}{dt} = K_1(q_{eq} - q) \quad (6)$$

$$\frac{dq}{dt} = K_2(q_{eq} - q)^2 \quad (7)$$

The adsorbent used in the column studies were then dried and tested for desorption using the standardized method of sequential extraction procedure for lake sediments developed by the Standards, Measurements, and Testing Programme of the European Commission (Ruban et al., 1998) to determine qualitatively the potential for the media to be further investigated for fertilizer potential.

Deliverable #5: Bioretention Design Spreadsheet Model

- Based on the outcomes of Deliverable #2, an EXCEL spreadsheet tool was developed to assist farmers with bioretention system design.
- A preliminary cost estimate was developed based on prior studies of bioretention applications for treatment of agricultural runoff.

A user-friendly EXCEL worksheet was developed to enable farmers to input key parameters and predict a bioretention system's N removal efficiency. The main inputs include irrigated acreage, irrigation rate (gpm), and runoff nutrient concentrations. The spreadsheet allows for the input of either default or user input values for flows and concentrations. Default flow values, based on typical sprinkler irrigation conditions at a container nursery, assume approximately 15,000 gallons of irrigation water per acre per day, equating to about 0.53 inches of water applied daily (Fulcher and Fernandez 2021). With an estimated 60-70% of this water becoming runoff, about 8,000 gallons of runoff per acre per day is the default value that requires treatment. Equations 8, 9 and 10 were used to calculate the nitrogen loading rate (NLR) and the nitrogen removal rate (NRR).

$$EBCT = \frac{V_{empty}}{Q} \quad (8)$$

$$NLR = \frac{C_0}{EBCT} \quad (9)$$

$$NRR = \frac{C_0 - C_{effluent}}{EBCT} \quad (10)$$

Default nitrogen concentrations were derived from average experimental values from this study (Richardson, 2024). By combining the runoff volume and nitrogen concentrations, NLR can be estimated using a regression model developed from experimental results to determine the NRR. The model allows farmers to explore different options for the size of the bioretention system and decide whether to include biochar. The spreadsheet then calculates the required bioretention surface area and predicts the nitrogen removal efficiency based on the input parameters. The

user-friendly interface guides farmers through the process with clear labels and instructions, ensuring easy data input and results interpretation. This tool aims to facilitate nitrogen management in irrigation runoff. The EXCEL Tool and other project data are available at https://github.com/natchayal/FDACS_Bioretenction_project.

RESULTS AND DISCUSSION

Deliverables #1 and #2

Runoff Characteristics and Bioretention Design Basis

Irrigation runoff from a Holmberg Farm greenhouse used to produce citrus trees was discharged through a culvert to a slough. Flow rate (Figure A14) and pollutant concentrations (Figure A15) were measured in the culvert during an irrigation event. Pollutant loading rates were calculated as a product of the flow rate and concentration and are shown in Figure A16. At the beginning of the irrigation period, soil in the planters absorbed most of the applied water. Therefore, runoff started flowing through the culvert approximately one hour after irrigation commenced. Over time, as the root zones became saturated, less water was retained, and more runoff was generated until steady-state flow rate of approximately 1.57 cfs was observed between 75 and 220 minutes. A decline in flow rate was observed from 220–270 minutes after irrigation ended. Pollutant loading rates initially increased over time and then decreased at the end of each irrigation event. pH steadily decreased over time from 8.32 to 7.75 while alkalinity remained consistent within a narrow range of 124 – 144 mg/L as CaCO₃. Data from runoff characteristic study were used to calculate the target HLR using Equation 1, assuming a field-scale bioretention unit would take up either 5%, 2.5% or 1% of the 12 acres greenhouse area. Runoff characterization also showed that nitrogen in the runoff was mainly in the form of NO_x (nitrate + nitrite) and that organic carbon concentrations were relatively low. Therefore, the bioretention systems were designed to incorporate wood chips into the IWSZ and test a higher elevation outlet. These were modifications from the Phase I FDACS study (Rahman et al., 2024).

Deliverable #3

Tracer Studies

Conservative tracer studies to determine the effect of biochar addition to the IWSZ and outlet elevation on mean residence time (MRT) and dispersion (Table A3). The parameters t₁₀, t₅₀, and t₉₀, represent the time it took for 10%, 50% and 90% of the tracer mass, respectively, to exit the system. The Morrill Dispersion Index (MDI) and “n” value for the tanks in series (TIS) model are related to the amount of mixing and short-circuiting in the units.

Tracer studies revealed key hydraulic differences between bioretention systems with and without biochar amendment in the IWSZ. The CBA unit (with biochar in the IWSZ) had a significantly longer MRT of 70 minutes compared with 46 minutes for the PBA unit (without biochar in the IWSZ), likely due to biochar's microporosity creating a more tortuous flow path and decreasing saturated hydraulic conductivity (Zhang et al. 2016). Shortcutting was less prevalent in the CBA unit (t₅₀/MRT = 0.89) compared to the PBA unit (t₅₀/MRT = 0.74), indicating that biochar reduced preferential flow, most likely due to the presence of macropores. The CBA unit exhibited more uniform flow, as shown in Figure A27 and Figure A28. The lower MDI, variance, and standard deviation, and a higher TIS n value, also provides supporting evidence of uniform flow in the presence of biochar. When the effluent exited through the upper outlet, differences between the two units were less pronounced (Figure A31). Biochar in

the PBA unit's IWSZ reduced shortcutting, preventing dead zones and leading to more uniform flow patterns. Raising the outlet height significantly increased the MRT multiplied by HLR, from 21.9 to 33.8 cm (154% increase) in the CBA unit and from 13.1 to 30.8 cm (234% increase) in the PBA unit. This indicates that MRT is more dependent on the time spent in the IWSZ than in the unsaturated layer. These findings highlight the influence of biochar on hydraulic behavior and the importance of outlet height adjustments for optimizing retention times and nutrient treatment.

Bioretention Monitoring

Average concentrations for all irrigation events are shown in Table 4 and statistical analysis of TIN and TP total load reductions are shown in Figures 4 through 6. Decreases in TIN, NO_3^- , PO_4^{3-} , and pH were observed between average influent and effluent samples. COD increased between the influent and effluent for 17% of all events for the CBA unit (events 3, 13, and 16), and for 72% of all events for the PBA unit (all events except 6, 7, 10, 11, 12). An increase in COD between influent and effluent may indicate leaching of organic carbon from the wood chips. Alkalinity increased during all events for both units, which is consistent with denitrification. Detailed results from the irrigation events studied are shown in Tables A7 through A15. Box and whisker plots comparing different bioretention configurations are shown in Figure A38 through Figure A43. In general, an average of 12g/m³-day and 10 g/m³-day removals were observed for TIN for CBA and PBA units, respectively, while PO_4^{3-} results are inconclusive due to potential desorption of phosphate from adsorbents during antecedent dry periods (ADP). NO_2^- concentrations increased slightly, indicating incomplete nitrification.

Biochar amendment in the IWSZ had a significant effect on TIN removal (Figure 6a; $p = 0.0308$). For PO_4^{3-} removal (Figure 6b), the comparison between CBA and PBA yielded a p -value of 0.0577, which is close to the threshold for significance. These results suggest that incorporating biochar in the IWSZ enhances removal of both N and P, most likely by increasing the retention time and enhancing microbial activity. As shown in Figure 4, there was no significant difference in TIN removal between Low and Medium HLR ($p > 0.05$), while there were significant differences between Low and High HLR ($p = 0.0224$) and between Medium and High HLR ($p = 0.0456$). For TP removal (Figure 5), there were no significant differences across all comparisons (Low vs. Medium, Low vs. High, Medium vs. High; $p > 0.05$). There was a small but not significant increase in TIN and PO_4^{3-} removals between low and high outlet heights ($p > 0.05$). These findings suggest that decreasing the HLR enhances TIN removal by increasing the residence time in the IWSZ and promoting denitrification but does not significantly affect P removal by adsorption.

Before planting Muhly grass, the two bioretention systems were operated at a high HLR of 0.55 cm/min and a low outlet height, resulting in average total load removals of 50% and -3% for TIN and PO_4^{3-} (Table A13). After planting, average total load removals of 33% and 22.4% for TIN and PO_4^{3-} were observed under similar testing conditions. The addition of plants improved P removal and unexpectedly decreased TIN removal (Figure 4 and Figure 5); however, during this period plants had not been fully established and some of the controlled release fertilizer may

have been present in the plant roots. Additional monitoring of the bioretention systems with plants continued throughout events 16–18 for the two bioretention systems operating at the “optimal” design of a medium HLR of 0.22 cm/min, upper outlet height and with established plants (Figure 4 and Figure 5). Under these conditions, the systems were able to achieve the highest overall total load reductions, 81.05% for TIN and 33.06% for PO_4^{3-} .

The above ground plant biomass from the two Muhly grass plants was measured in April 2024 after dismantling the units. Above ground biomass for the CBA unit was 1.6 lbs and for PBA unit was 0.6 lbs. As shown in the photos (Figure A63 and Figure A64), the below ground plant roots were still embedded in the potting mix and could not be removed for measurement without affecting the roots. Based on visual observations, there was greater Muhly Grass below ground biomass growth in the CBA system compared with the PBA system. The results show that biochar addition to the IWSZ enhanced both above and below ground plant growth.

Table 4: Average influent and effluent concentrations and corresponding removal percentages for runoff events 1 and 3-18.

Events (1, 3-18)	Influent	CBA		PBA		CBA and PBA	
Contaminants	Average Influent	Average Effluent	% Removal	Average Effluent	% Removal	Average Effluent	Average % Removal
$\text{NH}_4^+\text{-N}$ (mg/L)	0.23	0.30	-150.70	0.13	-58.62	0.34	-104.66
$\text{NO}_2^-\text{-N}$ (mg/L)	0.06	0.11	-121.13	0.22	-191.78	0.11	-156.46
$\text{NO}_3^-\text{-N}$ (mg/L)	2.55	0.30	85.33	1.28	61.96	0.52	73.65
TIN (mg/L)	2.84	0.71	70.15	1.63	54.59	0.97	62.37
NO_x (mg/L)	2.66	0.40	79.33	1.50	43.01	0.95	61.17
COD (mg/L)	32.5	27.61	13.16	33.52	-21.89	29.76	-4.36
Alkalinity (mg/L as CaCO_3)	123	152.80	-20.36	124.10	-17.46	197.35	-18.91
pH	7.81	7.20	7.76	7.25	7.39	6.96	7.57
$\text{PO}_4^{3-}\text{-P}$ (mg/L)	1.77	1.24	24.56	1.29	-12.46	1.40	6.05
$\text{SO}_4^{2-}\text{-S}$ (mg/L)	115	86.43	22.05	102.48	29.56	63.80	25.81
DO (mg/L)	1.66	0.26	13.58	0.35	13.75	0.00	13.67

*Negative % removal indicates effluent concentrations were higher than influent concentrations.

*Event 2 data not included as bioretention operations were interrupted by rain.

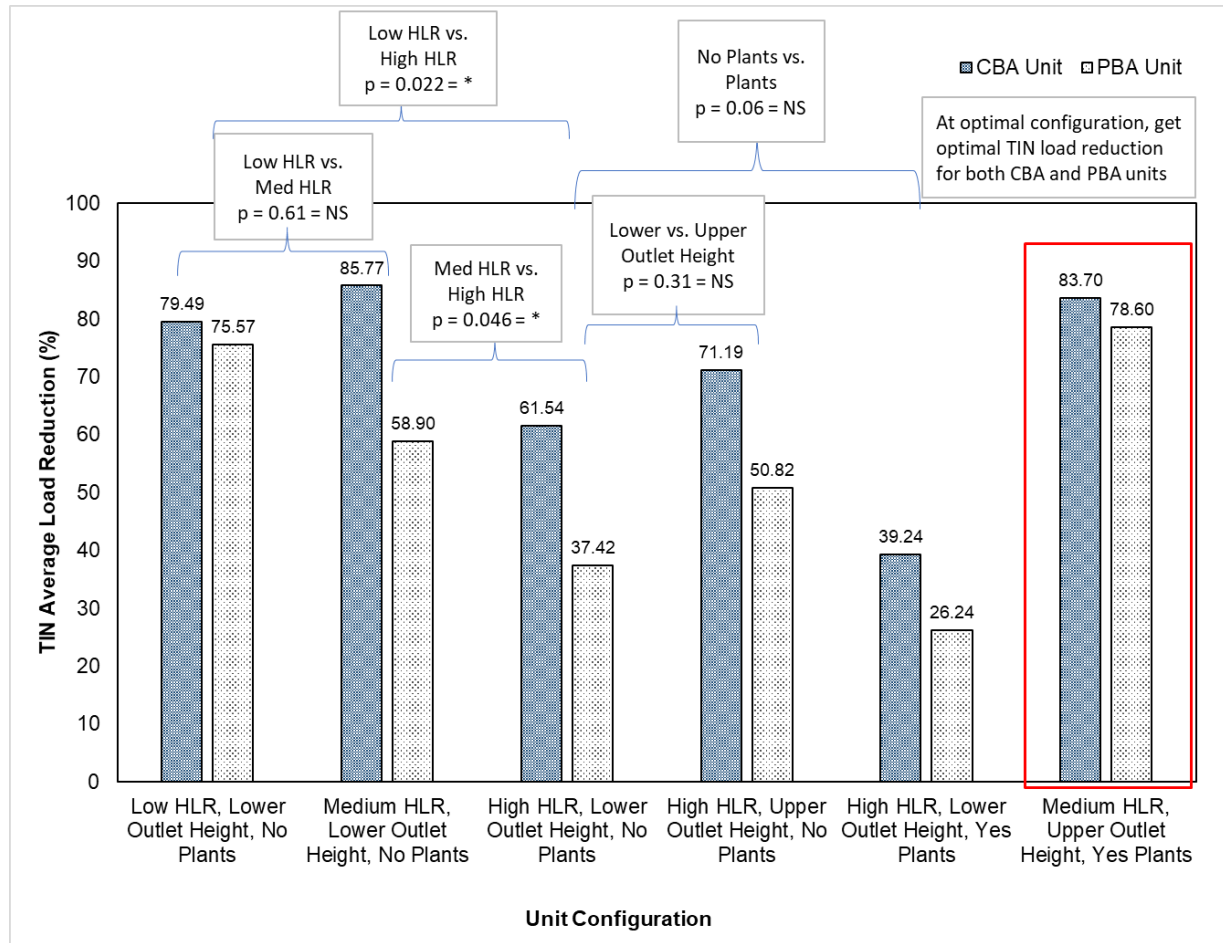


Figure 4: Average TIN load reductions for all events and units, with ANOVA p-values (* for $p < 0.05$, NS = not significant). The red box indicates a system operating under optimal configuration.

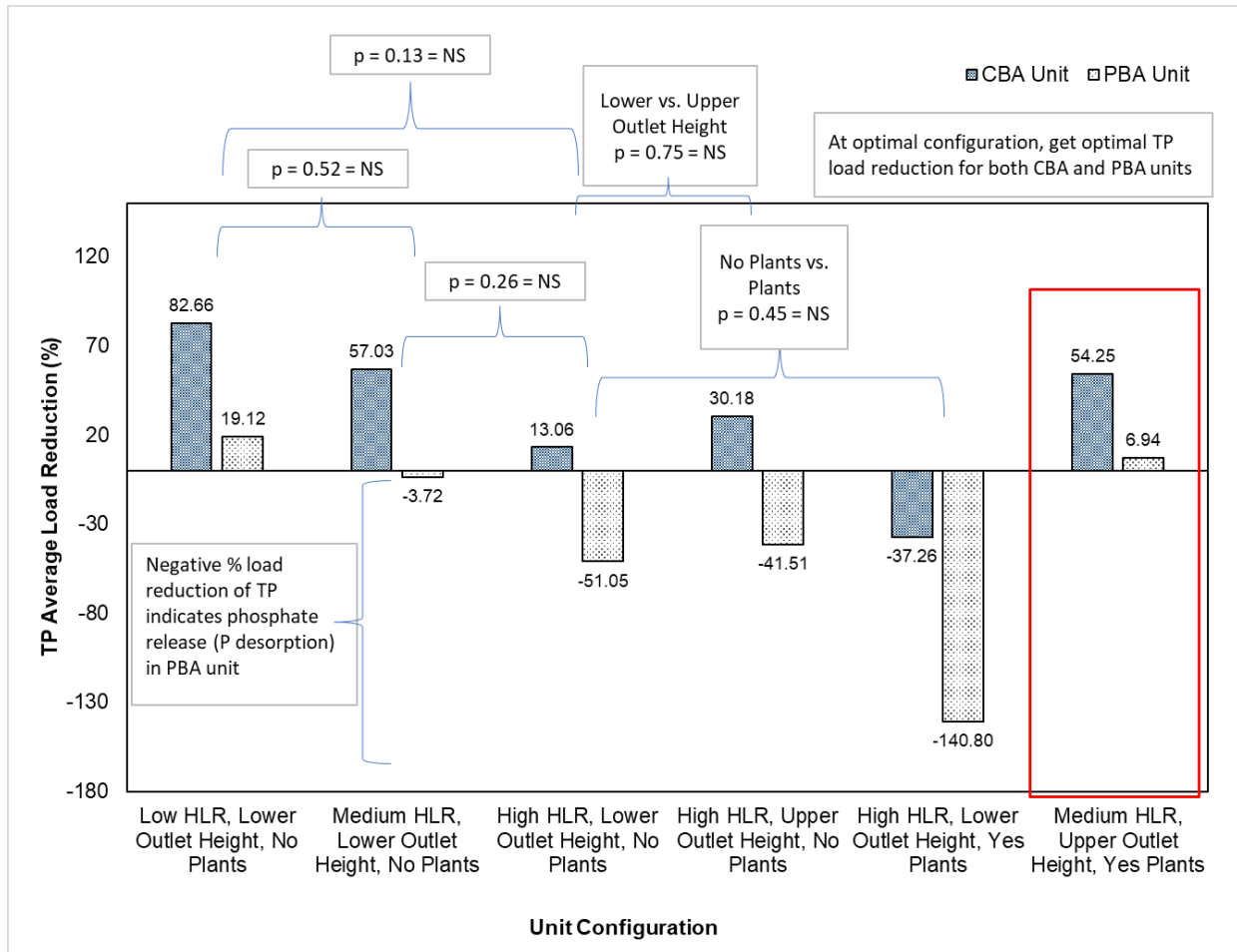


Figure 5: Average total phosphate load reductions for all events and units, with ANOVA p-values (* for $p < 0.05$, NS = not significant). The red box indicates a system operating at optimal configuration.

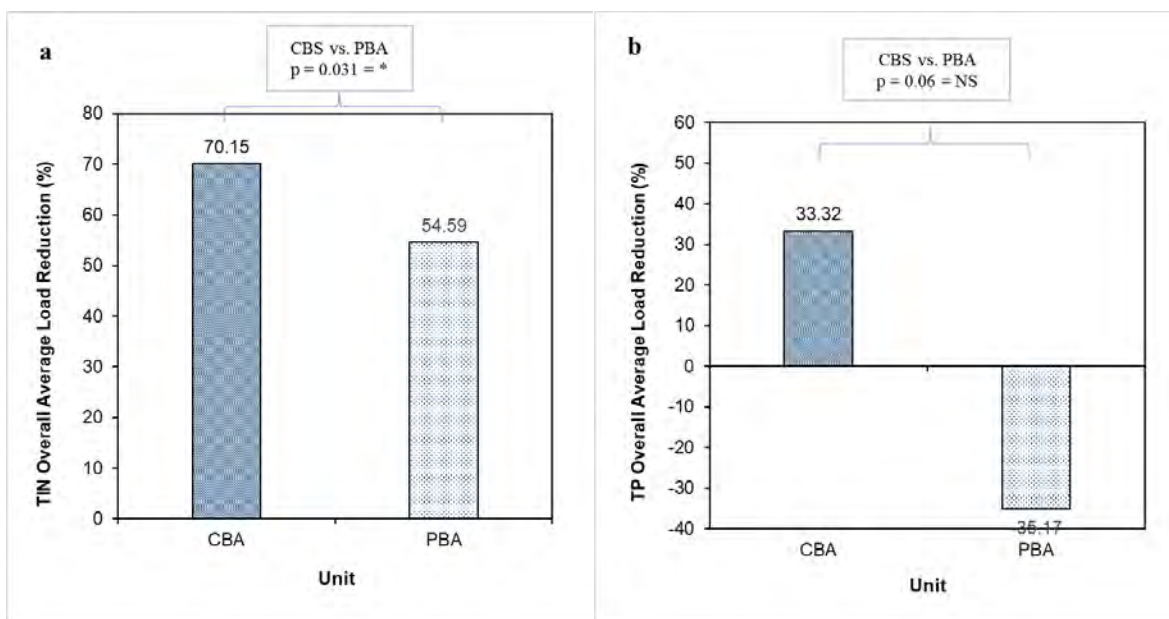


Figure 6: Bar chart comparing a) Total TIN load reductions and b) Total TP load reductions between CBA and PBA units. With ANOVA p-values (* for $p < 0.05$, NS = not significant).

Deliverable #4

P Removal and Recovery Using Adsorbents

Four adsorbents (Phosflow, HPS, DWT Residual, and UIUC biochar) stood out in small-batch kinetic (Figure 7) and capacity (Figure 8) experiments. With each of the four adsorbents, phosphorus removal increased with increasing contact time and concentration. Within the range of concentrations tested (0-40 mg/L), the capacity of HPS, DWT Residual, and UIUC Biochar was found to fit a linear isotherm function, while the capacity of Phosflow was found to fit the Freundlich isotherm. Isotherm fitting parameters are listed in Table 5. Small-batch (Figure A57) and capacity (Figure A58) results for other adsorbents tested are included in the appendix. The absorbance of crushed mixed seashells was lower than anticipated and highlighted the variability of P removal from waste material. Additional research will be carried out in the future to identify and test if other sources of seafood shell waste could be utilized as an adsorbent to recover P. Although PO_4Sponge had better P removal results than biochar and crushed mixed seashells, the product desorbed sulfate (data not shown) and is designed for a more energy-intensive recirculating treatment systems than a bioretention design. Therefore, it was not selected for additional experiments.

After 24 hours of contact time in the kinetic and capacity experiments, Phosflow had the greatest P adsorbed per gram, while the DWT Residual, the UIUC Biochar, and the HPS were not as consistent between the two experiments. Phosflow and HPS adsorbed more P than DWT Residual and UIUC Biochar in the kinetic experiments at each contact time. In the capacity experiments, the larger isotherm parameter (K) for Phosflow and DWT Residual illustrates a greater capacity for P adsorption than UIUC Biochar and HPS. The kinetics of these four

adsorbents were further investigated in additional batch (Figure 9) and column (Figure 10) experiments.

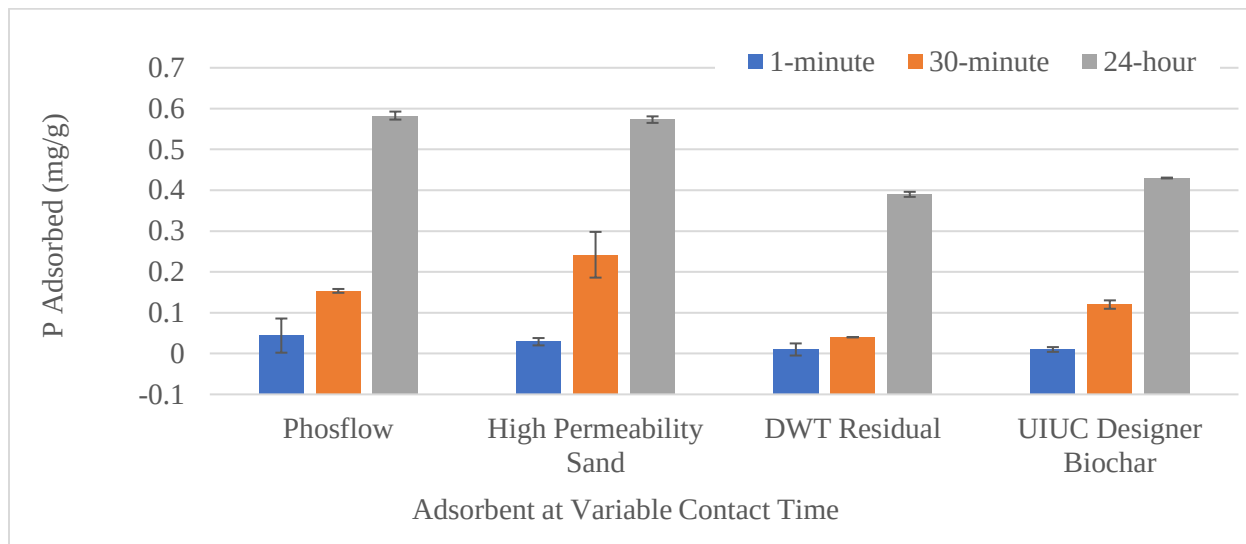


Figure 7: Average normalized phosphate removal of adsorbents selected for kinetic studies.

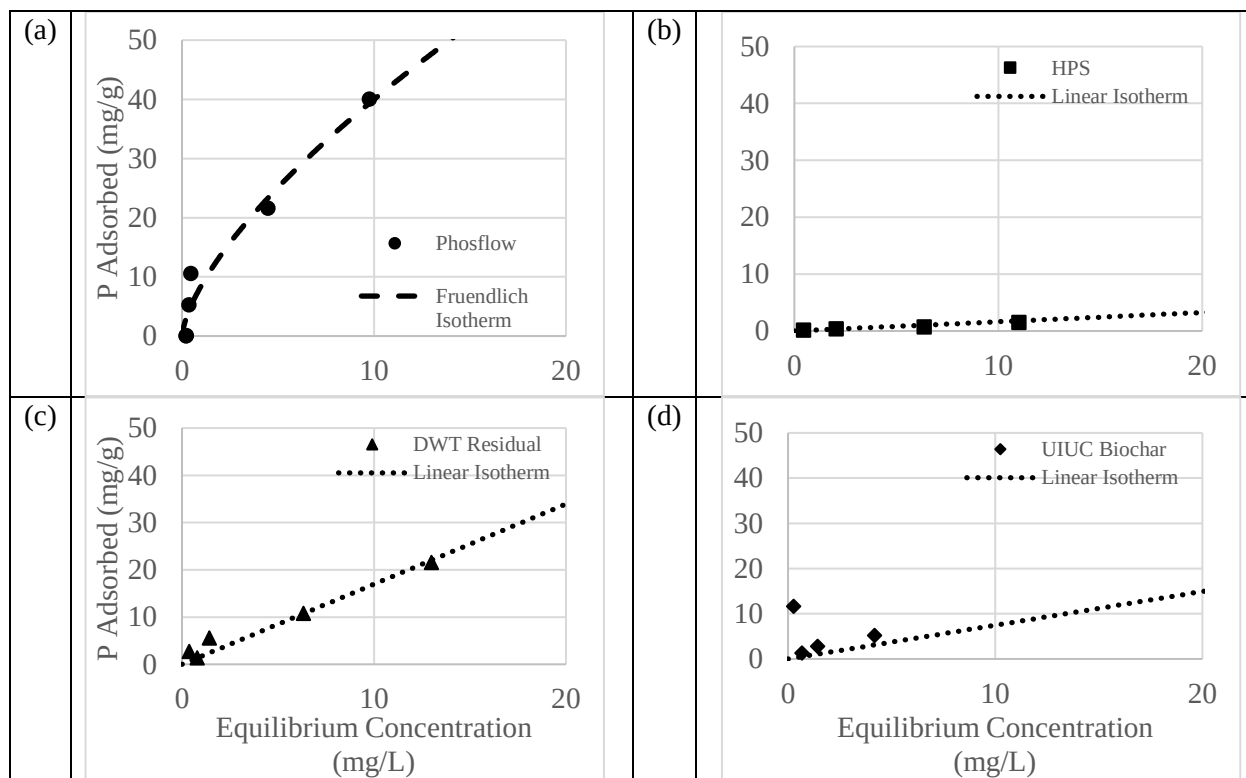


Figure 8: P adsorption capacity of (a) Phosflow (b) HPS (c) DWT Residual (d) UIUC Biochar.

Table 5: Isotherm parameters of phosphate on adsorbents selected for additional kinetic studies.

Media	Isotherm Type	Isotherm Fitting Parameters	R ²
Phosflow	Freundlich	$K = 8.534 \text{ (mg/L)}^n$; $n = 0.67$	0.9634
HPS	Linear	$K_L = 0.162 \text{ L/g}$	0.9685
DWT Residual	Linear	$K_L = 1.696 \text{ L/g}$	0.9467
UIUC Biochar	Linear	$K_L = 0.745 \text{ L/g}$	0.4431

All four adsorbents removed more than 85% of P in the large batch experiments. UIUC Biochar removed the least (86%) and DWT residual removed the most (93%). DWT residual was the only adsorbent to reach equilibrium, with data points beginning to approach an asymptote after 20 hours. Phosflow, HPS, and UIUC biochar continued to increase P removed throughout the 48-hour duration of the experiment without approaching an asymptote. PSO kinetic models had a better fit to experimental data than PFO for each adsorbent except UIUC Biochar. UIUC biochar removed P more slowly than the other adsorbents and had a slightly greater R² for a PFO kinetic model than PSO.

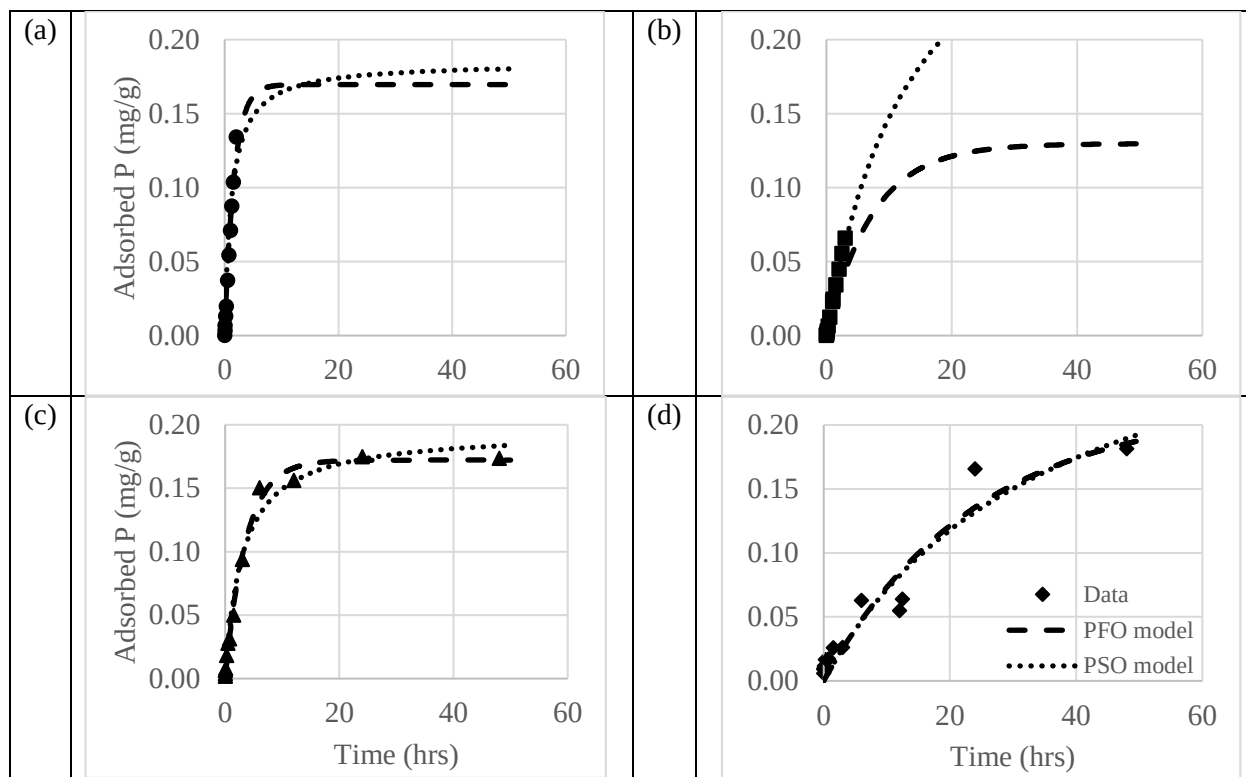
**Figure 9:** Phosphorus adsorption kinetics of (a) Phosflow (b) HPS (c) DWT Residual (d) UIUC Biochar in 48-hour batch kinetic experiment.

Table 6: Pseudo-first and second-order fitting parameters from 48-hour batch experiment.

Media	Pseudo-First Order			Pseudo-Second Order		
	q_e (mg/g)	K_1 (1/min)	R^2	q_e (mg/g)	K_2 (g/(mg/min))	R^2
Phosflow	0.1696	0.6103	0.9891	0.1847	4.4252	0.998
HPS	0.1595	0.1364	0.9832	0.1892	0.7980	0.993
UIUC Material	0.1722	0.2762	0.9927	0.1950	1.6761	0.986
DWT Residual	0.2143	0.0417	0.9165	0.3352	0.0810	0.915

Operational parameters for each column are listed in Table 6. Each adsorbent was tested for a variable run time as shown in Figure 10. Each of the four adsorbents tested has a different permeability in a flow-through treatment setting based on grain size and how closely the media packs together. Due to the small grain size of the DWT residual, it was necessary to mix the DWT with coarse sand to prevent clogging in the column studies. The DWT was mixed at 5% by volume with the sand in the column.

Similar to the large batch experiments, P removal increased over the run time for each adsorbent. P removal was greatest over time from UIUC Biochar, followed by DWT residual, Phosflow, and HPS. P adsorbed by the adsorbents in the flow-through experiments is greater than in the large batch experiments because a continuous P supply at a constant concentration is given in column studies, where P to adsorb in a batch experiment is limited.

Desorption results of the saturated adsorbents taken from the column (Table 8) show P released during the desorption test after exposure to NaOH and then HCl. Desorption of P after exposure to NaOH indicates P is bound to iron, aluminum, or manganese oxide. In contrast, desorption of P after HCl indicates P is bound to calcium or magnesium. Comparing desorption results to adsorption results within the column shows that P removed in the Phosflow, HPS, and DWT residual column was recovered after exposure to NaOH. While P was removed by UIUC Biochar, it was not recovered in the desorption test. This is likely because P reactions with calcium and magnesium are precipitation reactions, and P reacting with UIUC Biochar could be precipitating out of the column. If this is the case, P would not be recoverable for direct reuse from the adsorbent and would need a secondary treatment step to remove and recover P from the water.

Table 7: Operational parameters for each column.

Adsorbent	Adsorbent Mass (g)	Average Influent Concentration (mg/L)	Average Flowrate (mL/min)	Hydraulic Loading Rate (cm/min)	Hydraulic Residence Time (min)
Phosflow	506	4.99	159	7.88	1.29
HPS	1102	7.29	60	2.97	3.42
DWT Residual	50.02	4.89	166	8.21	1.24
UIUC Biochar	450	4.92	162	8.02	1.27

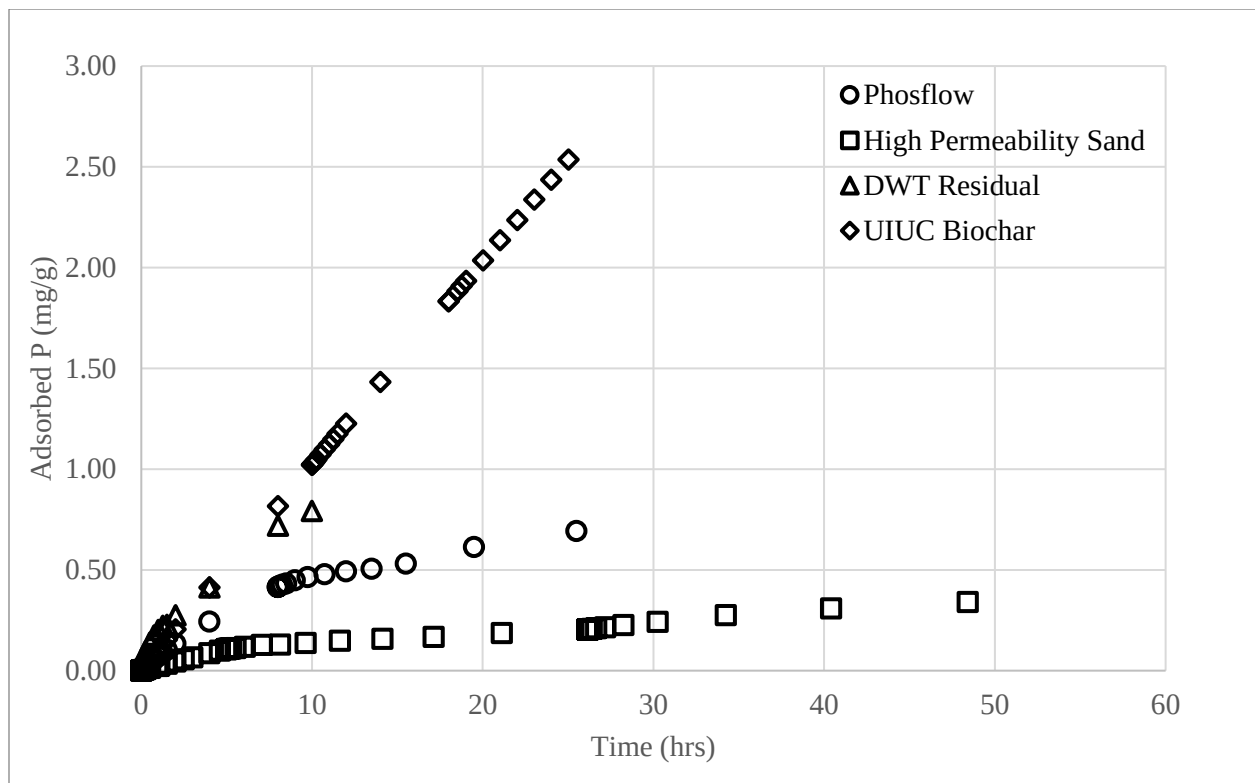


Figure 10: Adsorption kinetics of phosphate for selected material in a flow-through the column.

Table 8: Desorption of P before and after column experiment.

Adsorbent		NaOH-P Desorption (mg/g)	HCl-P Desorption (mg/g)
Phosflow	Pre-Column	Below Detection Limit	Below Detection Limit
	Post-Column	0.10	
HPS	Pre-Column	Below Detection Limit	
	Post-Column	0.08	
DWT Residual	Pre-Column	0.05	
	Post-Column	0.07	
UIUC Biochar	Pre-Column	Below Detection Limit	
	Post-Column		

Deliverable #5*Bioretention Design Spreadsheet Model*

An EXCEL spreadsheet tool was developed for preliminary design of a bioretention system to treat runoff from a commercial nursery that would be similar to the demonstration site in terms of runoff flows and water quality characteristics. At the request of FDACS staff, the model was developed to calculate the relationship between NO_x (NO₂⁻-N + NO₃⁻-N) loading rate and NO_x removal rates. Regressions were developed using experimental results from the CBA and PBA units (Figure 11 and Figure 12). Model inputs and outputs are shown in Figure 13 and Figure 14.

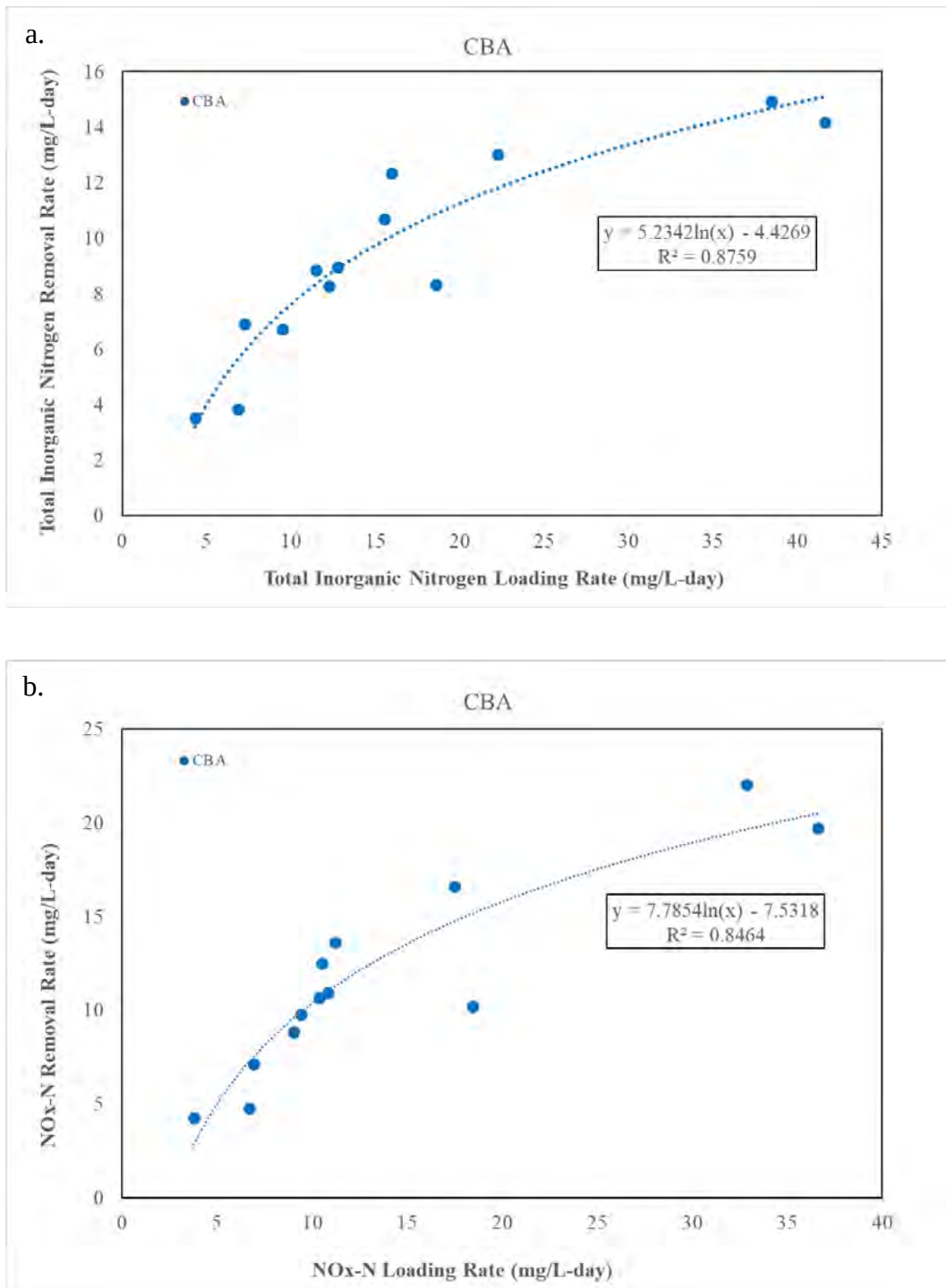


Figure 11: Regression for a) TIN Removal and b) NOx Removal for CBA Unit.

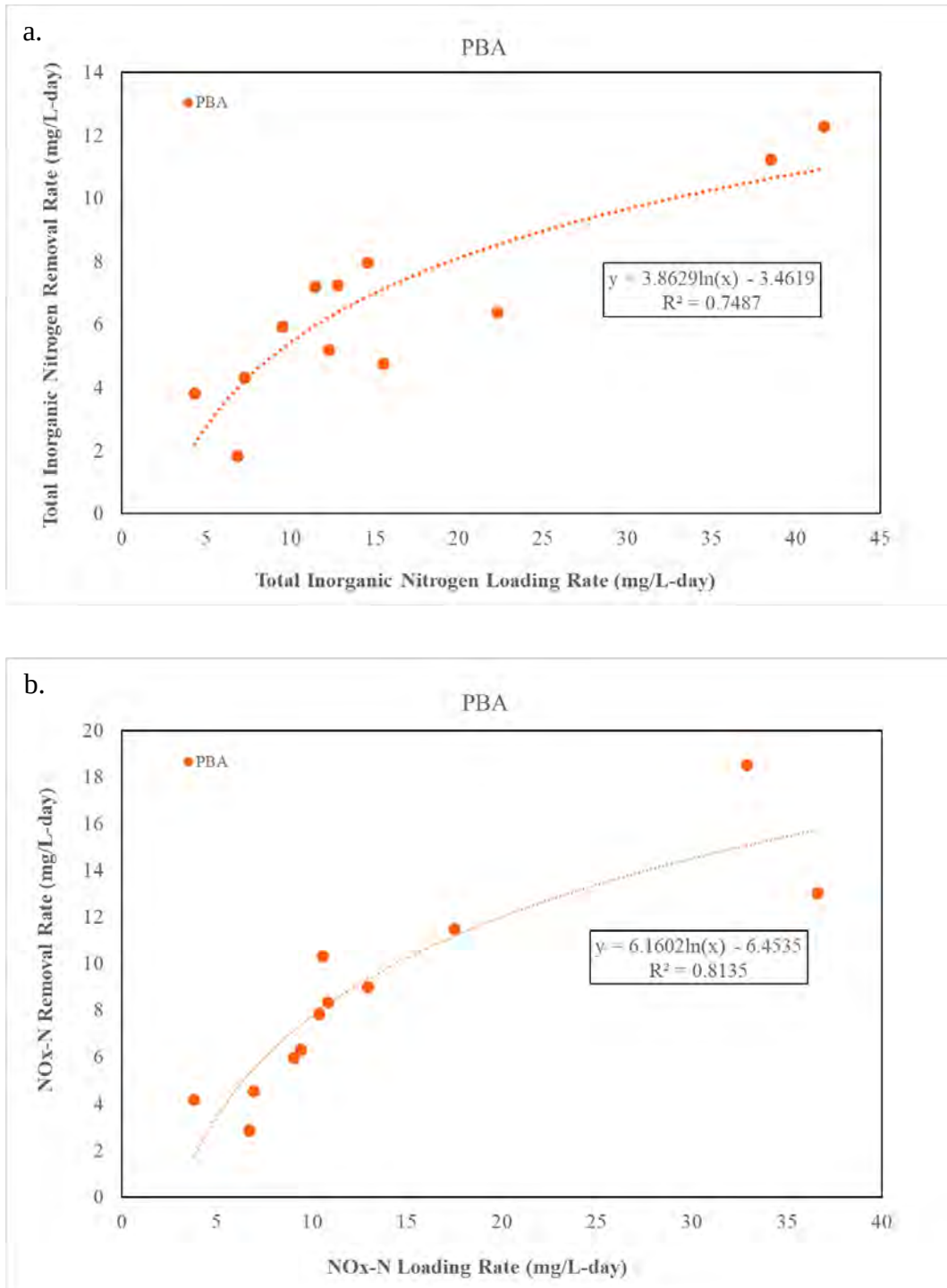


Figure 12: Regression for a) TIN Removal and b) NOx Removal for PBA Unit.

	A	B	C	D	E	F	G	H	I	J	K	L
1	BIORETENTION SYSTEM DESIGN AND COST INPUT											
2	STEP 1											
3	Manually input irrigated acreage		12	Select units acres								
4												
5	STEP 2											
6	Manually input irrigation rate (average runoff flow rate)		44.0	L/s <-- pick units								
7												
8	STEP 3											
9	Select Target Contaminant		TIN									
10												
11	STEP 3											
12	Manually input nutrient concentrations OR leave as "DEFAULT" value		2.8	mg/L <-- enter value otherwise DEFAULT value = 2.85 mg/L TIN								
13												
14	STEP 4											
15	Select Bioretention size (% area used for bioretention)		2.5%									
16												
17	STEP 5											
18	Select Bioretention Design Type		Completely Biochar Amended (CBA)									
19												
20	RESULTS											
21	To see results, go to the OUTPUT sheet.											
22												
23												
	Title Page INPUT OUTPUT Bioretention Constraints Lookup Tables											

Figure 13: User input tab.

	A	B	C	D	E	F	G	H
1	OUTPUT SUMMARY							
2	(Recommended Design for maximum nutrient removal)							
3		Parameter	Value Units					
4		Technology	BRT					
5		Target Contaminant	TN					
6		Design Type	CBA					
7		Average Flow	822 cm^3/min	=	0.82 L/min			
8		Hydraulic Loading rate	0.22 cm/min	=	Medium			
9		EBCT	358 min	=	6.0 hr			
10								
11		Outlet Height	79 cm	=	Upper			
12		Length	61 cm					
13		Width	61 cm					
14		Plant	Yes (e.g. Muhly grass)					
15								
16		Cinf	2.85 mg/L					
17		NLR	11.47 mg/(L-day)					
18		Ceff	0.78 mg/L					
19		NRR	8.35 mg/(L-day)					
20								
21		Removal Efficiency	73%	Note:				
22								
	<	>	Title Page	INPUT	OUTPUT	Bioretention Constraints	Lookup Tables	

Figure 14: Output tab showing recommended design for bioretention system and total TIN removal efficiency.

Cost Estimation

Bioretention has been previously implemented for both agricultural and urban stormwater management, to manage stormwater runoff and improve water quality. Bioretention design and installation includes design and engineering, plant selection, soil preparation, and maintenance. For the purpose of this project, a preliminary cost estimate was carried out using data from prior

agricultural bioretention applications (Christianson et al., 2017). Note that current costs in Florida may vary significantly from this application. Christianson et al. (2017) reported on the installation of a ditch diversion bioreactor (35×7.9×0.9 m) in November 2015 at a farm (soybean, lima bean, or sweet corn-spinach) in Caroline County, MD. The bioreactor treated a 35-ha drainage area and cost \$27,000 in total (\$770/ha of treated area). The authors also reported on the installation of three in-line in-ditch woodchip bioreactors (2 for 27×1.1×0.7m and 1 for 38×1.1×0.7m) in December 2015 at a farm (corn-wheat-soybean) in Somerset County, MD. The bioreactor treated 6.6-ha area and cost \$18,000 in total (\$2,700/ha). Generally, subsurface drainage woodchip bioreactor installation costs for agricultural areas ranged from less than \$5,000 to \$27,000 (Christianson et al., 2021). For urban lands, bioretention construction costs range from \$5,000 to \$10,000 for larger areas and \$3/ft² to \$15/ft² for small areas (UF IFAS Extension, 2008). Cost varies depending on soil characteristics. For example, installation costs ranged from \$1.5 to \$3 per ft² of bioretention area for highly permeable soils and \$4 to \$6 per ft² for slowly permeable soils (Jarrett, 2022). The bioretention installed in this study was designed to treat 12-acre of nursery at an optimal bioretention area of 2.5% of the greenhouse area (0.3 acres or 0.2 hectares). The estimated cost ranges from \$4,800 to \$17,000 in 2024, based on the cost of \$770/ha to \$2700/ha of treated area in 2017 (Christianson et al., 2017).

Deliverable #6

Research Outreach and Dissemination

A pre-webinar survey was distributed to gauge stakeholders' familiarity with BMPs for nurseries and their openness to various nutrient treatment methods. The respondents included 36 professionals from academia, engineering, management, farmers, and research sectors (Table A17). The majority of participants (64%) indicated familiarity with BMPs, though 36% did not, highlighting the need for increased educational outreach. Participants showed strong support for the use of bioretention and treatment wetlands for nutrient management. There were moderate levels of support for controlled release fertilizers and biochar amendments, though high "NA" responses (44% and 39%, respectively) suggested a need for more information. Additionally, respondents were least familiar with coagulation and hybrid wetland treatment. Key concerns about implementing new nutrient management technologies included saltwater intrusion (23.22%), lack of adoption at neighboring operations (18.66%), and difficulties in receiving government incentives (14.53%) (Figure A63). Other concerns included geographical limitations, land use area, maintenance, cost, and the effectiveness of nutrient removal. These findings highlight the importance of educational initiatives including community outreach, workshops, and detailed success stories to build confidence and facilitate the adoption of innovative nutrient management practices among stakeholders.

Additional key dissemination activities included:

- A poster presented at the *American Ecological Engineering Society* (AEES) meeting at USF on June 6th–9th, 2023.
- A poster presented at the *Association of Environmental Engineering and Science Professors* (AEESP) distinguished lecture at USF on February 22nd, 2024.

- The research was the subject of one USF Civil & Environmental Engineering MS Thesis, published in March 2024 entitled, “Nutrient Removal of Biochar Amended Modified Bioretention Systems Treating Nursery Runoff” (Richardson, 2024).
- A manuscript based on the Phase I project was published in the journal *Water*, entitled Treatment of dairy farm runoff in vegetated bioretention systems amended with biochar (Rahman et al., 2024).
- USF hosted a webinar to educate stakeholders on this research on April 5th, 2024. Detailed webinar flyer, presentation and attendees’ lists are provided in the Appendix.
- USF hosted a workshop at Holmberg Farms on April 15th, 2024, allowing nursery managers to learn about the systems on-site. Detailed webinar flyer, presentation and attendees’ lists are provided in the Appendix.
- Funding from this project supported a number of USF students, who received training in management of nutrients from agricultural runoff.
 - Graduate students: Nicholas Richardson, Rachael Cooper, Nusrat Sharmin, Natchaya Luangphairin.
 - Undergraduate students: Vicky Lopez, Brianna Nogoy, Olta Tarko and Stephanie Rodriguez.

CONCLUSIONS

The overall goal of this research was to understand nutrient removal mechanisms in bioretention systems amended with biochar for treatment of agricultural runoff. The research was carried out through bench- and field-scale studies with agricultural runoff from Holmberg farm’s nursery. The research team also developed an EXCEL spreadsheet tool that can be used for preliminary bioretention design and did a preliminary cost estimate. The research was disseminated to water managers, engineers, farmers, researchers, students, and other stakeholders through presentations and publications. The following are the major findings of this research:

Tracer studies:

- The microporosity of the biochar helps maintain moisture and increases the MRT of pollutants in the system.
- Biochar addition to the IWSA resulted in less dispersion, as shown by the lower MDI, and higher TIS for the CBA unit compared to PBA unit.
- The lower t_{50}/MRT value for the CBA unit showed that biochar addition to the IWSZ reduces shortcutting, providing more time for denitrification.

Field bioretention studies with varying biochar amendment strategies (CBA vs. PBA):

- Biochar amendment in both the unsaturated zone and the IWSZ (CBA) significantly improved TIN and TP removals compared with the bioretention system with only biochar in the unsaturated zone (PBA) ($p = 0.031$).

Field bioretention studies with varying HLRs:

- Nitrate load reductions (averaged of both units) increased as HLR decreased.
- HLR did not significantly impact TP load reduction (p-value > 0.05). This is due to the high variability of TP load reductions between each event, possibly due to phosphate desorption during the ADP.
- Increasing HLR from 0.11 cm/min to 0.22 cm/min, and from 0.22 cm/min to 0.55 cm/min did not have a significant impact on TIN load reduction (p-value > 0.05). However, the increase of HLR from 0.11 cm/min to 0.55 cm/min did have significant impact (p = 0.022). Lower HLRs provide more time for both nitrification in the unsaturated layer, and denitrification in the IWSZ but requires more land area for treatment.
- Increasing land area for bioretention from 1% (0.55 cm/min) of the greenhouse area to 2.5% (0.22 cm/min) improved total load reductions by an additional 22.4% for TIN and 45.8% for TP. However, increasing land area for bioretention from 2.5% (0.22 cm/min) of the greenhouse area to 5% (0.11 cm/min), only improved removal by 1.9% TIN and 19.8% TP.

Field bioretention studies with varying outlet height:

- Increasing outlet heights for both bioretention systems resulted in slight but not significant increases in TIN and TP total load reductions (p-value > 0.05).
- As noted in the tracer study, increasing the outlet height increases MRT, allowing more time for denitrification but decreases the available pore space for oxygen replenishment and nitrification in the unsaturated zone.

Field bioretention studies with and without Muhly Grass:

- In this study, addition of Muhly grass did not significantly improve TIN and TP load reductions (p-value > 0.05). This may have been due to the short growth period and the presence of residual controlled release fertilizer in the roots of the plants.

P Removal and Recovery:

- Preliminary research showed low P adsorption with unmodified and modified biochar. Therefore, the research objectives were broadened to evaluate additional readily available P adsorbent materials.
- Four adsorbent materials tested stood out to remove P at the low concentrations typical of nursery runoff. Two were commercially available adsorbents (Phosflow and HPS), one was a proprietary modified biochar (UIUC Biochar), and one was a recycled industrial material (iron-based DWT residuals). Of the four materials, Phosflow had the greatest P adsorption capacity, fastest uptake rate, and greatest amount of P recovered (desorbed). UIUC Biochar had the lowest capacity and kinetic coefficients, and no detectable recovered P.

Bioretention system cost estimation:

- A simple spreadsheet model can be used as a design tool to estimate bioretention system land requirements.
- Bioretention system installation approximately costs \$4,800 to \$17,000.

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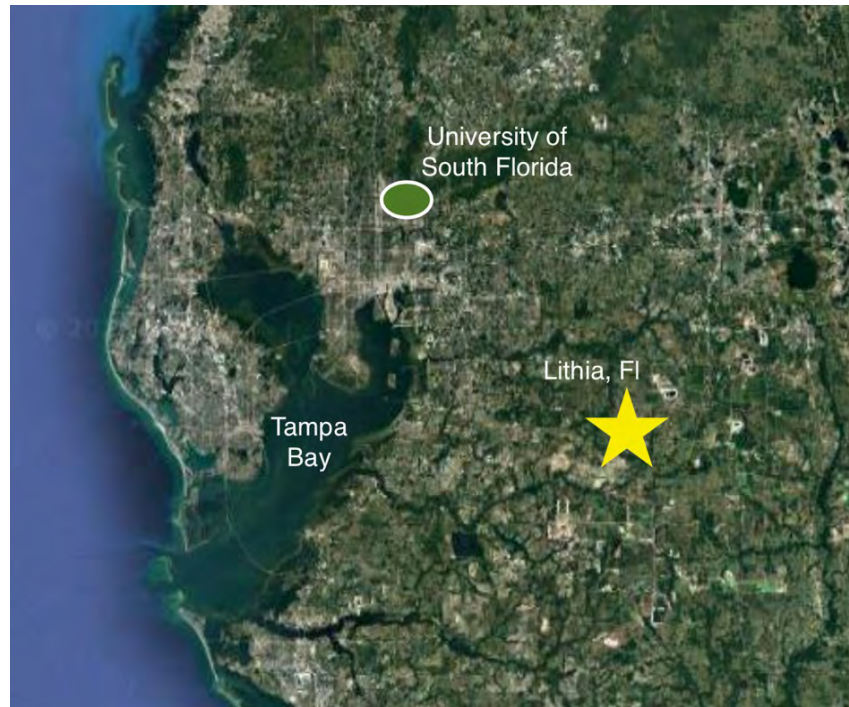


Figure A1: Site location, USF and Tampa Bay. Images from Google maps.





Figure A2: Drainage features at Holmberg Farms: (a) direction of drainage from the greenhouse and location of the slough (b) discharge pipe from the greenhouse and (c) slough (Lithia, FL).

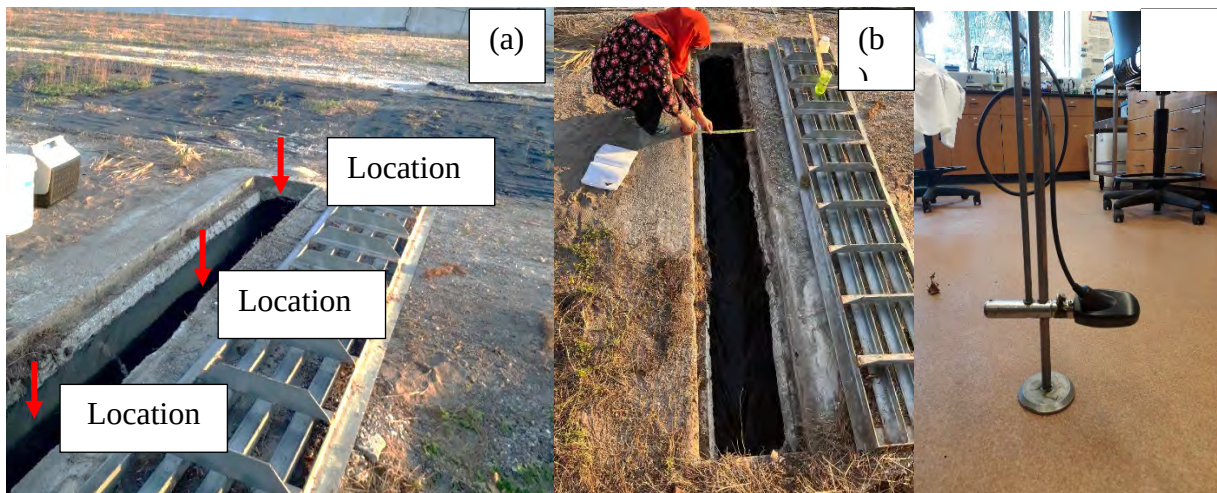


Figure A3: Flow measurement for partially full pipes: (a) location of velocity measurement (b) measuring diameter of the pipe and depth of flow (c) velocity meter.

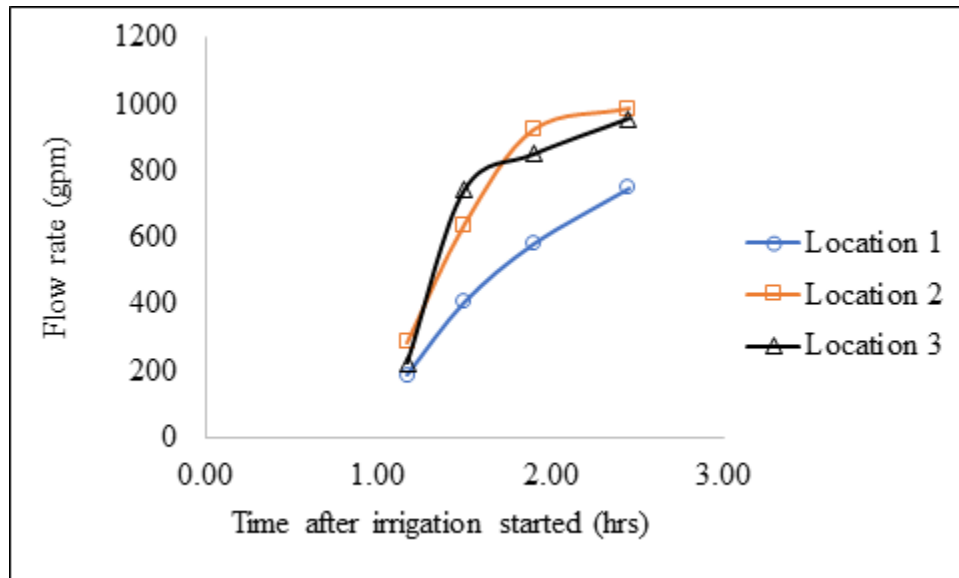


Figure A4: Change in irrigation return flow with time.



Figure A5: Empty bioretention tank showing internal PVC piping.



Figure A6: Bottom 2" gravel drainage layer.



Figure A7: Top of 18" woodchip + biochar + coarse sand (1:2:2 by volume) mixture.



Figure A8: Top of 10" biochar + coarse sand (1:1 by volume) mixture.



Figure A9: Top of 1" coarse sand layer.



Figure A10: Side view of completed demonstration bioretention tank.



Figure A11: Insect and weather-proof box for electrical components.



Figure A12: Double valved effluent pipe for irrigation runoff distribution.



Figure A13: Bioretention systems CBA (left) and PBA (right).

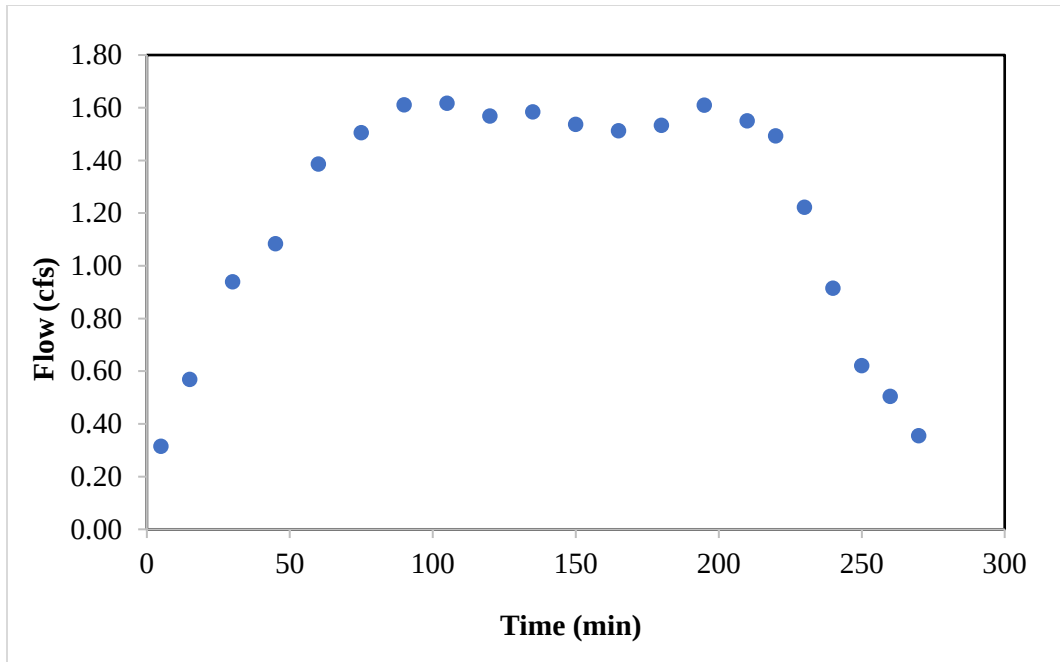


Figure A14: Runoff flow rate (in cfs) over time during an irrigation event. Steady state at 75–220 min and peaking at 1.574 cfs.

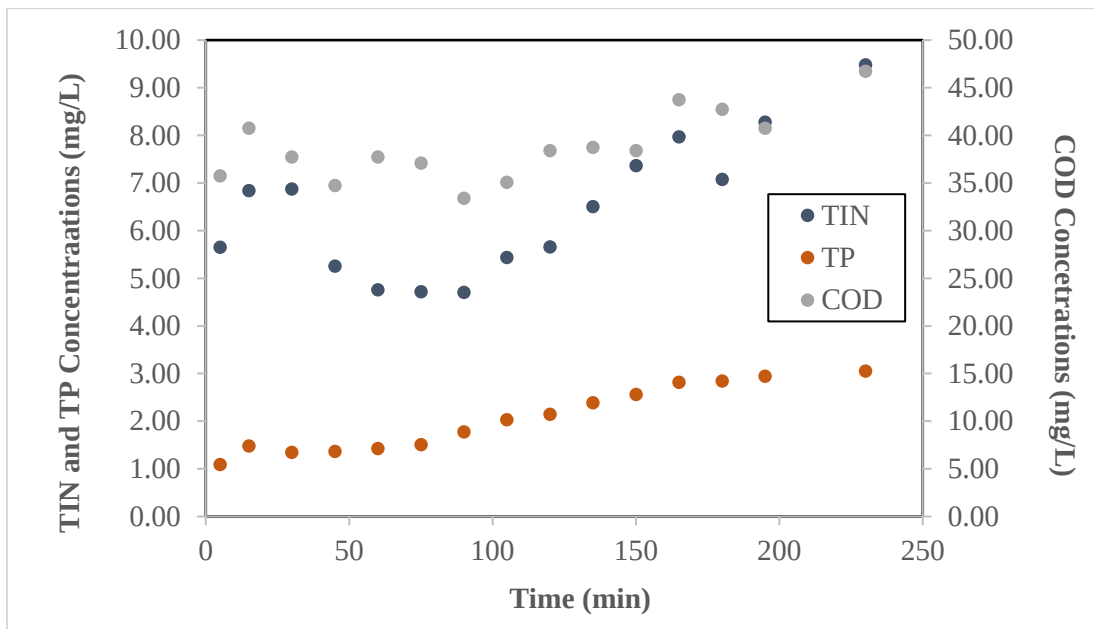


Figure A15: Runoff TIN (mg N/L), TP (mg P/L), and COD (mg/L) concentrations over time during an irrigation event.

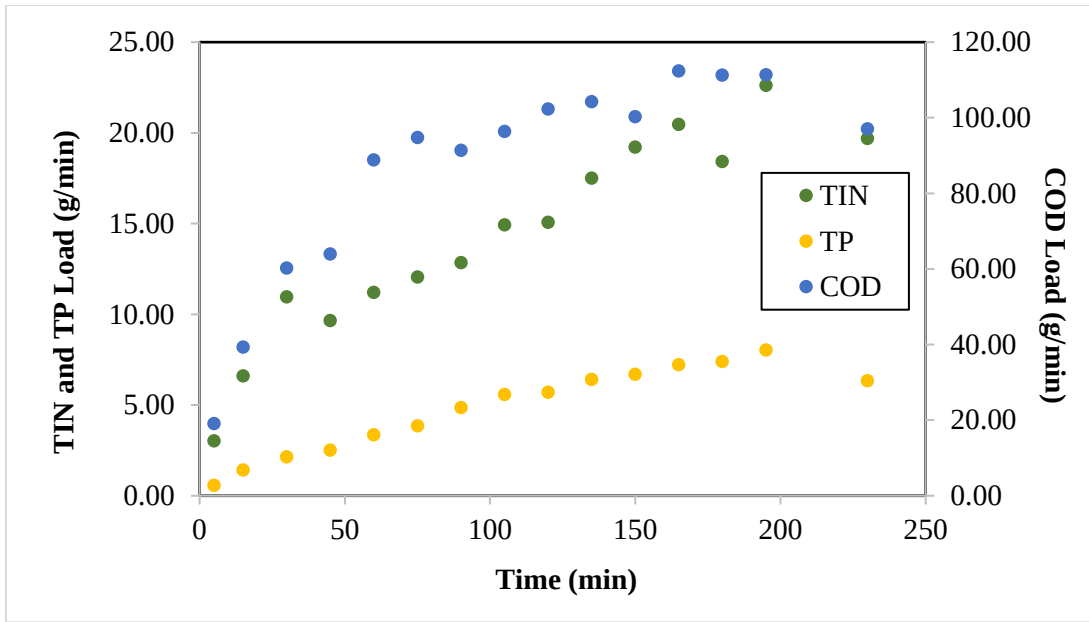


Figure A16: Runoff TIN, TP, and COD loads (g/min) over time during an irrigation event.

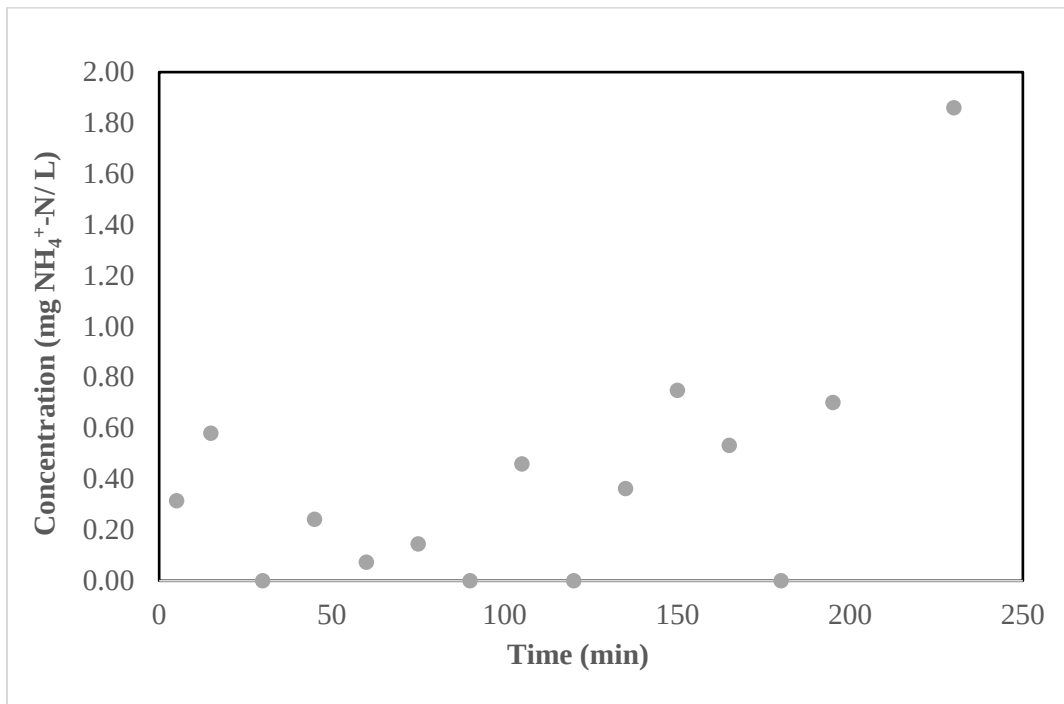


Figure A17: Influent runoff $\text{NH}_4^+ -\text{N}$ results.

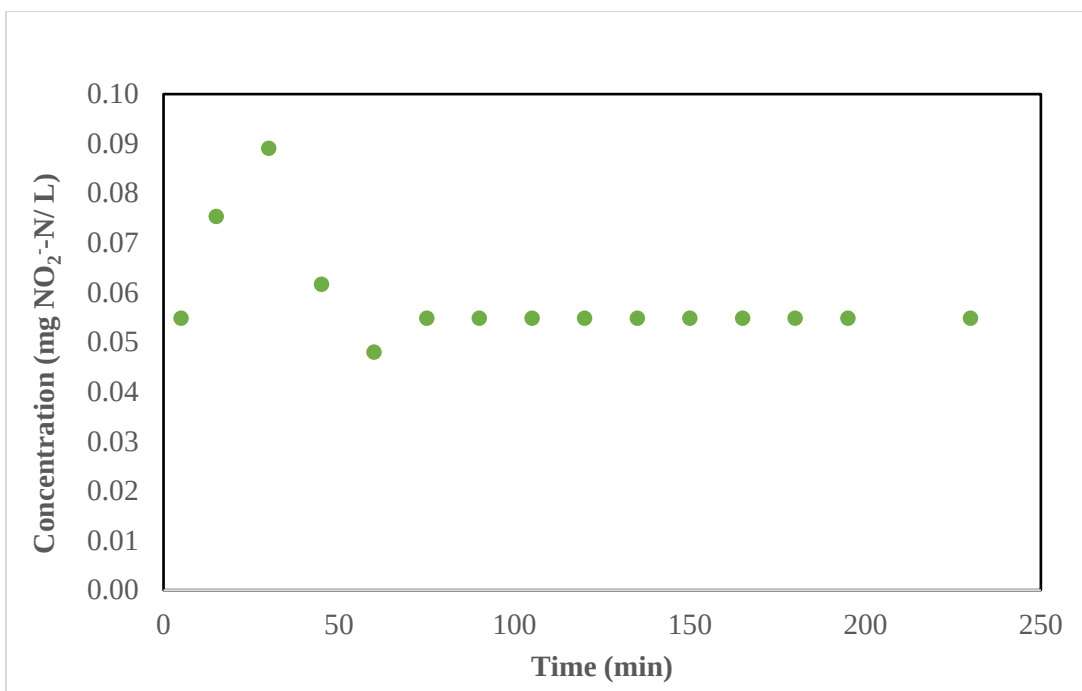


Figure A18: Influent runoff NO_2^- -N results.

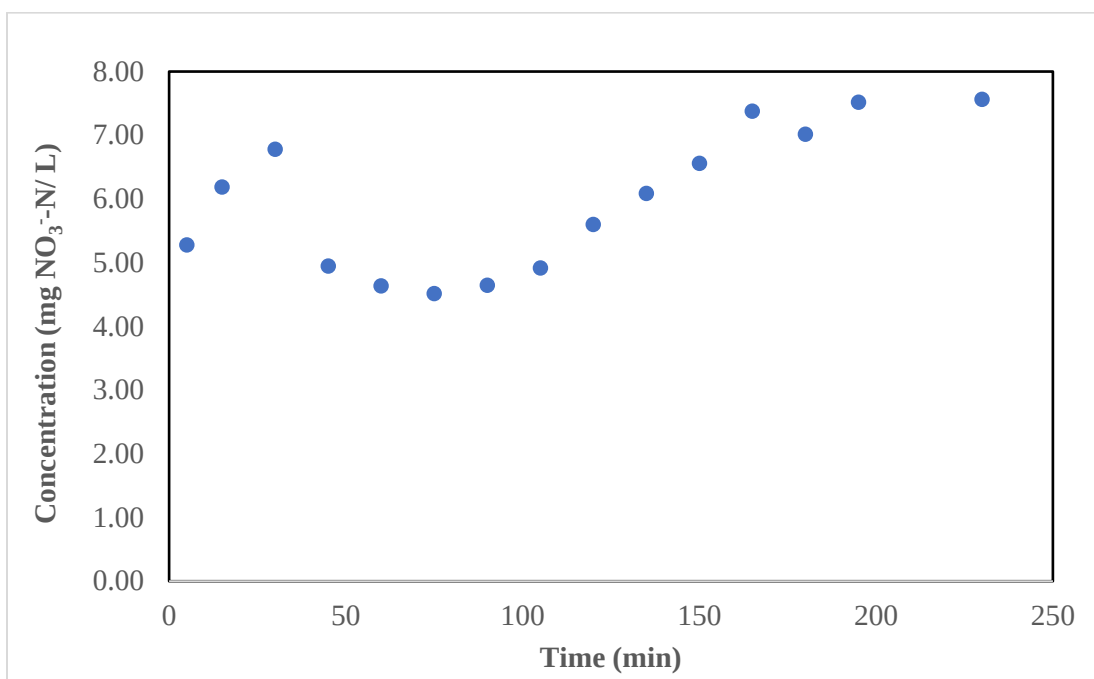


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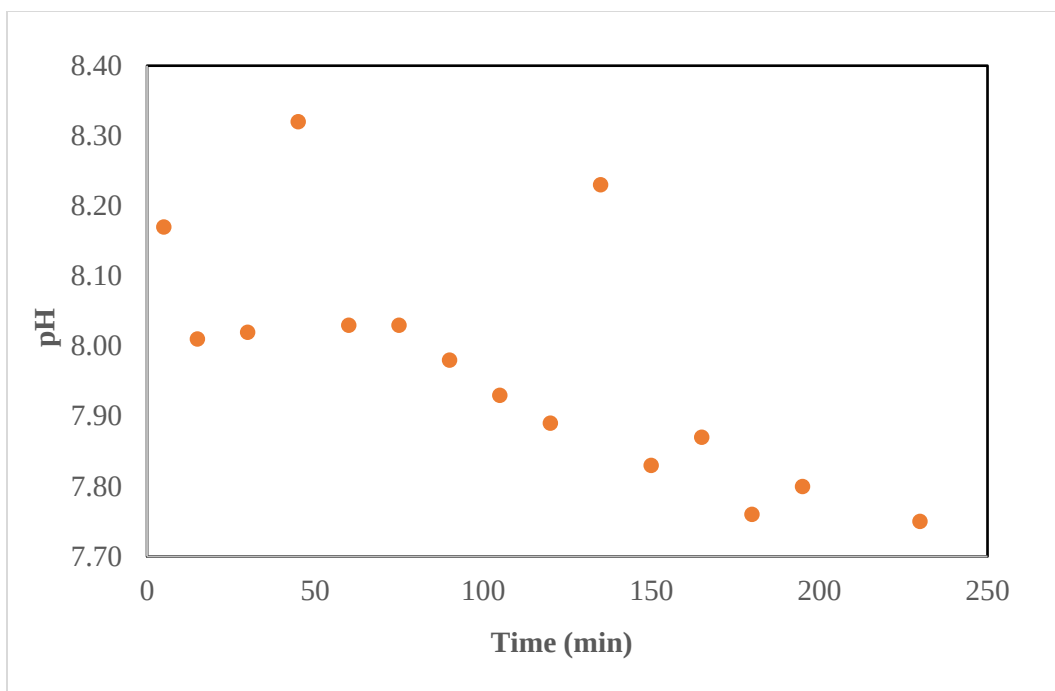


Figure A20: Influent runoff pH results.

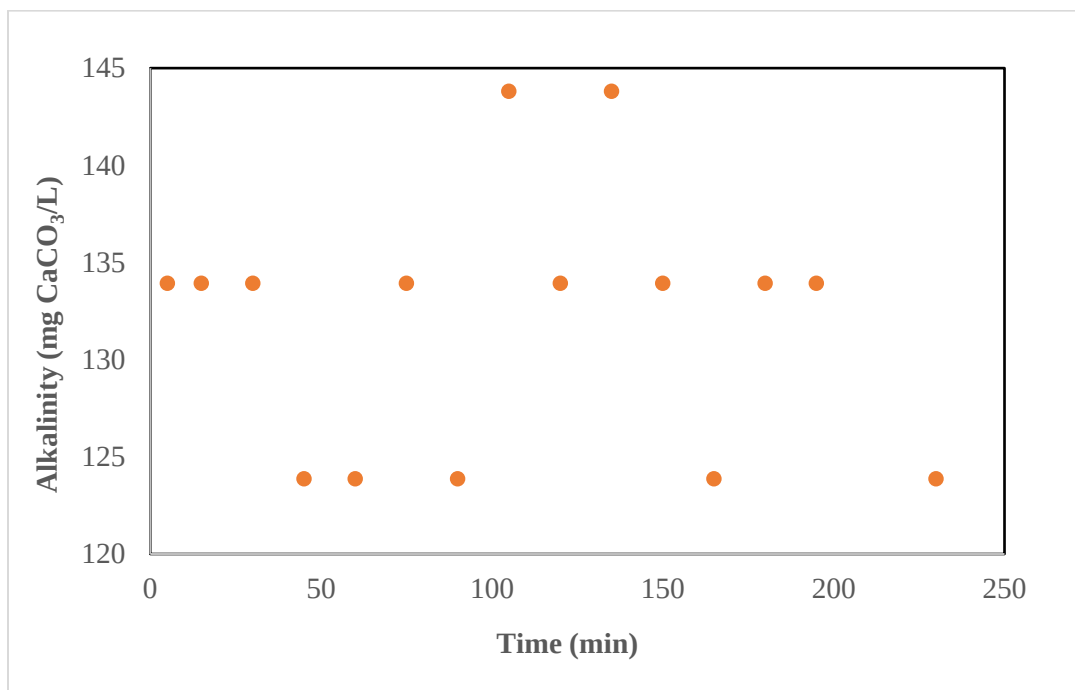


Figure A21: Influent runoff alkalinity results.



Figure A22: Muhly plant.



Figure A23: Muhly grass after being planted in CBA unit on 12/12/2023.



Figure A24: Muhly grass after being planted in PBA unit on 12/12/2023.

Table A1: Preliminary data (2022) for irrigation runoff chemical composition used for bioretention design (average concentrations for each sample period).

Visit on	TIN (mg/L N)	NH ₄ ⁺ -N (mg/L)	NO ₂ ⁻ -N (mg/L)	NO ₃ ⁻ -N (mg/L)	TP (mg/L PO ₄ ³⁻)	COD (mg/L)
2/3/2022	2.14	BDL	0.06	2.08	0.42	NM
2/25/2022	24.07	0.20	0.17	23.7	4.48	57
3/6/2022	NM	NM	0.04	4.00	NM	52
3/30/2022	6.86	0.87	0.09	5.90	3.40	NM

BDL = below detection limit; NM = not measured

Table A2: Preliminary data (2023) for irrigation runoff chemical composition (average bioretention monitoring results).

Sample	NH ₄ ⁺ -N (mg/L)	NO ₂ ⁻ -N (mg/L)	NO ₃ ⁻ -N (mg/L)	PO ₄ ³⁻ (mg/L)	TIN (mg/L N)	TP (mg/L PO ₄ ³⁻)	COD (mg/L)	pH	Alk (as CaCO ₃)
Average influent	0.82	0.07	5.54	2.17	29.46	1.65	12.03	7.91	176.75
Average effluent	0.46	0.06	3.32	1.37	17.33	1.22	15.81	7.17	294.98

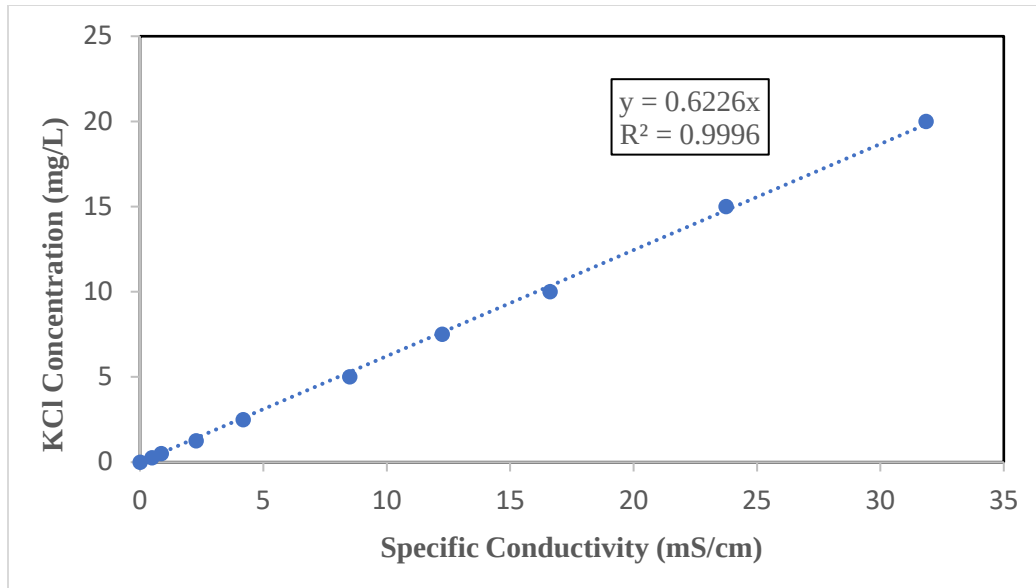


Figure A25: Standard curve for converting specific conductivity (mS/cm) to KCl concentration (mg/L).

Table A3: Tracer study data analysis on the bioretention systems.

	BC unit (CBA) lower outlet	Unsaturated BC unit (PBA) lower outlet	BC unit (CBA) upper outlet	Unsaturated BC unit (PBA) upper outlet
HLR (cm/min)	0.312	0.287	0.558	0.499
MRT (min)	70.11	45.62	60.47	61.65
MDI (t_{90}/t_{10})	3.53	6.67	2.98	2.41
t_{50}/MRT	0.89	0.74	0.85	0.89
Variance (min)	1019	1064	712	510
Standard Deviation (min)	32	33	27	23
Variance (Θ)	0.21	0.51	0.19	0.13
Standard Deviation (Θ)	0.46	0.72	0.44	0.37
TIS n	5.82	2.95	6.13	8.45
Mass Recovered (%)	99.9	98.0	104.2	102.2

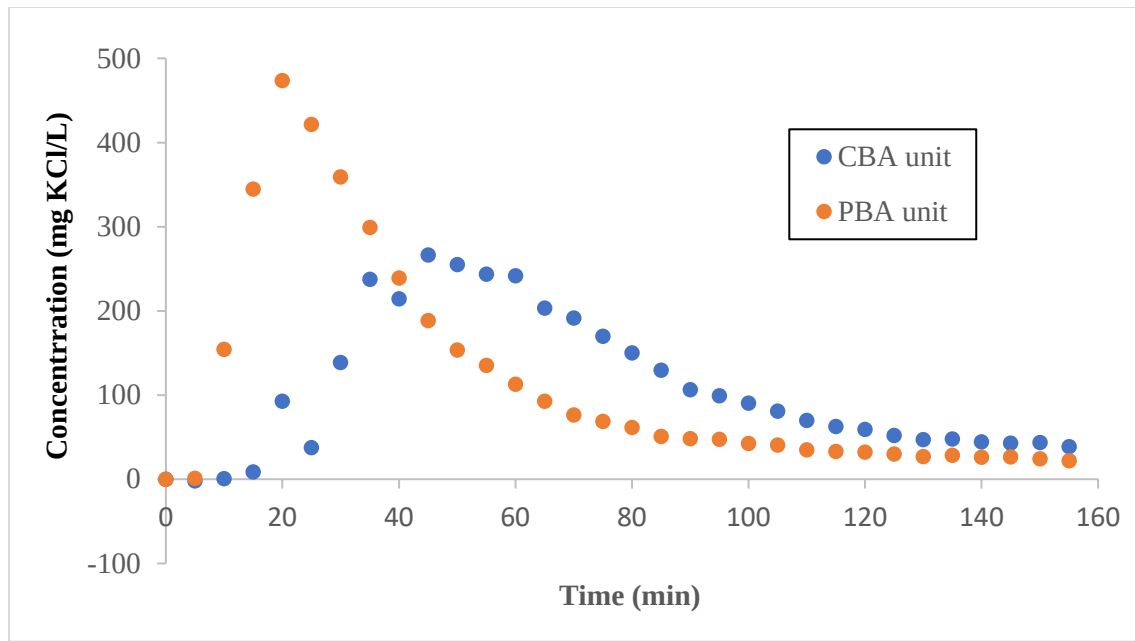


Figure A26: Tracer concentration (mg KCl/L) over time (minutes) for bioretention systems configured at their lower outlets.

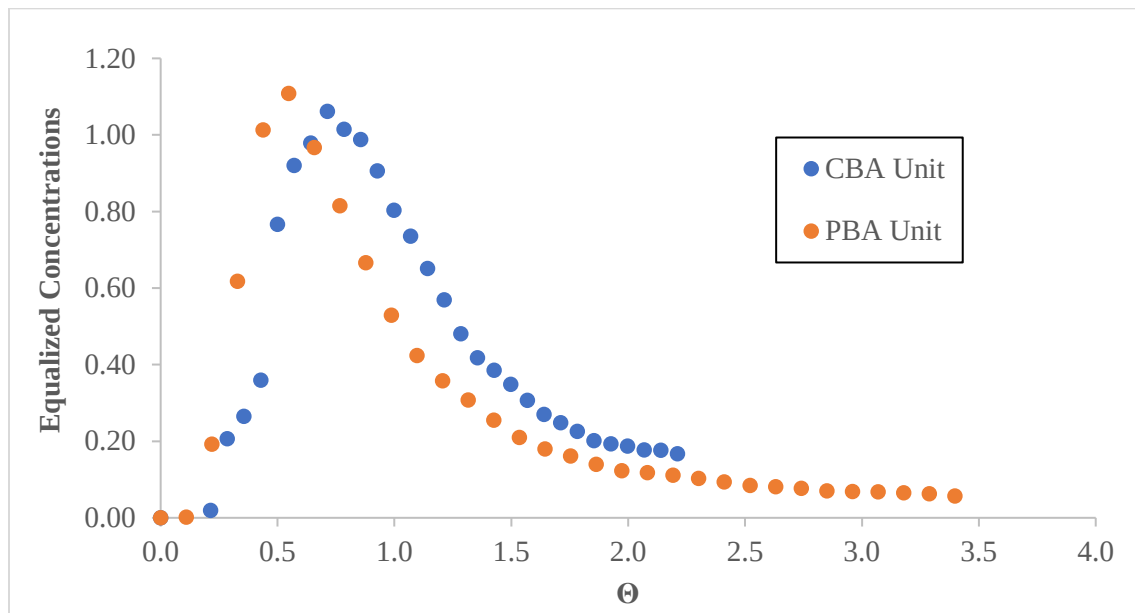


Figure A27: Tracer equalized concentration (-) over Θ (-) for bioretention systems configured at their lower outlets.

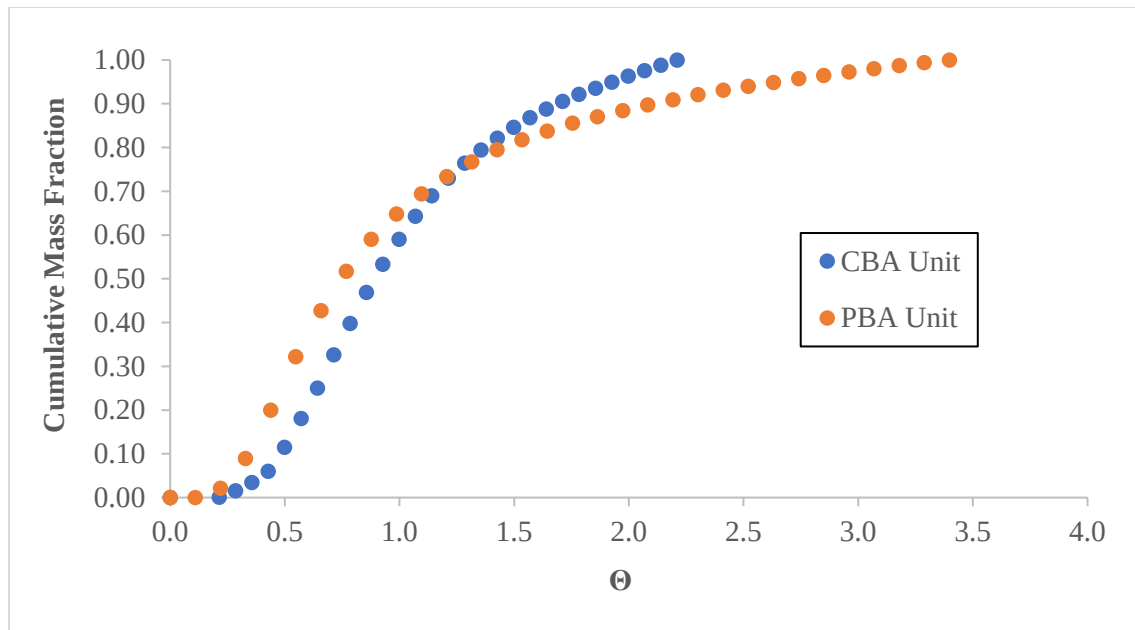


Figure A28: Cumulative tracer mass fraction exported (-) over Θ (-) for bioretention systems configured at their lower outlets.

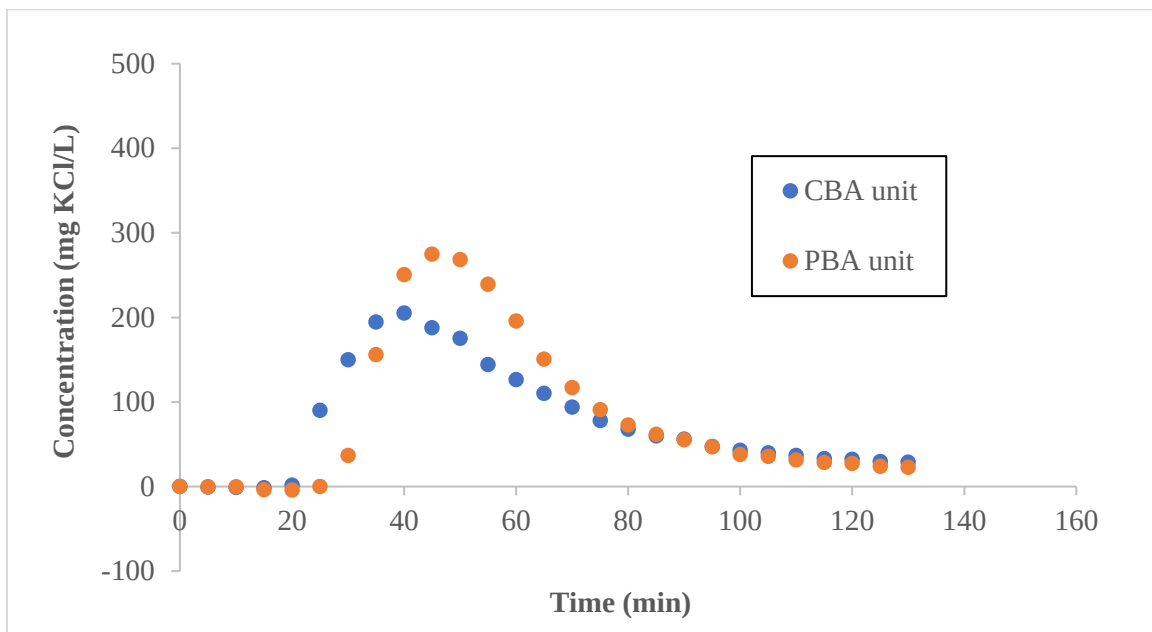


Figure A29: Tracer concentration (mg KCl/L) over time (minutes) for bioretention systems configured at their upper outlets.

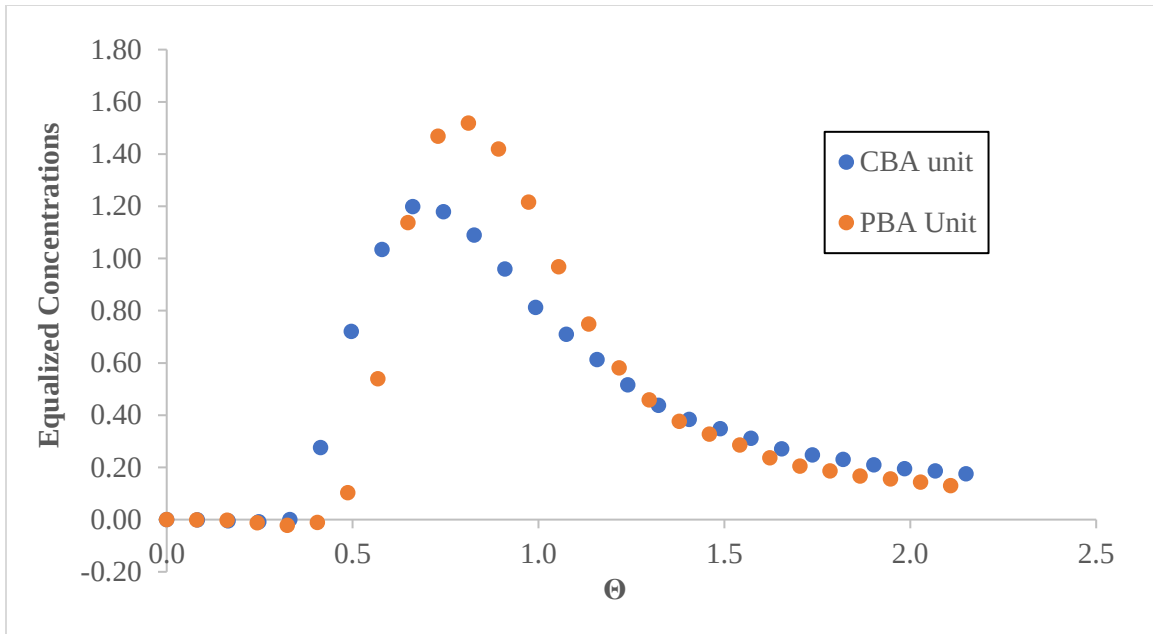


Figure A30: Tracer equalized concentration (-) over Θ (-) for bioretention systems configured at their upper outlets.

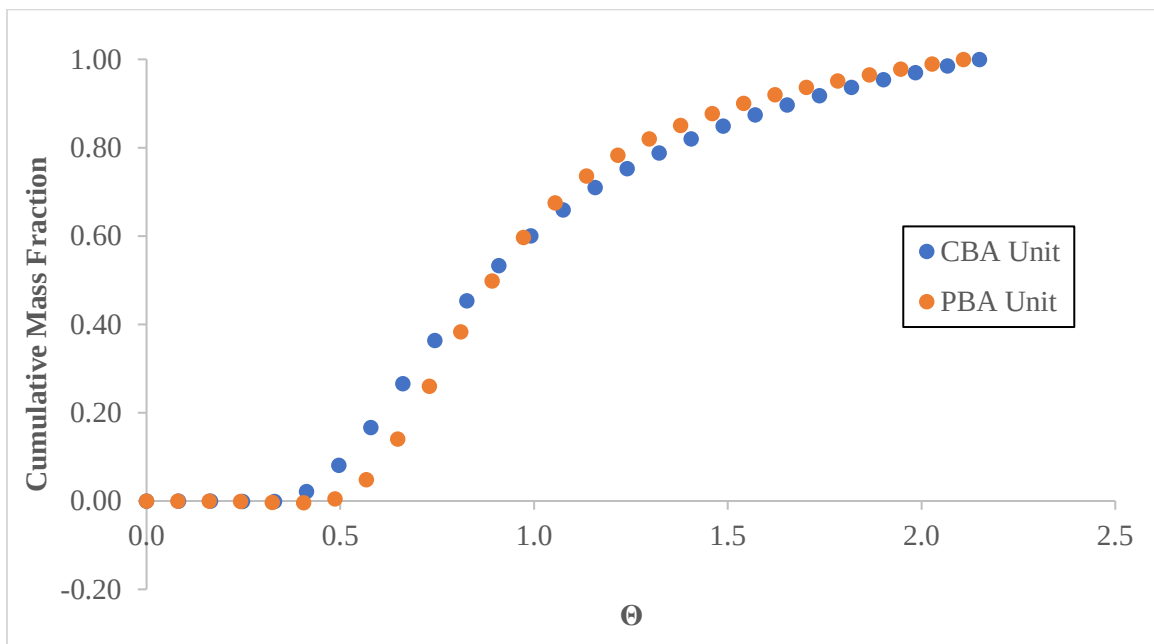


Figure A31: Tracer cumulative mass fraction exported (-) over Θ (-) for bioretention systems configured at their upper outlets.

Table A4: Summary of bioretention monitoring runoff events.

Event #	Date	HLR (cm/min)	Estimated % area covered	Outlet used	Muhly grass presence
1	6-Sep-23	0.109	5	Lower	No
2	12-Sep-23	0.109	5	Lower	No
3	26-Sep-23	0.109	5	Lower	No
4	18-Oct-23	0.217	2.5	Lower	No
5	25-Oct-23	0.217	2.5	Lower	No
6*	31-Oct-23	0.217	2.5	Lower	No
7	3-Nov-23	0.544	1	Lower	No
8	8-Nov-23	0.544	1	Lower	No
9*	16-Nov-23	0.544	1	Lower	No
10*	20-Nov-23	0.544	1	Upper	No
11	29-Nov-23	0.544	1	Upper	No
12	6-Dec-23	0.544	1	Upper	No
13*	22-Jan-24	0.544	1	Lower	Yes
14	31-Jan-24	0.544	1	Lower	Yes
15	8-Feb-24	0.544	1	Lower	Yes
16	20-Mar-24	0.217	2.5	Upper	Yes
17	10-Apr-24	0.217	2.5	Upper	Yes
18	18-Apr-24	0.217	2.5	Upper	Yes

* Represents that the influent runoff was leftover water in the slough backed up by the weir



Figure A32: Preliminary batch adsorption studies with bioretention media materials.

Table A5: Media characteristics of batch media tests.

Vial Name	Media Characteristics
HPS + BC	20 cm ³ of HPS + 20 cm ³ of biochar
HPS	40 cm ³ of HPS
HPS + WC	20 cm ³ of HPS + 15 cm ³ of woodchips

Table A6: Total and drainable porosities of media.

	SM+BC	BC+SM+WC
Total Volume used (L)	4	4.5
weight of water added (g)	1926	2020
volume added (mL)	1926	2020
Water sieved out (g)	974	976
Volume sieved out (mL)	974	976
Total Porosity (cm ³ /cm ³)	0.48	0.45
Drainable porosity (cm ³ /cm ³)	0.24	0.22

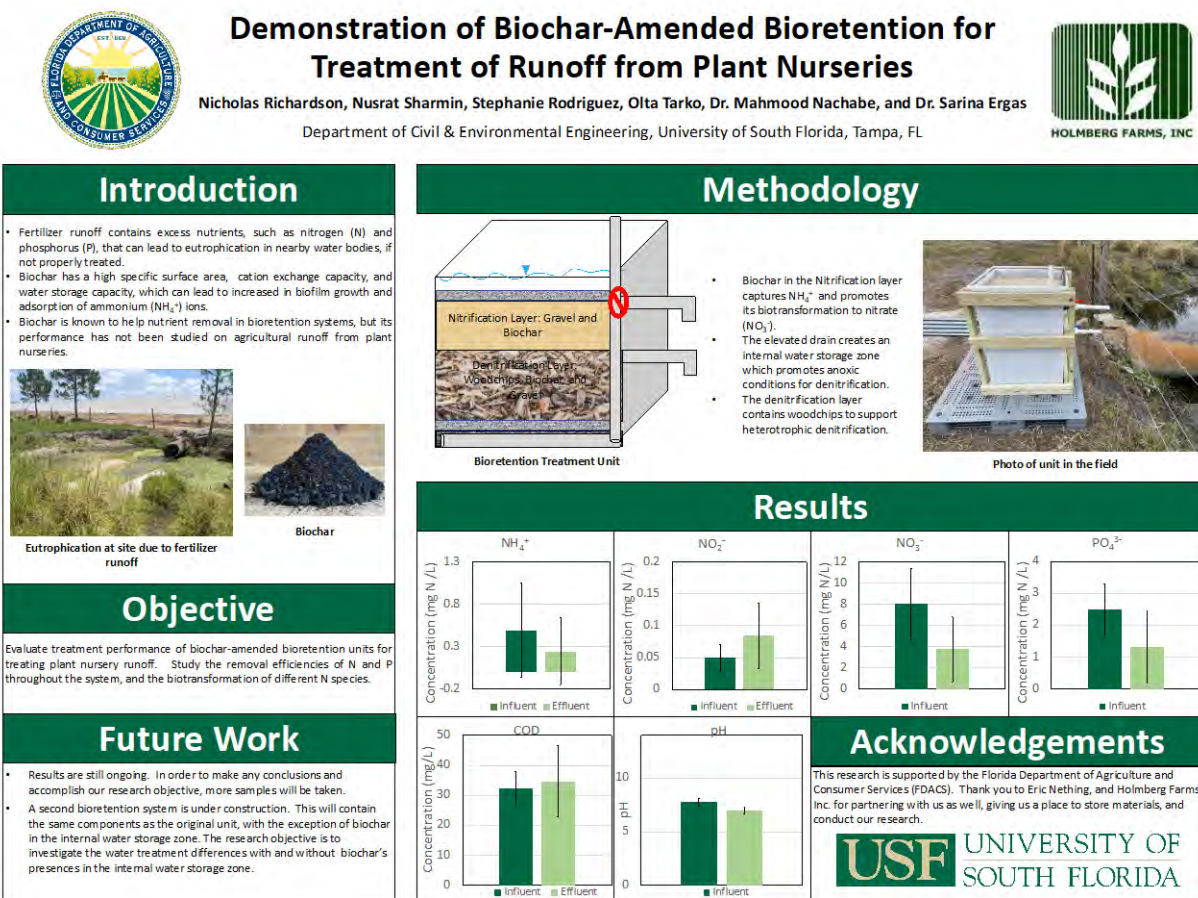


Figure A33: Poster from the American Ecological Engineering Society's annual meeting.



Figure A34: Muhly grass plants in both units on 1/2/2024.

Table A7: Average influent concentrations of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, Inorganic N, TP, COD, pH, and Alkalinity from all runoff Events 1-18.

Influent Concentrations								
Event #	$\text{NH}_4^+\text{-N}$ (mg/L)	$\text{NO}_2^-\text{-N}$ (mg/L)	$\text{NO}_3^-\text{-N}$ (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO_3)	pH
1	0.38	0.07	3.51	3.95	1.70	32.73	195.64	7.85
2	0.31	0.06	2.56	2.93	1.55	31.83	102.74	7.74
3	0.18	0.02	1.29	1.49	0.65	22.59	114.80	7.57
4	0.08	0.04	1.52	1.64	0.95	57.73	111.34	7.72
5	0.06	0.02	1.18	1.25	1.09	29.35	122.41	7.94
6	0.28	0.07	2.16	2.51	0.34	48.59	83.32	7.38
7	0.14	0.04	0.71	0.88	1.25	28.40	126.75	8.16
8	0.01	0.03	0.44	0.47	1.25	24.54	126.75	8.17
9	0.33	0.05	1.15	1.54	0.56	35.69	100.15	7.44
10	0.29	0.04	0.91	1.24	0.36	36.54	94.11	7.58
11	0.50	0.10	0.96	1.57	1.36	24.02	119.46	8.10
12	0.48	0.06	1.07	1.61	1.38	34.02	123.82	7.97
13	0.60	0.04	2.23	2.87	0.07	31.73	96.33	7.23
14	0.13	0.02	2.50	2.65	1.63	29.92	128.19	7.86
15	0.01	0.03	1.25	1.28	0.80	28.11	121.58	7.76
16	0.03	0.06	6.02	6.11	1.17	28.59	152.91	8.20
17	0.00	0.08	8.56	8.64	1.50	28.88	135.24	8.14
18	0.28	0.30	8.06	8.64	0.84	36.83	114.21	7.79

Table A8: CBA unit average effluent concentrations of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, Inorganic N, TP, COD, pH, and Alkalinity from all runoff Events 1-18.

CBA Unit Average Concentrations								
Event #	$\text{NH}_4^+\text{-N}$ (mg/L)	$\text{NO}_2^-\text{-N}$ (mg/L)	$\text{NO}_3^-\text{-N}$ (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO_3)	pH
1	0.63	0.11	0.16	0.90	0.05	18.88	320.59	6.93
2	0.70	0.06	0.16	0.92	0.39	29.69	119.46	6.97
3	0.25	0.00	0.01	0.27	0.21	26.69	135.31	7.02
4	0.36	0.04	0.07	0.48	0.16	48.87	109.08	7.15
5	0.04	0.00	0.02	0.06	0.39	23.88	128.11	7.23
6	0.12	0.03	0.06	0.21	0.26	30.92	122.12	7.16
7	0.14	0.05	0.08	0.26	0.66	23.69	113.64	7.60
8	0.06	0.06	0.09	0.21	0.97	22.45	120.95	7.62
9	0.25	0.16	0.23	0.64	0.73	27.92	120.54	7.40
10	0.15	0.13	0.12	0.40	0.37	28.02	103.10	7.32
11	0.18	0.13	0.18	0.49	0.65	17.45	123.85	7.32
12	0.13	0.11	0.12	0.36	0.81	22.30	126.69	7.13
13	0.54	0.19	1.16	1.89	0.20	35.21	125.86	7.24
14	0.33	0.18	1.11	1.62	0.92	26.88	133.11	7.18
15	0.13	0.13	0.45	0.71	0.60	24.16	123.82	7.09
16	1.44	0.04	0.03	1.51	0.49	32.54	217.81	7.10
17	0.00	0.23	0.79	1.02	0.54	28.30	212.57	6.92
18	0.22	0.31	0.55	1.07	0.50	36.02	185.31	7.15

Table A9: PBA unit average effluent concentrations of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, Inorganic N, TP, COD, pH, and Alkalinity from all runoff Events 1-18.

PBA Unit Average Concentrations								
Event #	$\text{NH}_4^+\text{-N}$ (mg/L)	$\text{NO}_2^-\text{-N}$ (mg/L)	$\text{NO}_3^-\text{-N}$ (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO_3)	pH
1	0.22	0.14	1.10	1.46	0.77	61.73	244.60	6.84
2	0.32	0.35	1.52	2.19	1.40	39.92	109.09	6.94
3	0.13	0.03	0.02	0.18	0.76	70.11	184.70	6.81
4	0.01	0.18	0.43	0.62	0.45	58.97	117.48	7.19
5	0.04	0.16	0.31	0.51	0.87	38.97	135.54	7.13
6	0.17	0.29	0.67	1.14	0.63	34.97	121.05	7.26
7	0.08	0.04	0.27	0.38	0.80	28.35	122.23	7.62

8	0.07	0.04	0.24	0.35	1.73	28.11	129.00	7.70
9	0.35	0.12	0.62	1.10	1.40	41.73	118.74	7.37
10	0.11	0.18	0.42	0.71	0.82	29.78	103.02	7.30
11	0.56	0.10	0.43	1.09	0.93	18.30	126.36	7.36
12	0.15	0.03	0.15	0.33	1.80	25.30	133.45	7.16
13	0.43	0.07	1.53	2.02	0.40	42.21	122.97	7.38
14	0.12	0.05	1.70	1.88	0.98	34.83	130.42	7.43
15	0.07	0.07	0.89	1.03	0.77	29.59	120.76	7.18
16	0.34	0.14	0.20	0.68	1.30	38.11	220.07	7.03
17	0.00	0.23	1.08	1.31	1.04	31.40	195.61	6.98
18	0.14	0.50	2.63	3.28	0.83	37.78	118.34	7.35

Table A10: CBA unit average % load reduction of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, Inorganic N, TP, COD, pH, and Alkalinity from all runoff Events 1-18.

CBA Unit % Load Reduction								
Event #	$\text{NH}_4^+\text{-N}$	$\text{NO}_2^-\text{-N}$	$\text{NO}_3^-\text{-N}$	TIN	TP	COD	Alkalinity (as CaCO_3)	pH
1	-66.0	-59.7	95.5	77.3	97.3	42.3	-63.9	11.8
2	-124.7	-9.4	93.9	68.6	75.0	6.7	-16.3	9.9
3	-38.9	78.9	98.8	81.7	68.1	-18.1	-17.9	7.3
4	-330.3	-19.2	95.2	70.8	83.3	15.4	2.0	7.3
5	31.8	73.9	98.3	94.9	64.5	18.7	-4.7	8.9
6	56.9	61.0	97.1	91.6	23.3	36.4	-46.6	2.9
7	0.0	-35.3	88.9	70.2	47.4	16.6	10.3	6.8
8	-525.0	-130.6	80.5	55.9	21.9	8.5	4.6	6.8
9	25.2	-201.3	80.3	58.5	-30.1	21.8	-20.4	0.4
10	46.4	-219.0	86.9	67.4	-2.9	23.3	-9.6	3.5
11	63.4	-24.8	81.7	68.8	52.0	27.4	-3.7	9.6
12	72.4	-70.8	88.4	77.3	41.5	34.4	-2.3	10.5
13	9.9	-354.0	47.8	34.0	-180.5	-11.0	-30.7	-0.1
14	-153.7	-772.0	55.5	38.9	43.2	10.2	-3.8	8.7
15	-1780.0	-340.0	63.8	44.9	25.5	14.1	-1.8	8.7
16	-4120.0	33.3	99.5	75.3	57.7	-13.8	-42.4	13.4
17	NA	-189.5	90.7	88.2	64.0	2.0	-57.2	15.0
18	20.7	-1.9	93.2	87.6	41.1	2.2	-62.2	8.2

Table A11: PBA unit average % load reduction of $\text{NH}_4^+\text{-N}$, $\text{NO}_2^-\text{-N}$, $\text{NO}_3^-\text{-N}$, Inorganic N, TP, COD, pH, and Alkalinity from all runoff Events 1-18.

PBA Unit % Load Reduction								
Event #	$\text{NH}_4^+\text{-N}$ (mg/L)	$\text{NO}_2^-\text{-N}$ (mg/L)	$\text{NO}_3^-\text{-N}$ (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO_3)	pH
1	41.7	-102.6	68.6	63.0	55.1	-88.6	-25.0	12.9
2	-3.5	-489.1	40.7	25.3	9.5	-25.4	-6.2	10.3
3	31.5	-73.7	98.4	88.2	-16.8	-210.4	-60.9	10.0
4	90.9	-371.2	71.4	62.4	53.3	-2.1	-5.5	6.8
5	31.8	-869.6	73.8	59.5	20.6	-32.8	-10.7	10.2
6	38.5	-308.0	68.8	54.8	-85.1	28.0	-45.3	1.5
7	44.2	-7.8	62.5	56.9	35.8	0.2	3.6	6.6
8	-575.0	-55.6	45.8	26.7	-38.8	-14.6	-1.8	5.8
9	-7.1	-127.3	46.3	28.7	-150.1	-16.9	-18.6	1.0
10	60.0	-343.1	54.2	42.4	-125.1	18.5	-9.5	3.8
11	-11.6	6.9	55.3	30.7	31.6	23.8	-5.8	9.1
12	69.2	47.2	85.8	79.4	-31.0	25.6	-7.8	10.1
13	28.9	-62.0	31.4	29.5	-465.9	-33.0	-27.7	-2.1
14	5.3	-164.0	32.1	29.3	39.5	-16.4	-1.7	5.5
15	-900.0	-134.3	28.5	19.9	3.9	-5.3	0.7	7.6
16	-908.0	-139.1	96.7	88.9	-11.7	-33.3	-43.9	14.2
17		-192.6	87.3	84.8	30.6	-8.7	-44.6	14.2
18	48.3	-66.3	67.4	62.1	2.0	-2.6	-3.6	5.7

Table A12: Average concentrations from runoff as influent into both units.

Influent Concentrations											
Event #s	Configuration	HLR (cm/min)	Outlet Height	NH ₄ ⁺ -N (mg/L)	NO ₂ ⁻ -N (mg/L)	NO ₃ ⁻ -N (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO ₃)	pH
1 and 3	Low HLR, Lower Outlet Height, No Plants	0.11	Lower	0.29	0.05	2.45	2.79	1.30	29.05	137.73	7.72
4 - 6	Medium HLR, Lower Outlet Height, No Plants	0.22	Lower	0.14	0.04	1.62	1.80	0.80	45.23	105.69	7.68
7 - 9	High HLR, Lower Outlet Height, No Plants	0.55	Lower	0.16	0.04	0.77	0.96	1.02	29.54	117.88	7.92
10 - 12	High HLR, Upper Outlet Height, No Plants	0.55	Upper	0.42	0.07	0.98	1.47	1.03	31.53	112.46	7.88
13 - 15	High HLR, Lower Outlet Height, Yes Plants	0.55	Lower	0.25	0.03	1.99	2.27	0.83	29.92	115.37	7.62
16 - 18	Medium HLR, Upper Outlet Height, Yes Plants	0.22	Upper	0.10	0.15	7.55	7.80	1.17	31.43	134.12	8.04

Table A13: Average % reduction from influent runoff load to effluent load from both units.

CBA and PBA Unit Average % Total Load Reduction											
Event #s	Configuration	HLR (cm/min)	Outlet Height	NH₄⁺-N (mg/L)	NO₂⁻-N (mg/L)	NO₃⁻-N (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO₃)	pH
1 and 3	Low HLR, Lower Outlet Height, No Plants	0.11	Lower	-9.38	-64.58	86.52	74.19	62.15	-60.34	-42.57	10.53
4 - 6	Medium HLR, Lower Outlet Height, No Plants	0.22	Lower	12.44	-181.36	83.81	72.12	42.35	12.81	-15.65	6.35
7 - 9	High HLR, Lower Outlet Height, No Plants	0.55	Lower	0.55	-104.27	67.10	49.30	-2.94	2.82	-2.52	4.69
10 - 12	High HLR, Upper Outlet Height, No Plants	0.55	Upper	49.03	-63.13	75.91	61.62	13.13	25.38	-6.18	7.85
13 - 15	High HLR, Lower Outlet Height, Yes Plants	0.55	Lower	-9.48	-276.36	42.73	32.78	22.41	-7.43	-9.35	4.85
16 - 18	Medium HLR, Upper Outlet Height, Yes Plants	0.22	Upper	-244.55	-64.79	88.33	81.04	33.01	-8.26	-42.87	11.85

Table A14: Average % reduction from influent runoff load to effluent load from the BC Unit.

CBA Unit Average % Load Reduction											
Event #s	Configuration	HLR (cm/min)	Outlet Height	NH₄⁺-N (mg/L)	NO₂⁻-N (mg/L)	NO₃⁻-N (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO₃)	pH
1 and 3	Low HLR, Lower Outlet Height, No Plants	0.11	Lower	-76.54	3.28	96.07	75.86	80.11	10.31	-32.67	9.65
4 - 6	Medium HLR, Lower Outlet Height, No Plants	0.22	Lower	-80.53	38.56	96.85	85.77	57.03	23.46	-16.40	6.36
7 - 9	High HLR, Lower Outlet Height, No Plants	0.55	Lower	-166.60	-122.38	83.23	61.54	13.06	15.63	-1.81	4.69
10 - 12	High HLR, Upper Outlet Height, No Plants	0.55	Upper	60.74	-104.86	85.68	71.19	30.18	28.37	-5.18	7.89
13 - 15	High HLR, Lower Outlet Height, Yes Plants	0.55	Lower	-641.25	-488.67	55.70	39.24	-37.26	4.43	-12.11	5.75
16 - 18	Medium HLR, Upper Outlet Height, Yes Plants	0.22	Upper	0.00	-52.69	94.48	83.70	54.25	-3.22	-53.95	12.19

Table A15: Average % reduction from influent runoff load to effluent load from the Unsaturated BC Unit.

PBA Unit Average % Load Reduction											
Event #s	Configuration	HLR (cm/min)	Outlet Height	NH₄⁺-N (mg/L)	NO₂⁻-N (mg/L)	NO₃⁻-N (mg/L)	TIN (mg/L)	TP (mg/L)	COD (mg/L)	Alkalinity (mg/L as CaCO₃)	pH
1 and 3	Low HLR, Lower Outlet Height, No Plants	0.11	Lower	23.23	-221.78	69.26	58.83	15.91	-108.13	-30.70	11.07
4 - 6	Medium HLR, Lower Outlet Height, No Plants	0.22	Lower	53.75	-516.24	71.35	58.90	-3.72	-2.30	-20.51	6.15
7 - 9	High HLR, Lower Outlet Height, No Plants	0.55	Lower	-179.29	-63.56	51.53	37.42	-51.05	-10.44	-5.59	4.46
10 - 12	High HLR, Upper Outlet Height, No Plants	0.55	Upper	60.00	-343.10	54.21	42.42	-125.09	18.50	-9.47	3.80
13 - 15	High HLR, Lower Outlet Height, Yes Plants	0.55	Lower	-288.61	-120.10	30.69	26.24	-140.80	-18.22	-9.58	3.66
16 - 18	Medium HLR, Upper Outlet Height, Yes Plants	0.22	Upper	0.00	-132.69	83.81	78.60	6.94	-14.88	-30.72	11.36

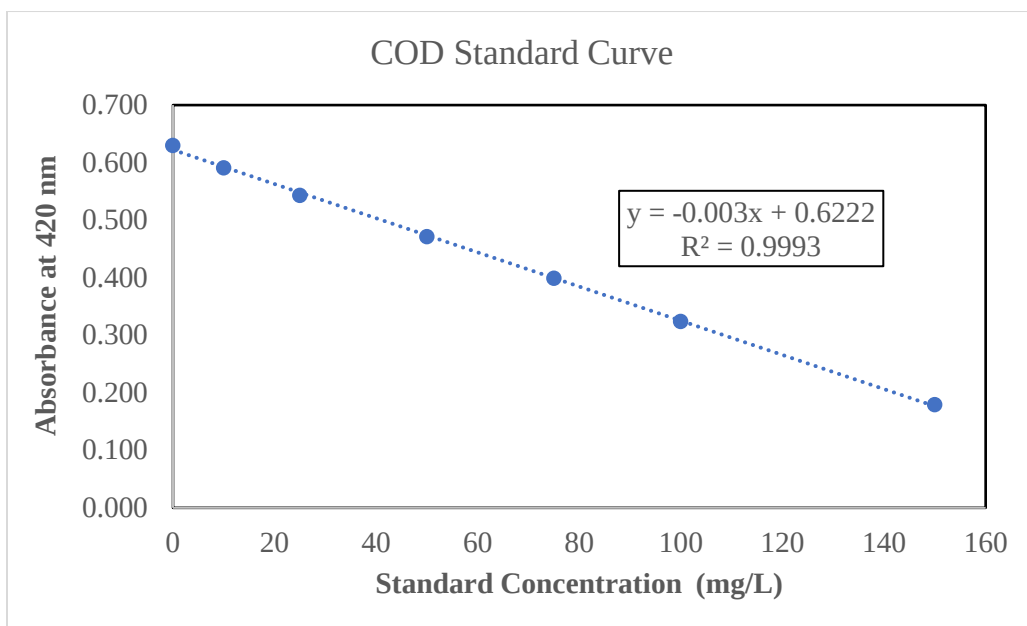


Figure A35: Standard curve for converting absorbance at 420nm to COD concentration (mg/L).

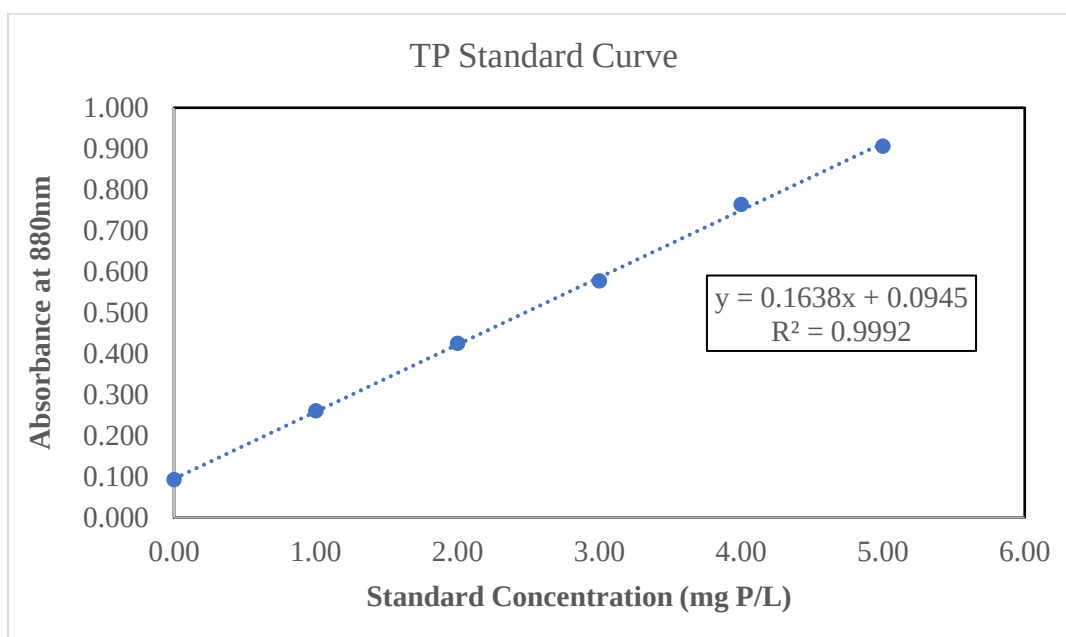


Figure A36: Standard curve for converting absorbance at 880nm to TP concentration (mg P/L).

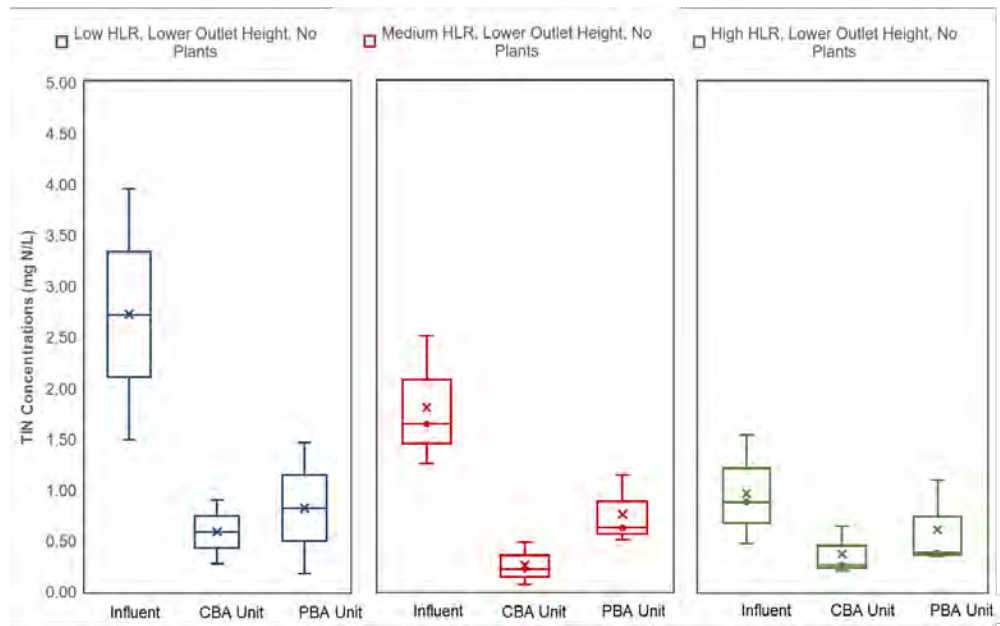


Figure A37: Box and Whisker plot for TIN Concentrations with low, medium, and high HLR; lower outlet height; for influent, CBA and PBA units without plants during events (1-9). Medium HLR shows best performance considering the cost benefit (greatest TIN reduction; lowest TIN concentrations in unit effluent). See ANOVA for statistical analysis.

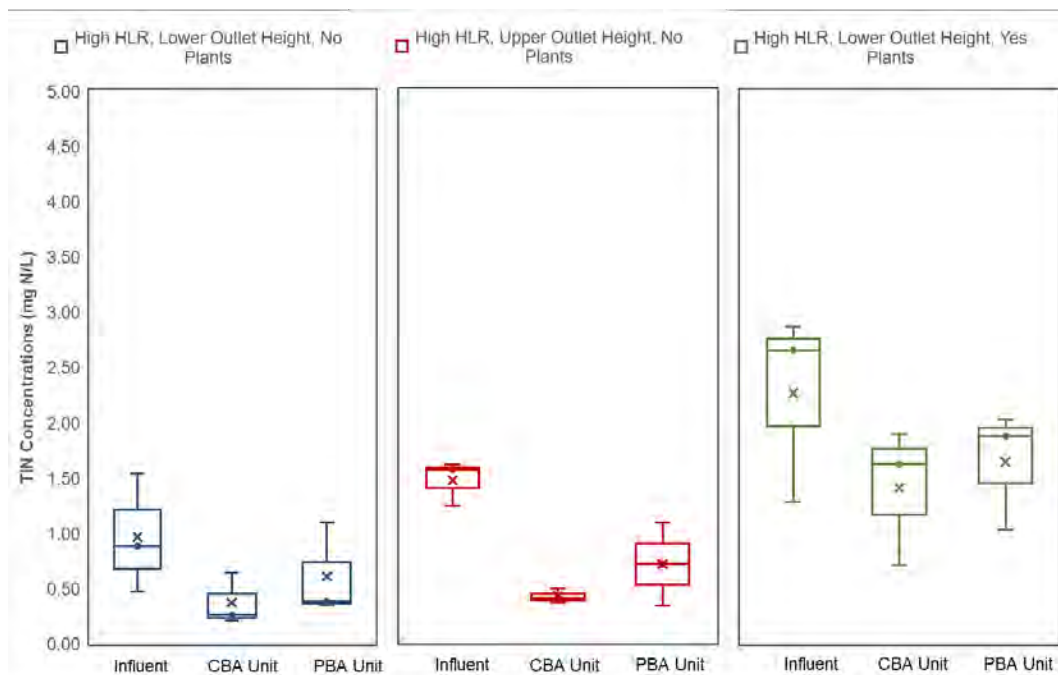


Figure A38: Box and Whisker plot for TIN Concentrations with high HLR; lower and upper outlet height; for influent, CBA and PBA units without plants during events (7-15). Results show higher TIN concentration (less load reduction) with plant addition. Most likely due to the presence of TIN in fertilizer in plants. Requires further investigation.

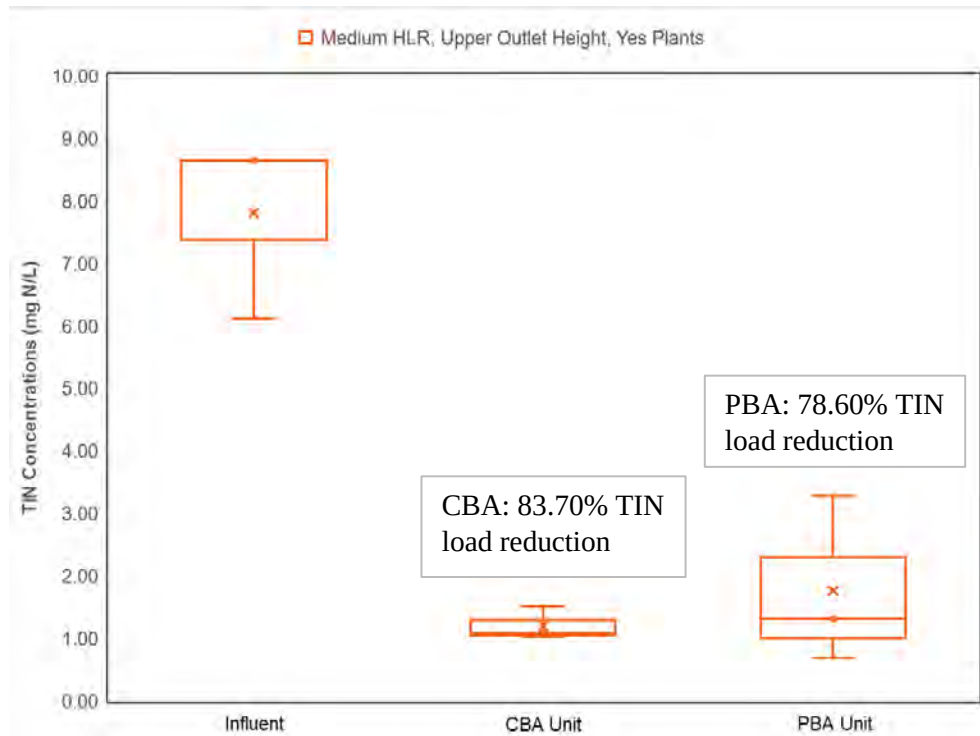


Figure A39: At the optimal configuration (Med HLR, Upper outlet height, with plants) events (16-18), CBA shows greatest TIN reduction (83.70 %), with lowest effluent TIN concentrations.

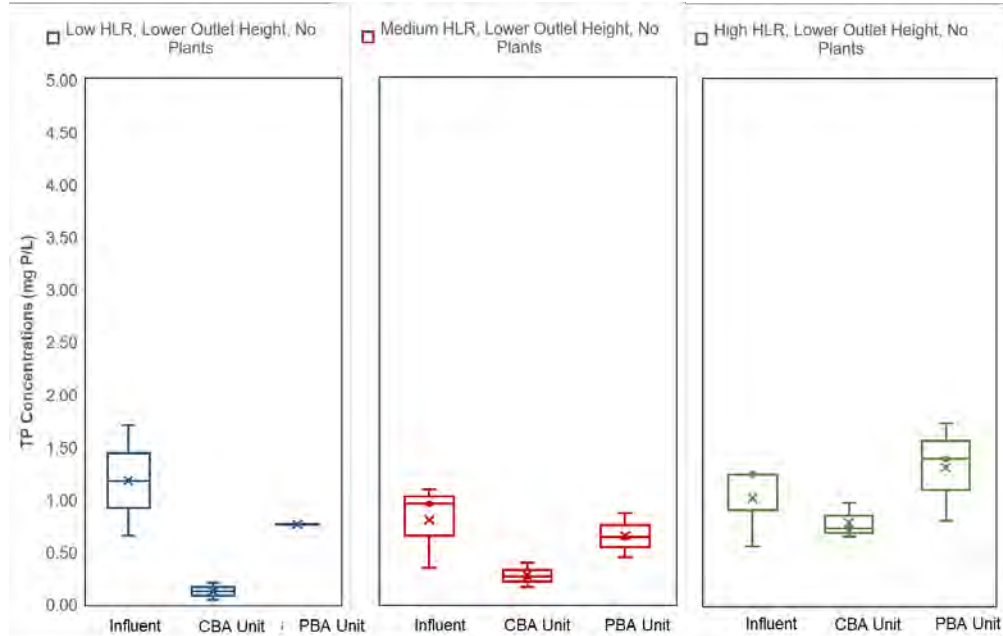


Figure A40: Box and Whisker plot for TP Concentrations with low, medium, and high HLR; lower outlet height; for influent, CBA and PBA units without plants during events (1-9). Medium HLR shows best performance considering the cost benefit (greatest TP reduction; lowest TP concentrations in unit effluent). See ANOVA for statistical analysis.

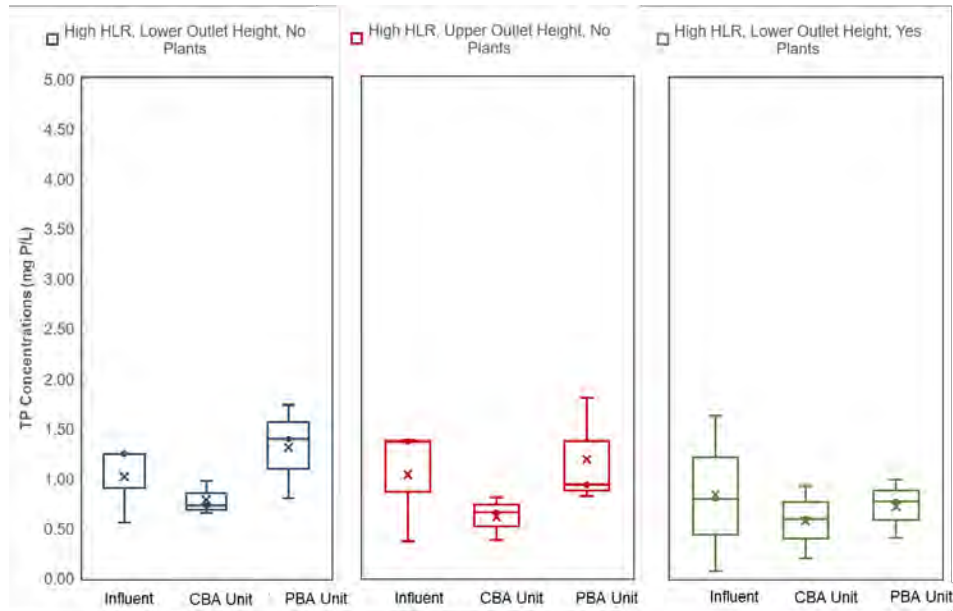


Figure A41: Box and Whisker plot for TP Concentrations with high HLR; lower and upper outlet height; for influent, CBA and PBA units without plants during events (7-15). Results show higher TP concentration (less load reduction) with plant addition. Most likely due to the presence of TP in fertilizer in plant. Requires further investigation.

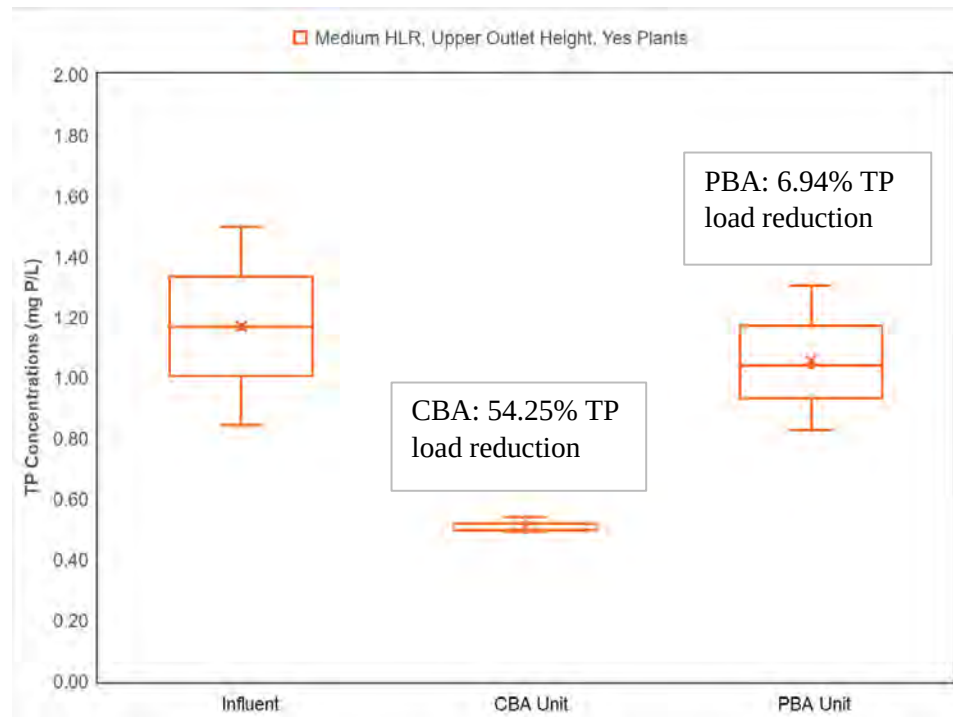


Figure A42: At the optimal configuration (Med HLR, Upper outlet height, with plants), CBA shows greatest TP reduction (54.25 %), with lowest effluent TP concentrations.

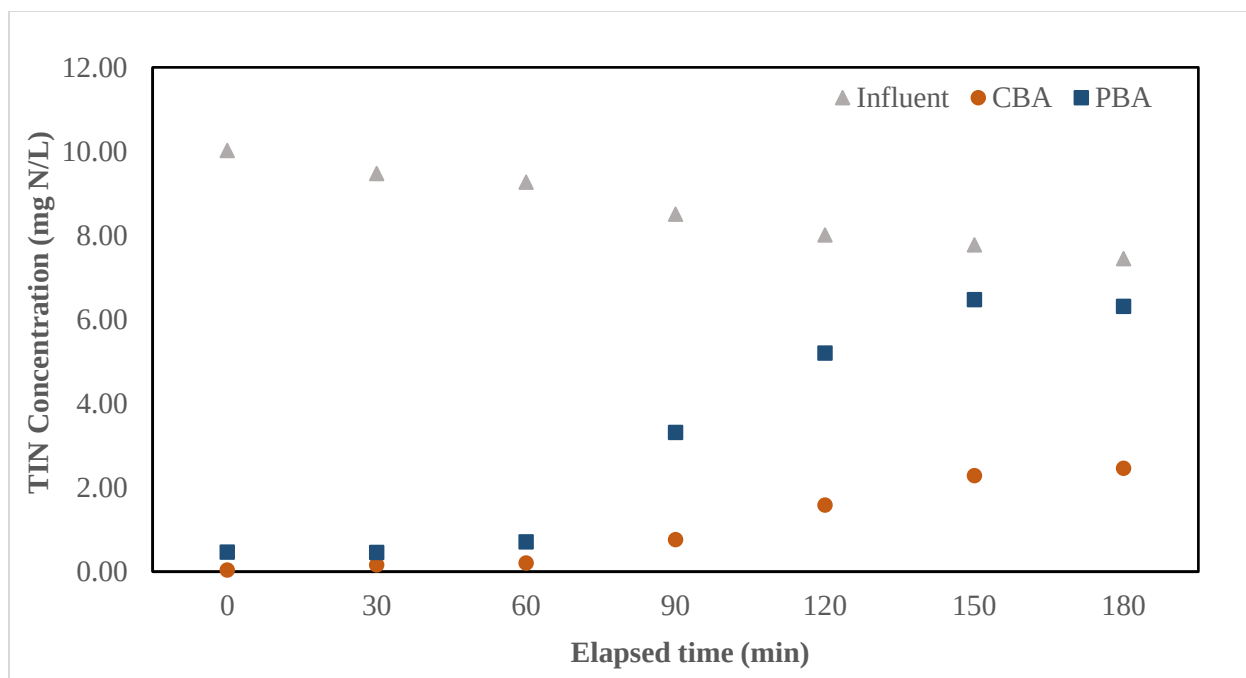


Figure A43: TIN loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

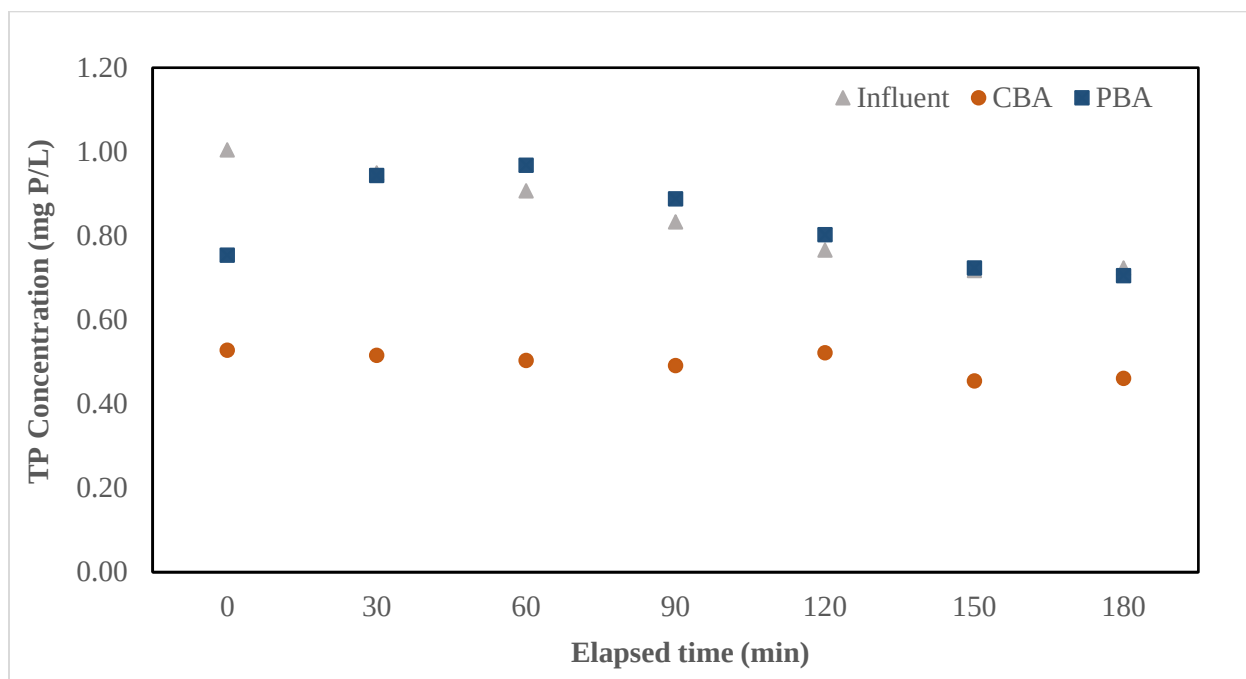


Figure A44: TP loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

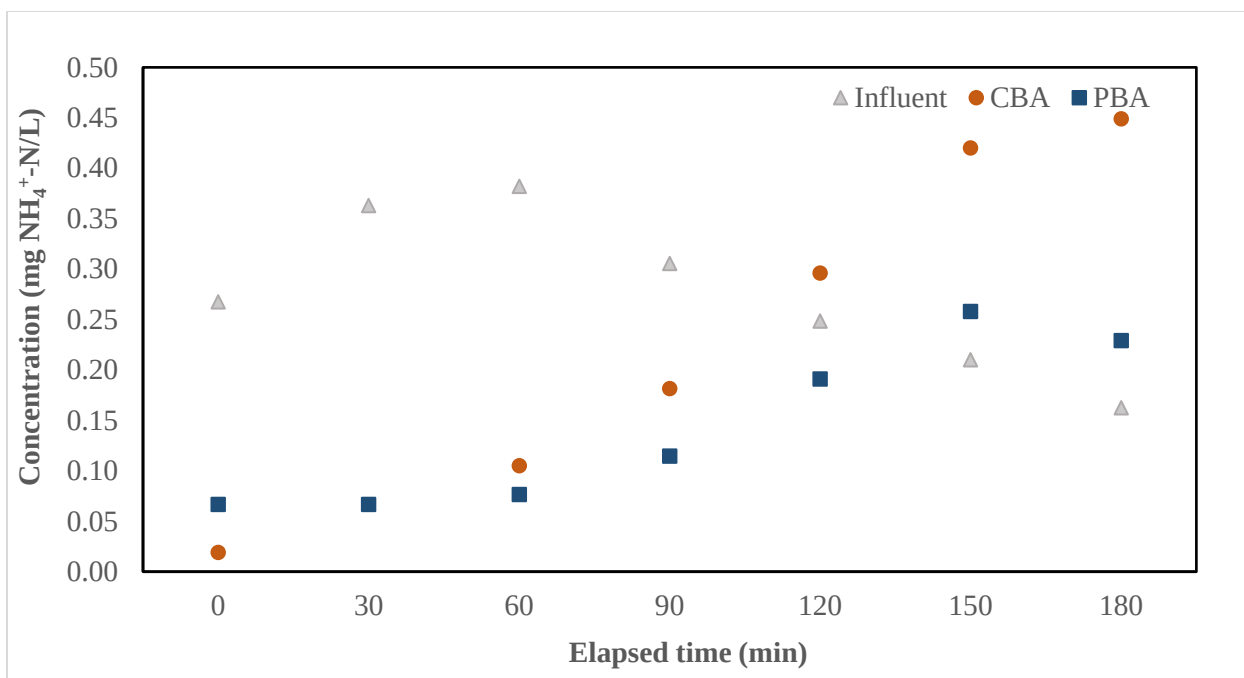


Figure A45: NH₄⁺ loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

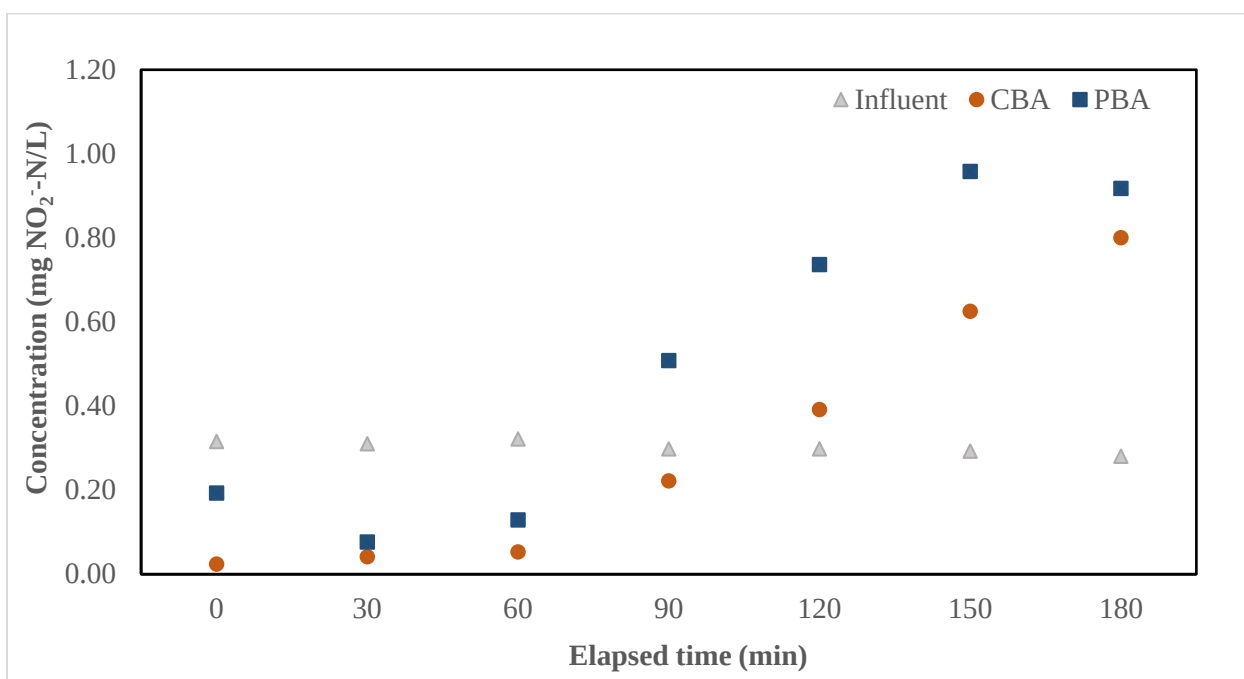


Figure A46: NO₂⁻ loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

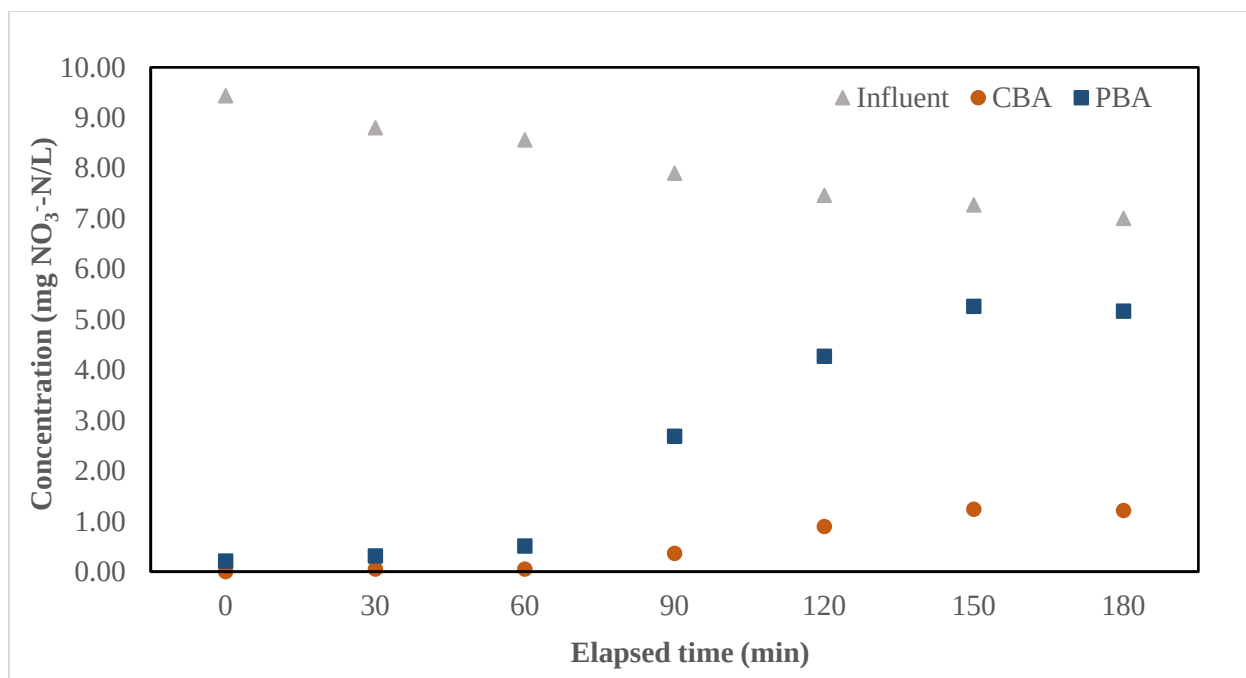


Figure A47: NO₃⁻ loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

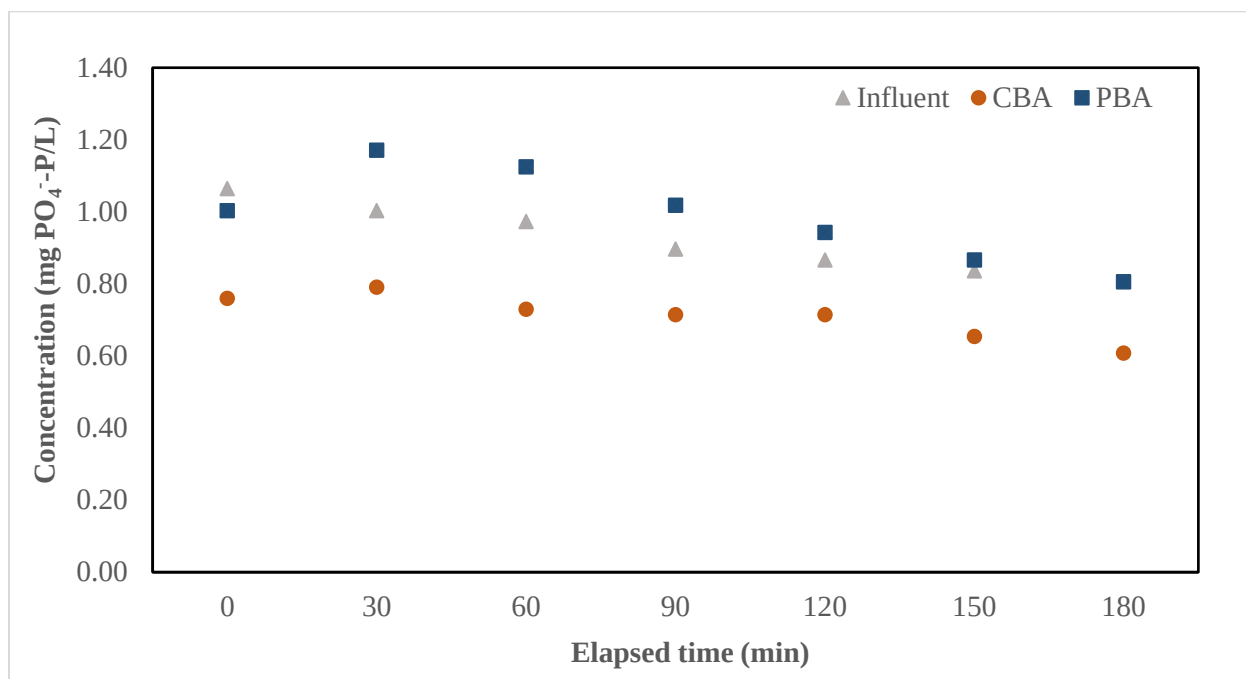


Figure A48: PO₄⁻ loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

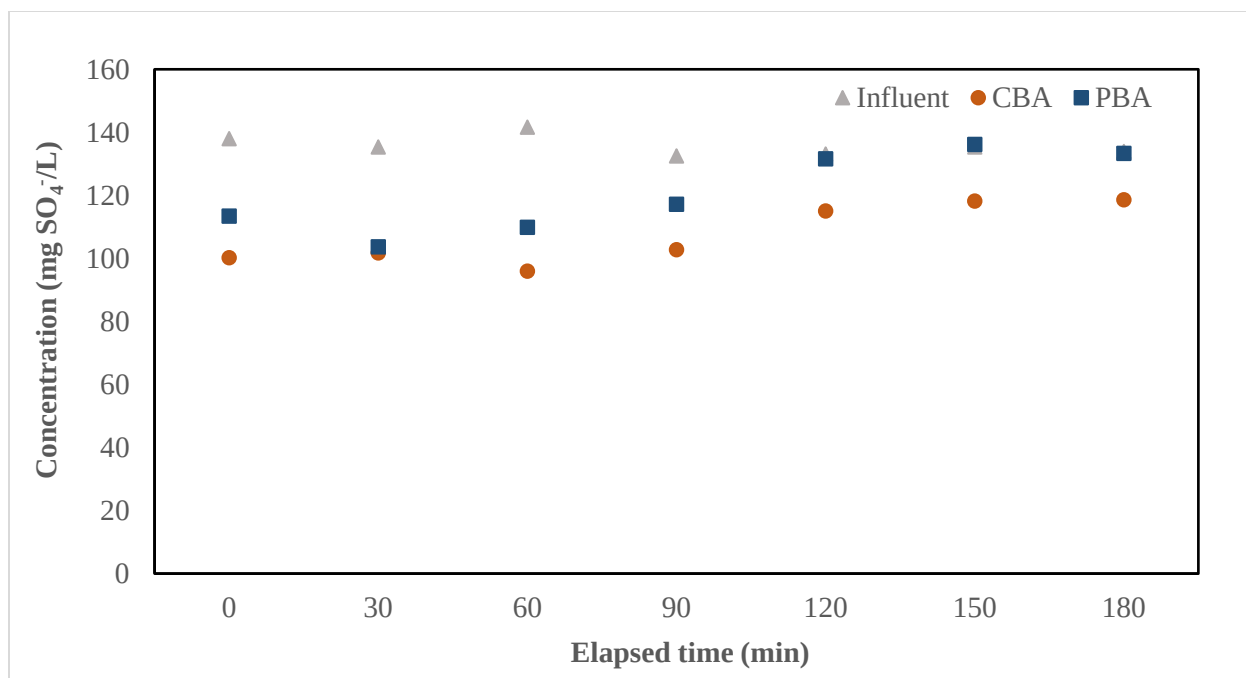


Figure A49: SO_4^{2-} loading concentrations at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

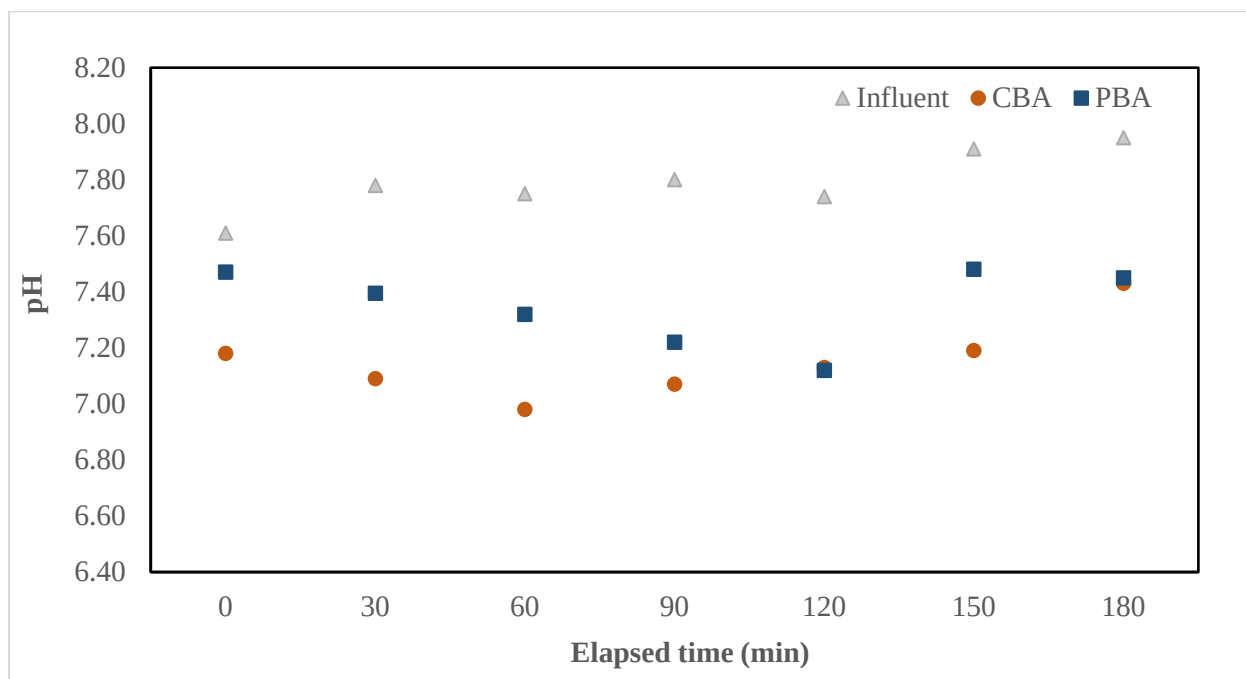


Figure A50: Influent pH at the optimal configuration (Med HLR, Upper outlet height, with plants) events event 18.

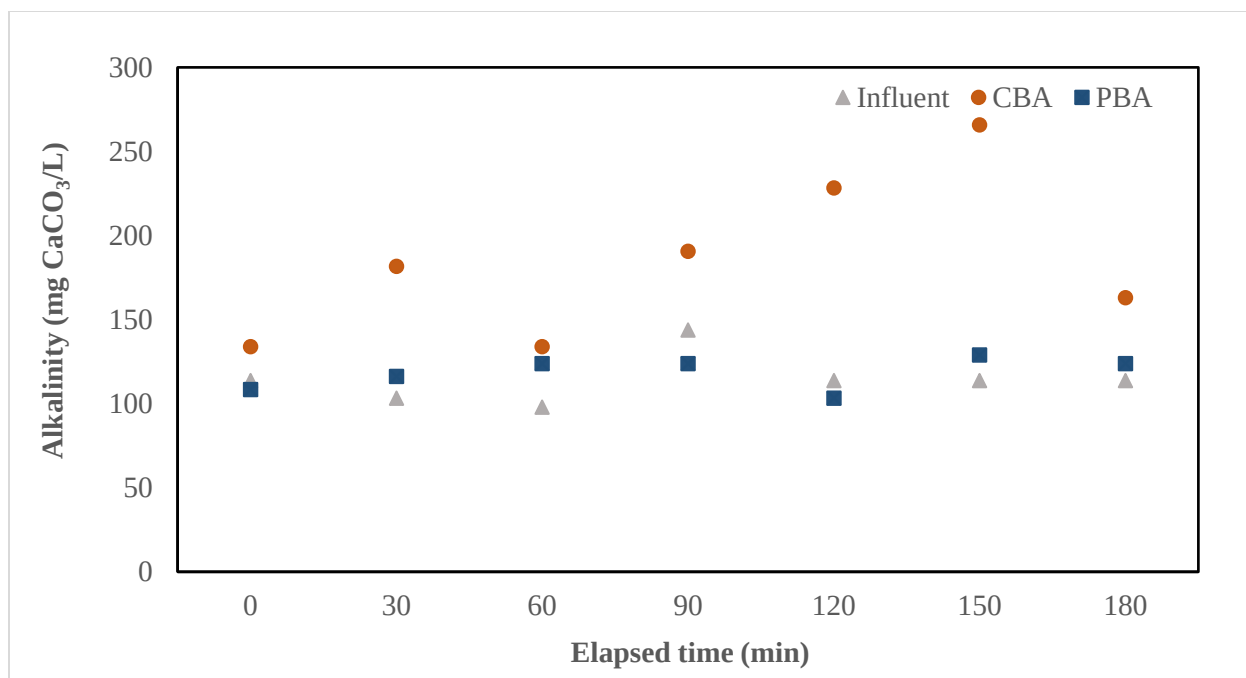


Figure A51: Influent Alkalinity at the optimal configuration (Med HLR, Upper outlet height, with plants) event 18.

Table A16: Average phosphate removal in a one-inch column with a thickness of 4 inches (n=4).

Hydraulic Loading Rate (cm/s)	0.166 (0.162 – 0.167)
Contact Time (s)	28.6 (28.3 – 29.2)
Effluent Concentration (mg PO ₄ ³ -P/L)	0.04 (0.03 – 0.04)

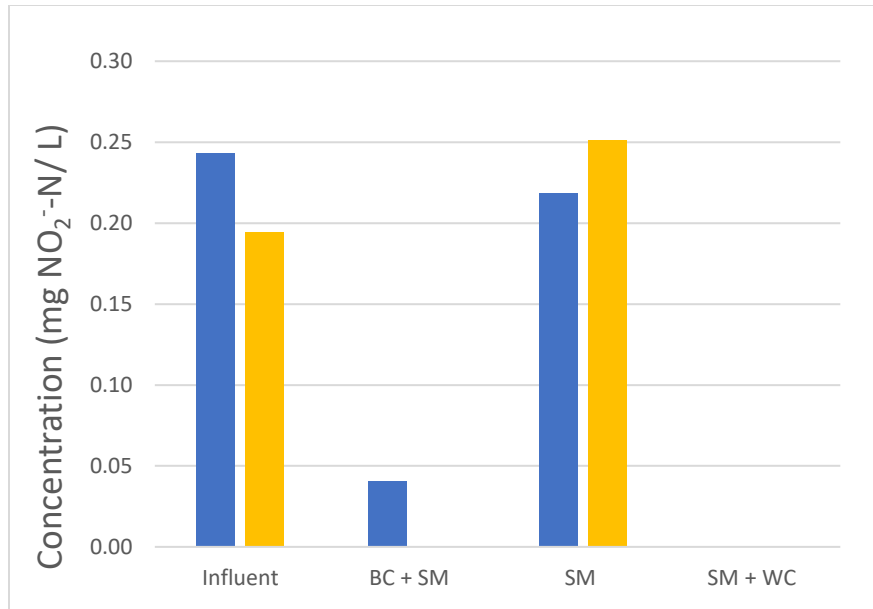


Figure A52: Batch media test $\text{NO}_2^- \text{-N}$ results.

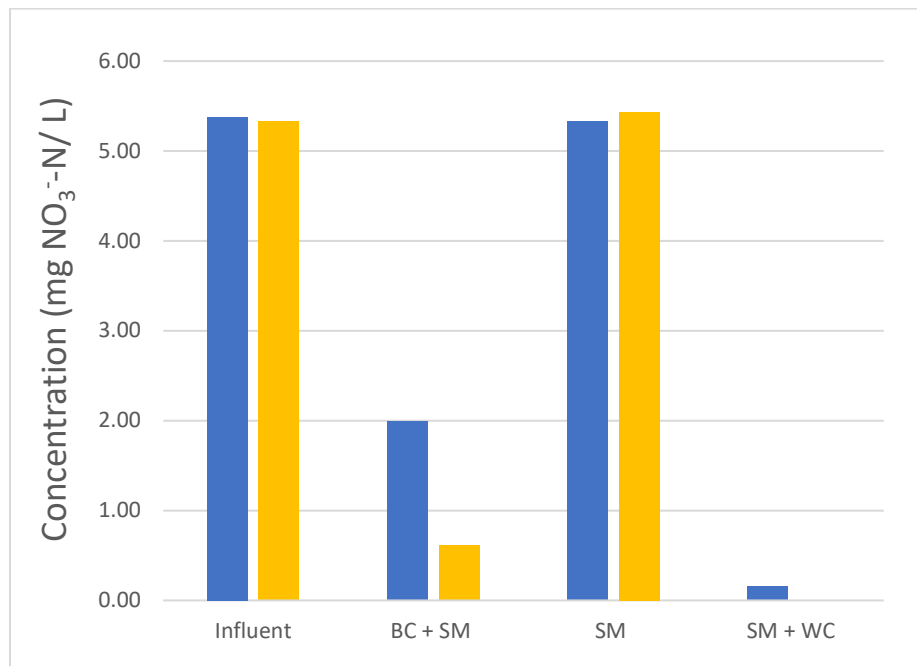


Figure A53: Batch media test $\text{NO}_3^- \text{-N}$ results.

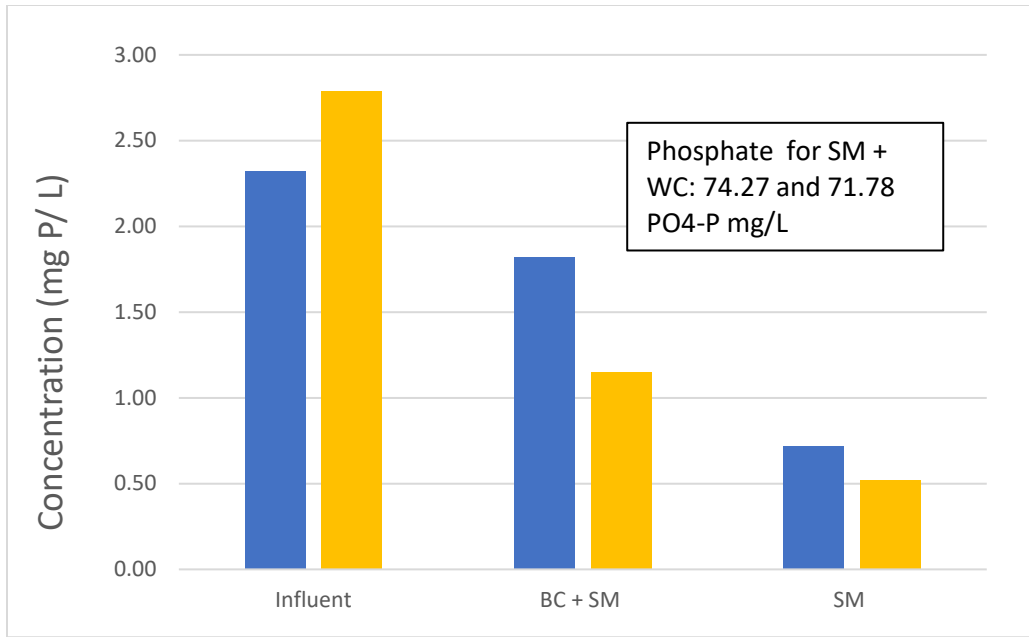


Figure A54: Batch media test PO₄- results.

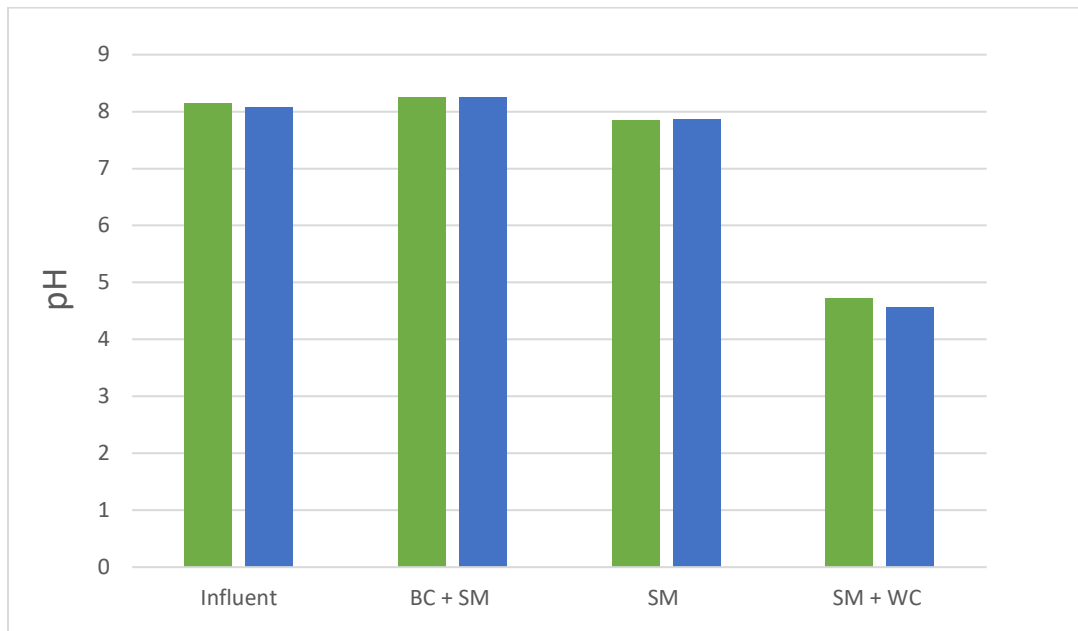


Figure A55: Batch media test pH results.

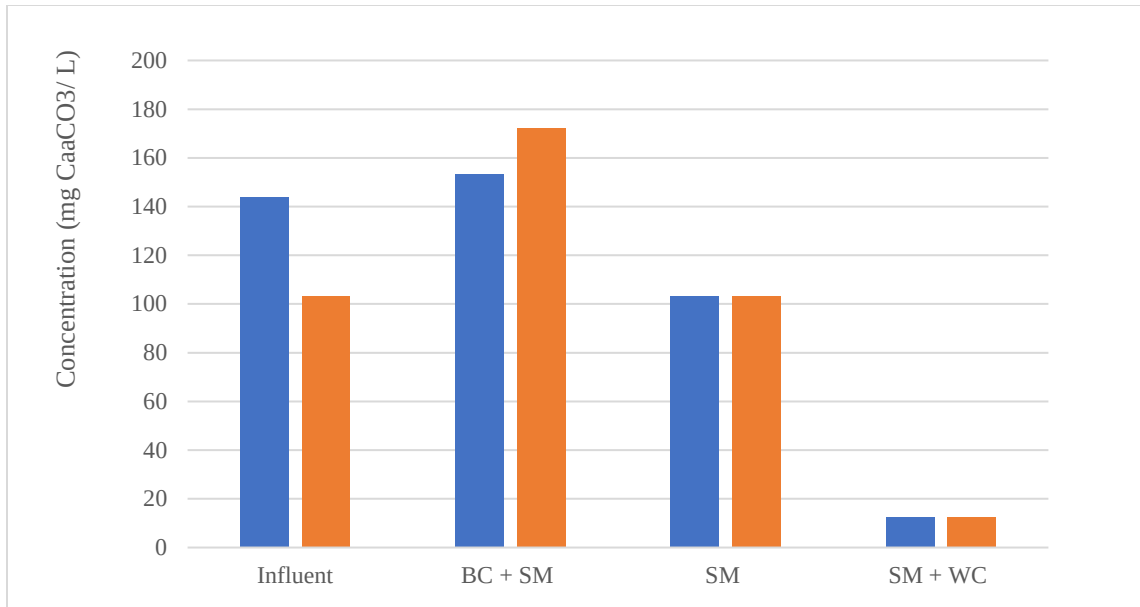


Figure A56: Batch media test alkalinity results.

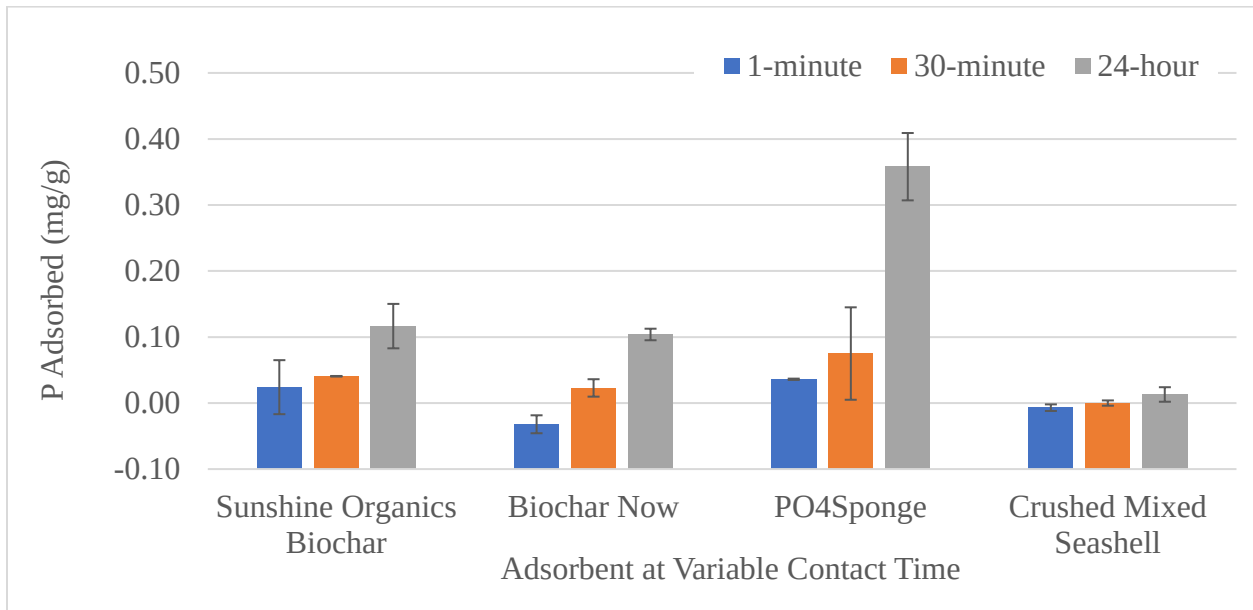


Figure A57: Average normalized phosphate removal of adsorbents not selected for additional kinetic studies.

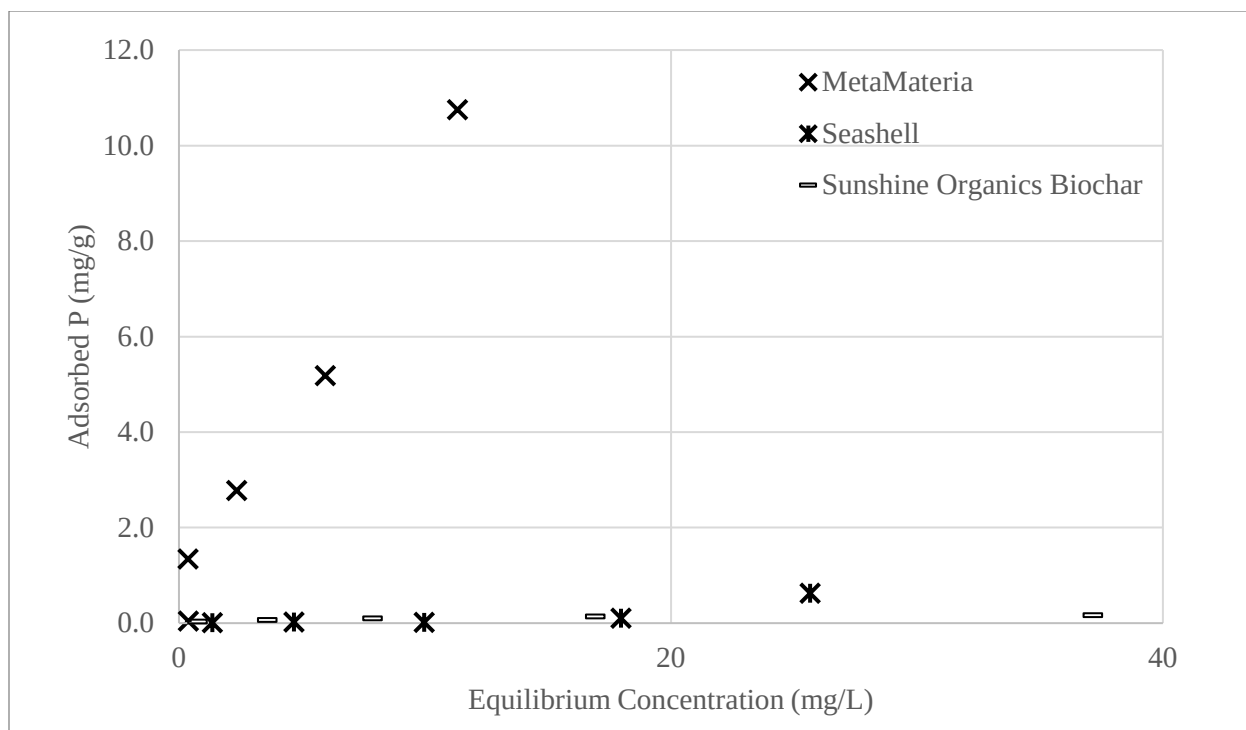


Figure A58: Adsorption capacity of phosphate after 24-hours on adsorbents not selected for additional kinetic studies.



Figure A59: Photos from workshop at the study site in April 2024.



Figure A60: Image of study site post-dismantling.



Figure A61: Image of roots of Muhly plant from CBA unit still embedded with the potting mix.



Figure A62: Image of roots of Muhly plant from PBA unit still embedded with the potting mix.

Infographic (analysis of pre-webinar survey responses)

On 03/06/24 pre-webinar surveys were distributed (results shared here). 36 responses observed.

On 04/05/24 University of South Florida hosted a webinar entitled, “Biochar Amended Bioretention Systems for Nutrient Management from Agricultural and Urban Runoff”.

On 04/15/2024, The University of South Florida hosted an in-person workshop on the bioretention systems at Holmberg Farms.

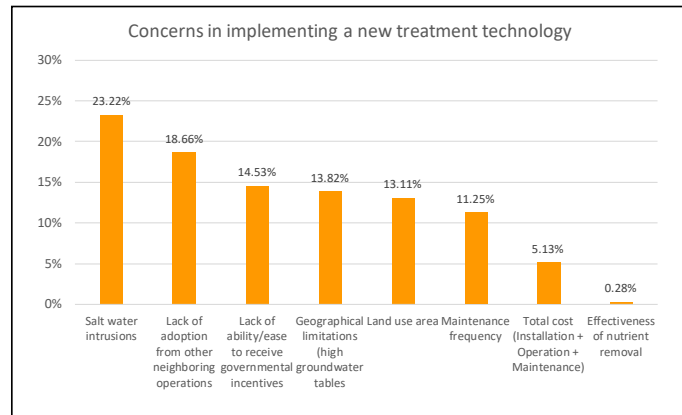
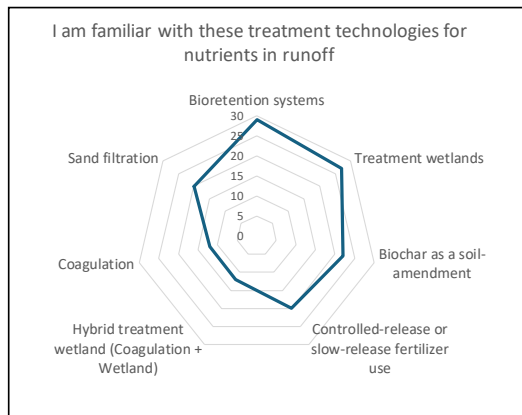
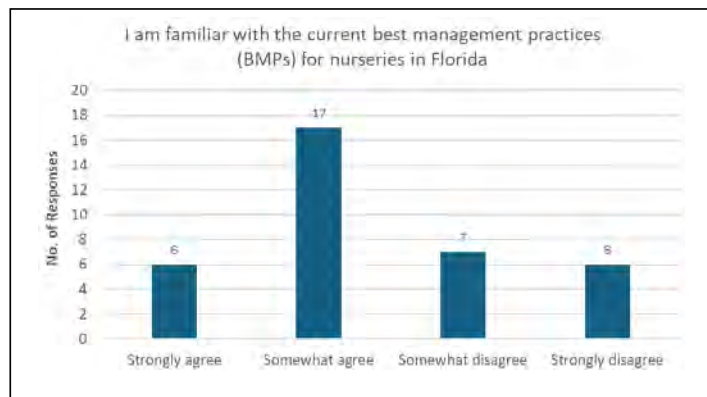
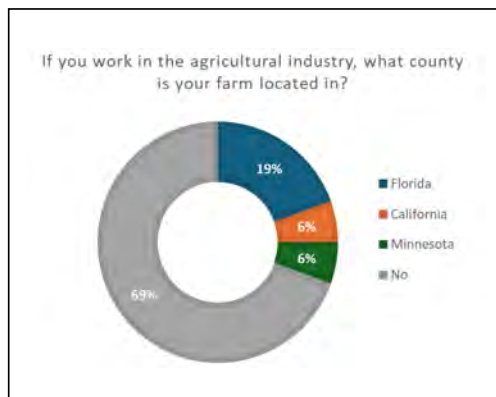
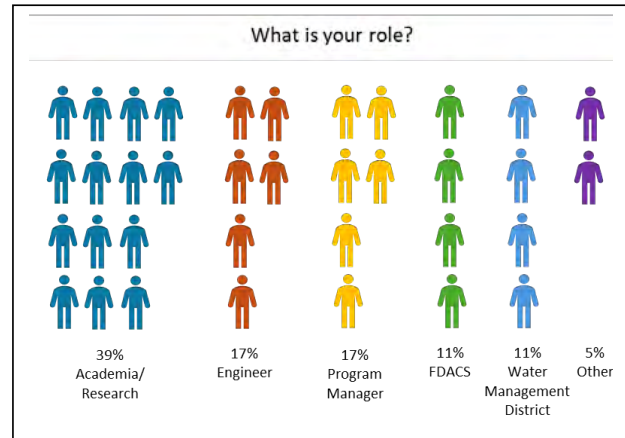


Figure A63: Infographic of webinar attendance and survey responses.

Table A17: Survey responses on familiarity of different treatment technology and openness to recommend or implement the technology ranking from top to bottom, strongly agree to strongly disagree.

Survey Statement	Strongly Agree	Somewhat Agree	Somewhat Disagree	Strongly Disagree	NA	Total
Familiarity with current BMPs for nurseries in Florida	6	17	7	6	0	36
Openness to/recommend implementing a treatment wetland to treat nutrients in runoff	15	12	0	0	9	36
Openness to/recommend controlled release fertilizers to prevent nutrients in runoff	13	6	1	0	16	36
Openness to/recommend bioretention system to treat nutrients in runoff	12	15	0	0	9	36
Openness to/recommend amending soil with biochar to treat nutrients in runoff	9	13	0	0	14	36
Openness to/recommend implementing a hybrid wetland treatment to treat nutrients in runoff	7	5	0	0	24	36
Openness to/recommend using coagulation to treat nutrients in runoff	4	6	1	0	25	36
Openness to/recommend implementing a sand filter to treat nutrients in runoff	7	10	3	1	15	36